GSSSB · STAFF NURSE EXAM

Child Health Nursing (Paediatrics)

Complete Colourful Notes · 3D Diagrams · Mnemonics · Exam Points



19 Chapters | 3D Coloured Diagrams | Growth · Immunization · IMNCI | Systemic disorders | Paediatric drugs & dosage | Exam P

Hinglish explanation + English exam terms



Sister Asha
Aapki Nursing teacher

Index – Kya padhenge?

1 **Introduction to Child Health Nursing & National Programmes**
Age groups, child health indicators, paediatric nurse roles, child rights, RBSK, RKSK, HBNC, HBYC, IMNCI, ICDS, POSHAN, Mission Indradhanush, important days

2 **Growth & Development**
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17 Behavioural & Developmental Disorders, Child Abuse, Accidents & Poisoning
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18 Paediatric Emergencies & Life Support
ETAT triage, paediatric BLS/ CPR, choking, shock, respiratory distress, oxygen therapy, convulsions, hypoglycaemia, anaphylaxis, emergency drugs & doses, raised ICP, GCS

19 Quick Revision – Master Tables & 150 One-Liners
All paediatric numbers in one place – growth, milestones, vitals, vaccines, doses, signs, eponyms, programmes; 150 exam one-liners

★ **Key Points** – sabse important facts

Asha Didi ki Tip – samjhane wali baat

! Nursing Alert – galti mat karna

✎ **Exam Point** – PYQ style, baar baar aata hai

♫ **Mnemonic** – yaad karne ki trick

D Definition – exact definition

1 Introduction to Child Health Nursing & National Programmes

1.1 Basic Concepts

D Child Health (Paediatric) Nursing

Specialised nursing care of children from **birth to adolescence** – promotion of growth & development, prevention of illness, care during illness and rehabilitation, with **family-centred care** as the core philosophy. Father of Paediatrics – **Abraham Jacobi**.

Age group	Period
Neonate (newborn)	**Birth – 28 days** (early neonatal 0–7 days, late 8–28 days)
Infant	**Birth – 1 year**
Toddler	**1 – 3 years**
Preschooler	**3 – 6 years**
School-age	**6 – 12 years**
Adolescent	**10 – 19 years** (WHO) · teenager 13–19
Perinatal period	**28 weeks gestation – 7 days after birth**
Under-five	0 – 5 years (most vulnerable)

- **Children (0–14 yrs) ≈ 31 % of India's population (Census 2011)** – a large vulnerable group.
- **Concept of "child health"** – not just absence of disease but complete physical, mental, social wellbeing + optimal growth & development.
- **Modern trends**: family-centred care, **atraumatic care** (minimise psychological & physical distress – EMLA, play, parents present), rooming-in, **day-care surgery**, home-based newborn care, preventive & promotive focus, evidence-based practice, child rights, **"Child-friendly hospital"**.

Roles of a paediatric nurse

- Direct **caregiver**, **health educator** (parents), **advocate** (child rights), **counsellor**, **coordinator/ collaborator**, researcher, manager, **case finder** (screening), **play therapist**, consultant, protector (child abuse detection).

1.2 Child Health Indicators (India – SRS 2021)

Indicator	Definition	India
IMR	Infant deaths (< 1 yr) per **1000 live births**	**27**
NMR	Neonatal deaths (< 28 days) per 1000 live births	**19**
U5MR	Under-5 deaths per 1000 live births	**31**
Perinatal mortality rate	Still births (≥ 28 wks) + early neonatal deaths per 1000 **total births**	–

- **SDG 3.2 (by 2030)**: **NMR ≤ 12**, **U5MR ≤ 25** per 1000 live births.
- **Leading causes of neonatal death** – **prematurity/ LBW (most common)**, birth asphyxia, neonatal sepsis, congenital anomalies.
- **Leading causes of under-5 death** – neonatal causes, **pneumonia**, **diarrhoea**, malnutrition (underlying ~45 %), injuries.
- **IMR is the most sensitive indicator** of health status of a community.

1.3 Child Rights & Laws

Law/ Convention	Key point
UNCRC (UN Convention on Rights of the Child)	Adopted **20 Nov 1989**; India ratified **1992**; 4 core rights – **survival, development, protection, participation**
Universal Children's Day	**20 November**
Children's Day (India)	**14 November** (Pandit Nehru's birthday)
POCSO Act 2012	Protection of Children from Sexual Offences – child = **< 18 yrs**; **mandatory reporting** (nurse also)
Juvenile Justice (Care & Protection) Act 2015	Child in conflict with law/ in need of care; **16–18 yrs heinous crimes** – may be tried as adult
Right to Education (RTE) Act 2009	Free & compulsory education **6–14 yrs** (Article 21A)
Child & Adolescent Labour (Prohibition) Act 1986 (amended 2016)	No employment **< 14 yrs**; adolescents 14–18 not in hazardous work
Prohibition of Child Marriage Act 2006	Girls < 18, boys < 21 = child marriage
PCPNDT Act 1994	Bans sex selection/ sex determination
Infant Milk Substitutes (IMS) Act 1992 (amended 2003)	Promotion of formula/ feeding bottles banned (breastfeeding protection)
CHILDLINE	**1098** – 24×7 free emergency helpline for children
NCPCR	National Commission for Protection of Child Rights (2007)

1.4 National Child Health Programmes

Programme	Year	Key points
EPI (Expanded Programme on Immunization)	**1978**	Started immunization
UIP (Universal Immunization Programme)	**1985**	Now 12 VPDs (+ JE in endemic districts)
Pulse Polio	**1995**	OPV to all < 5 yrs on NIDs; India **polio-free certified 27 March 2014** (last case **13 Jan 2011**, Howrah, WB)
ICDS (Integrated Child Development Services)	**2 Oct 1975**	Anganwadi; beneficiaries – **children 0–6 yrs, pregnant & lactating women, adolescent girls**; **6 services** – supplementary nutrition, immunization, health check-up, referral, pre-school non-formal education, nutrition & health education

Programme	Year	Key points
IMNCI	**2003** (India)	Integrated Management of Neonatal & Childhood Illness – 0–2 months & 2 months–5 yrs
NRHM → NHM	2005 / 2013	**RMNCH+A** strategy (2013) – continuum of care
JSSK	2011	Free care for sick infants (now up to 1 yr) incl. transport
HBNC (Home Based Newborn Care)	**2011**	ASHA home visits – institutional delivery: **day 3, 7, 14, 21, 28, 42** (6 visits); home delivery – **+ day 1** (7 visits)
HBYC (Home Based Young Child care)	**2018**	ASHA visits at **3, 6, 9, 12, 15 months** – nutrition, ORS/ zinc, development, WASH
RBSK (Rashtriya Bal Swasthya Karyakram)	**Feb 2013**	**"4 D's"** – **D**efects at birth, **D**eficiencies, **D**iseases, **D**evelopmental delays incl. disability; 0–18 yrs; mobile health teams; **DEIC** (District Early Intervention Centre)
RKSK (Rashtriya Kishor Swasthya Karyakram)	**Jan 2014**	Adolescents **10–19 yrs**; 6 themes – nutrition, sexual & reproductive health, NCDs, substance misuse, injuries & violence, mental health; **AFHC** (Adolescent Friendly Health Clinics)
Mission Indradhanush	**25 Dec 2014**	Full immunization of left-out/ drop-out children; **IMI** (Intensified MI) 2017 onwards
MAA (Mothers' Absolute Affection)	**2016**	Breastfeeding promotion
POSHAN Abhiyaan (National Nutrition Mission)	**8 Mar 2018** (Jhunjhunu, Rajasthan)	Reduce stunting, undernutrition, anaemia, LBW; **Poshan Maah – September**
Anaemia Mukt Bharat	**2018**	6×6×6 strategy; IFA supplementation by age (see Ch. 3)
National Deworming Day	2015	**10 Feb & 10 Aug** – albendazole 1–19 yrs
SAANS	**2019**	Social Awareness & Action to Neutralise Pneumonia Successfully
IDCF / STOP Diarrhoea campaign	2014 / 2024	ORS + zinc distribution (pre-monsoon)
PM POSHAN (Mid-day meal)	**1995** (renamed 2021)	Hot cooked meal for school children
LaQshya, SNCU/ NBSU/ NBCC	–	Facility newborn care – **NBCC** (every delivery point), **NBSU** (FRU/ CHC), **SNCU** (district hospital)
Gujarat – Bal Sakha Yojana	2009	Newborn care of BPL babies by private paediatricians (PPP)

1.5 Important Days

Day	Date
National Girl Child Day	**24 January**
National Deworming Day	**10 February** & 10 August
National Vaccination Day	**16 March**

Day	Date
World Immunization Week	**Last week of April**
ORS Day	**29 July**
World Breastfeeding Week	**1–7 August**
National Nutrition Week	**1–7 September**
International Day of the Girl Child	11 October
Children's Day (India)	**14 November**
Newborn Care Week	**15–21 November**
World Prematurity Day	**17 November**
Universal Children's Day	**20 November**

✎ PYQ style

- Neonatal period – **first 28 days**; infancy – **first year**; toddler – **1–3 yrs**.
- Most sensitive indicator of community health – **IMR**.
- Commonest cause of neonatal mortality – **prematurity/ LBW**.
- UNCRC adopted – **20 Nov 1989**; child helpline – **1098**.
- ICDS launched – **2 Oct 1975**; worker – **Anganwadi worker**.
- RBSK covers – **4 D's**, age **0–18 yrs**; RKSK age – **10–19 yrs**.
- HBNC visits (institutional delivery) – **6 visits: 3, 7, 14, 21, 28, 42 days**.
- India declared polio-free – **27 March 2014**.
- Mission Indradhanush launched – **25 Dec 2014**.
- Children's Day – **14 Nov**; ORS Day – **29 July**.

2 Growth & Development

2.1 Definitions & Principles

D Growth vs Development

Growth – increase in **size/ mass** (physical, **quantitative**, measurable – weight, height). **Development** – increase in **skill & function** (**qualitative**, functional maturity). **Maturation** – genetically determined unfolding. Growth stops in adulthood; development continues lifelong.

- **Principles of growth & development**:
- Continuous & orderly process; **sequence is same for all children, rate differs**.
- **Cephalocaudal** – head → toe (head control before sitting/ walking).
- **Proximodistal** – centre → periphery (shoulder control before fingers).
- **General → specific** (mass → specific activity), **simple → complex**.
- Different organs grow at different rates – **brain/ nervous tissue fastest early** (brain 25 % adult weight at birth, ~75 % by 2 yrs, ~90 % by 5 yrs); **lymphoid tissue** peaks before puberty; **reproductive organs** at puberty.
- **Critical periods** – development most vulnerable at certain times.
- Development is **predictable** and interrelated (physical, mental, social).
- **Periods of rapid growth** – **infancy** (fastest) and **adolescence (growth spurt)**. Slowest – school age (**latent** period).
- **Factors**: **Genetic** (heredity, sex, race), **environmental** – prenatal (maternal nutrition, infections, drugs, radiation), postnatal (**nutrition – most important**, illness, hormones – GH, thyroid, sex hormones, insulin), socioeconomic status, family, culture, climate, emotional factors (**psychosocial dwarfism** – emotional deprivation).

2.2 Physical Growth – Anthropometry

Parameter	Birth	Milestones
Weight	**~3 kg** (India avg 2.8–3 kg)	Physiological loss **up to 10 % in first week**, regained by **10–14 days**; **doubles at 5 months**, **triples at 1 yr**, **quadruples at 2 yrs**, 5 × at 3 yrs, 6 × at 5 yrs, 10 × at 10 yrs
Length/ height	**~50 cm**	**75 cm at 1 yr** (+25 cm), 87 cm at 2 yrs, **100 cm at 4 yrs (double)**, ~150 cm at 12 yrs (triple)
Head circumference (OFC)	**33–35 cm**	**+2 cm/month first 3 months**, 1 cm/month 3–6 months, 0.5 cm/month 6–12 months → **45–47 cm at 1 yr**; 48–49 cm at 2 yrs; ~50 cm at 5 yrs
Chest circumference	**~31–33 cm** (2–3 cm **less than head**)	**Equals head at 1 yr**, then exceeds head
MUAC	~11 cm	**6 months – 5 yrs ≈ constant (> 13.5 cm normal; < 11.5 cm SAM)**

Parameter	Birth	Milestones
Upper : lower segment ratio	**1.7 : 1**	1.3 at 3 yrs, **1 : 1 at 7–10 yrs**

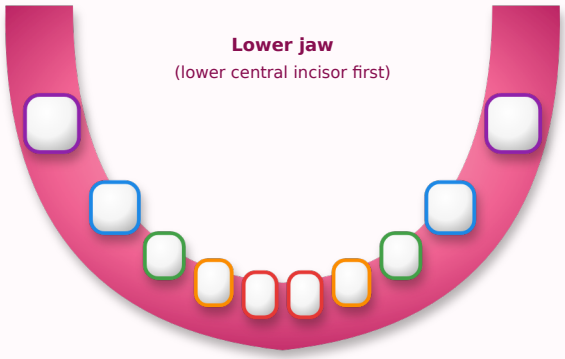
- **OFC measured** – tape over **occipital protuberance & supra-orbital ridges** (largest circumference).
- **Length** measured **lying (infantometer)** < 2 yrs; **standing (stadiometer)** > 2 yrs.
- **Anterior fontanelle** closes **12–18 months** (diamond); **posterior** – **6–8 weeks** (triangular). **Large AF/ delayed closure** – rickets, hypothyroidism, hydrocephalus, Down syndrome. **Early closure** – microcephaly, craniosynostosis.

Weight & height formulas (IAP – Weech's)

Age	Formula
3–12 months – weight (kg)	**(Age in months + 9) ÷ 2**
1–6 years – weight (kg)	**Age (yrs) × 2 + 8**
7–12 years – weight (kg)	**(Age (yrs) × 7 − 5) ÷ 2**
2–12 years – height (cm)	**Age (yrs) × 6 + 77**

- E.g. **5-year child**: weight = 5 × 2 + 8 = **18 kg**; height = 5 × 6 + 77 = **107 cm**.
- **Growth charts** – **WHO growth standards (2006)** used in MCP card – **weight-for-age** (underweight), **length/ height-for-age** (**stunting – chronic malnutrition**), **weight-for-height** (**wasting – acute malnutrition**), BMI-for-age. **Z-score < −2 SD = moderate, < −3 SD = severe**.
- **Road to Health chart** (growth chart, **David Morley**) – weight-for-age; flattening/ downward curve = danger.

Primary (Milk) Teeth – Eruption Sequence (20 teeth)



- **Central incisors** – 6–8 months
- **Lateral incisors** – 8–11 months
- **First molars** – 12–16 months
- **Canines** – 16–20 months
- **Second molars** – 20–30 months

Order: incisors → 1st molar → canine → 2nd molar

Formula: No. of milk teeth (up to 2 yrs) = Age in months − 6 · All 20 by 2½ yrs

Permanent: 32 teeth · first permanent tooth = 1st molar at 6 yrs · 3rd molars 17–25 yrs

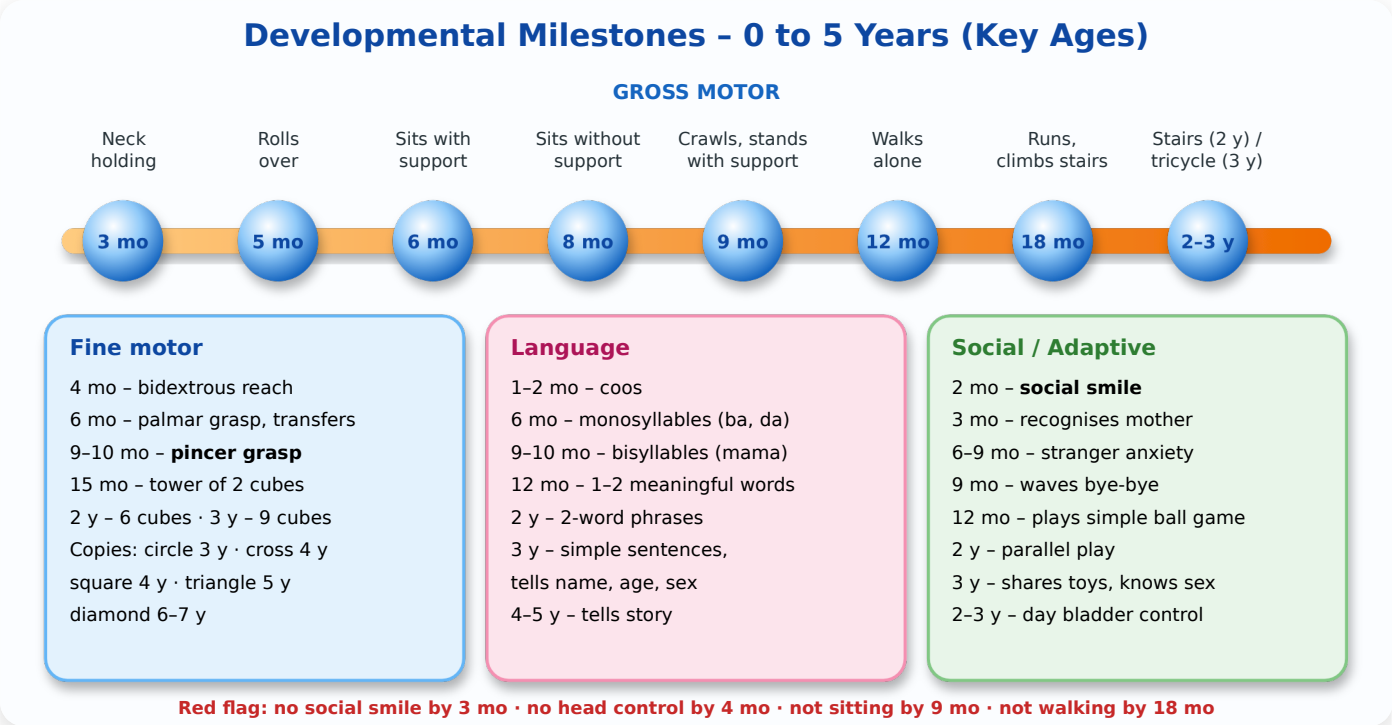
Delayed dentition (no tooth by 13 months): rickets, hypothyroidism, Down syndrome, idiopathic

Primary teeth – eruption order & formula

Dentition

- **First tooth** – **lower central incisor at 6–7 months**; **all 20 milk teeth by 2½ yrs (24–30 months)**.
- **No. of teeth (6–24 months) = age in months − 6**.
- **Permanent teeth – 32**; **first permanent tooth = first molar at 6 yrs**; last – third molar (17–25 yrs).
- **Delayed dentition (> 13 months)**: rickets, hypothyroidism, Down syndrome, malnutrition.
- **Teething** – drooling, irritability, gum rubbing; **does NOT cause high fever/ diarrhoea** – look for other causes.
- **Natal teeth** (at birth) / neonatal teeth (first month).

2.3 Developmental Milestones



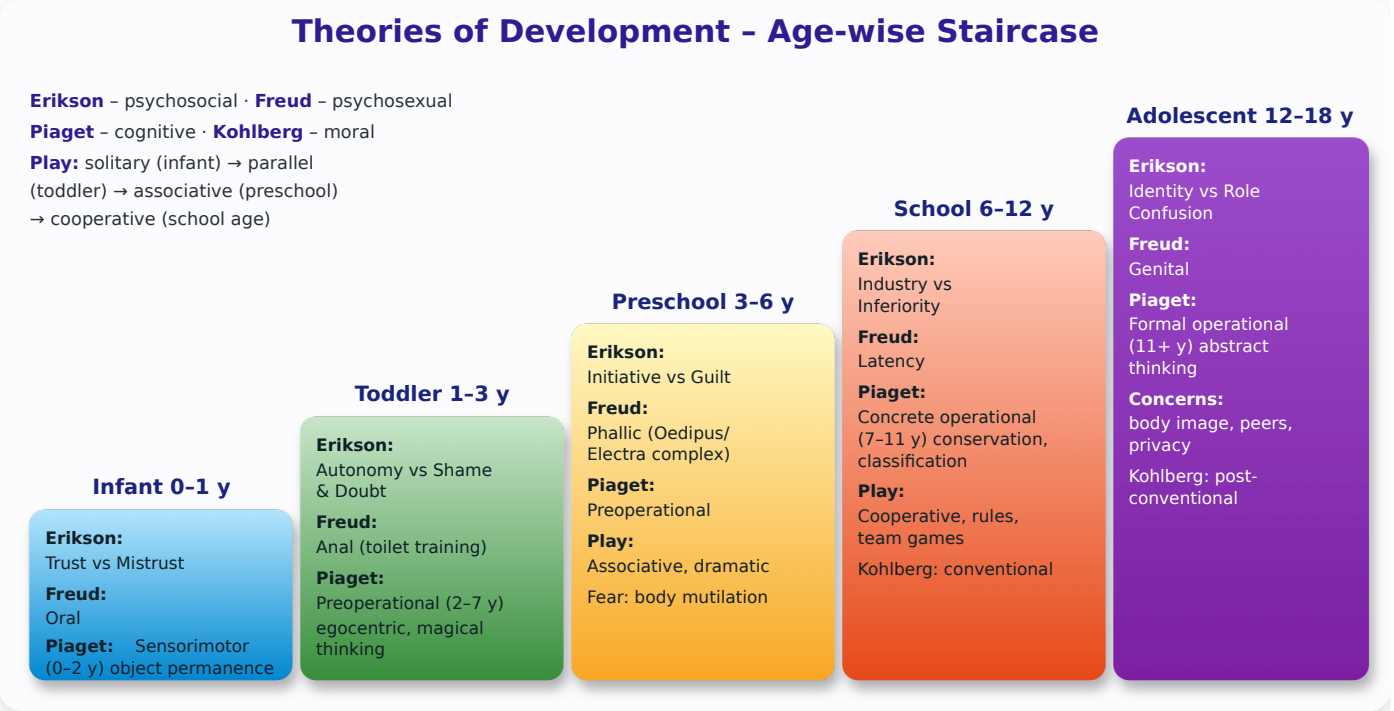
Key milestones – gross motor, fine motor, language & social

Age	Gross motor	Fine motor	Language	Social/ adaptive
1 month	Lifts head momentarily (prone)	Hands fisted	Startles to sound	Watches face
2 months	Head up 45° in prone	–	**Coos**	**Social smile (6–8 wks)**
3 months	**Neck holding (head control)**	Hands open, holds rattle	Laughs	**Recognises mother**
4–5 months	**Rolls over** (front → back 4 m, back → front 5 m)	**Bidextrous reach**	Laughs loud	Enjoys mirror
6 months	**Sits with support** (tripod)	**Palmar grasp, transfers** objects hand to hand	**Monosyllables** (ba, da)	**Stranger anxiety** begins (6–9 m)
8 months	**Sits without support**	–	–	–

Age	Gross motor	Fine motor	Language	Social/ adaptive
9 months	**Crawls, stands with support**	**Pincer grasp (immature 9 m, mature 10–12 m)**	**Bisyllables** (mama, dada)	**Waves bye-bye**, plays peek-a-boo
12 months	**Walks alone/ with one hand held** (10–15 m)	Releases object	**1–2 meaningful words**	Plays simple ball game
15 months	Walks well, creeps upstairs	**Tower of 2 cubes**, scribbles	4–6 words	Points to wants
18 months	**Runs**, walks upstairs (hand held)	Tower of 3–4 cubes, turns 2–3 pages	10–20 words	Feeds self with spoon, copies parents
2 years	**Walks up & down stairs (2 feet/ step)**, kicks ball	**Tower of 6–7 cubes**, turns single page	**2-word phrases**, uses "I, me"	**Parallel play**, dresses with help
3 years	**Rides tricycle**, climbs stairs alternate feet	**Tower of 9 cubes**, **copies circle**	**Sentences**, tells name, age, sex	**Day + night bladder control** (most), shares toys
4 years	**Hops on one foot**	**Copies cross (+) & square**	Tells story	Plays cooperatively, goes to toilet alone
5 years	**Skips**	**Copies triangle**	Fluent speech	Dresses/ undresses

- **Red flags (refer)**: no social smile by **3 months**, no head control by **4 months**, not sitting by **9 months**, **not walking by 18 months**, no words by **16–18 months**, no 2-word phrases by 2 yrs, **any loss of skills (regression) at any age**.
- **Developmental screening tools**: **DDST-II (Denver Developmental Screening Test, 0–6 yrs)**, **Trivandrum Developmental Screening Chart (TDSC)**, **Baroda Development Screening Test**, **Gesell**, Bayley scales, **Vineland Social Maturity Scale**, **Goodenough Draw-a-Man test** (IQ in children), **Stanford-Binet** (IQ).
- **Toilet training** – readiness **18–24 months** (bladder control signs); **bowel control before bladder**; **day bladder by 2–3 yrs, night by 3–5 yrs**; **enuresis** diagnosed only **after 5 yrs**.

2.4 Theories of Development



Erikson, Freud, Piaget & play – age-wise

Age	Erikson (psychosocial)	Freud (psychosexual)	Piaget (cognitive)
0–1 yr	**Trust vs Mistrust** (virtue – hope)	**Oral**	**Sensorimotor (0–2 yrs)** – **object permanence (~9 months)**, reflexes → purposeful
1–3 yrs	**Autonomy vs Shame & Doubt** (will)	**Anal** (toilet training)	**Preoperational (2–7 yrs)** – **egocentrism**, **magical thinking**, **animism**, centration, irreversibility, symbolic play
3–6 yrs	**Initiative vs Guilt** (purpose)	**Phallic** (Oedipus/ Electra complex)	Preoperational
6–12 yrs	**Industry vs Inferiority** (competence)	**Latency**	**Concrete operational (7–11 yrs)** – **conservation**, reversibility, classification, seriation
12–18 yrs	**Identity vs Role confusion** (fidelity)	**Genital**	**Formal operational (11+ yrs)** – **abstract**, hypothetical reasoning
Young adult	Intimacy vs Isolation	–	–
Middle adult	Generativity vs Stagnation	–	–
Old age	Integrity vs Despair	–	–

- **Kohlberg (moral development)** – **pre-conventional** (punishment–obedience, < 9 yrs) → **conventional** (good boy/ girl, law & order – school age/ adolescence) → **post-conventional** (social contract, universal ethics).
- **Sullivan** – interpersonal theory; **Bandura** – social learning (modelling); **Bowlby** – attachment; **Gesell** – maturational theory.
- **Fears**: infant – strangers (6–9 m); **toddler – separation**; **preschool – body mutilation, dark, ghosts**; school – failure, loss of control, death; adolescent – body image, peers.

2.5 Play

- **Functions**: sensorimotor development, intellectual, socialisation, creativity, self-awareness, **therapeutic value** (expresses fears), moral.

Type of play	Age	Example
Solitary	**Infant**	Plays alone – rattles, mobiles, mirror
Onlooker	Toddler	Watches others play
Parallel	**Toddler**	**Plays beside, not with** others – push-pull toys, blocks
Associative	**Preschool**	Plays together without rules/ organisation – dress-up, **make-believe/ dramatic**
Cooperative	**School age**	Organised, **rules, team games**, competition – board games, sports

- **Suitable toys**: infant – bright mobiles, rattles, soft toys (large, non-toxic, no small parts); toddler – **push-pull toys**, blocks, picture books; preschool – **dolls, doctor set, crayons, tricycle**, puzzles; school – **board games, collections, books, bicycle**, hobbies; adolescent – music, sports, computer.
- **Therapeutic play** in hospital – **doll play/ medical play** to reduce fear (explaining procedure on doll).

Erikson order

"**T**rust **A**unty **I**n **I**ndia **I**s **I**ntelligent, **G**reat **I**dea" – Trust, Autonomy, Initiative, Industry, Identity, Intimacy, Generativity, Integrity.

PYQ style

- Birth weight doubles at – **5 months**; triples – **1 year**; quadruples – **2 years**.
- Length at 1 yr – **75 cm**; doubles birth length at – **4 years**.
- Head = chest circumference at – **1 year**.
- Anterior fontanelle closes – **12–18 months**; posterior – **6–8 weeks**.
- Weight of 5-year child (IAP) – **2n + 8 = 18 kg**.
- First tooth – **lower central incisor, 6 months**; all milk teeth – **2½ yrs**.
- Social smile – **6–8 weeks**; neck holding – **3 months**; sits without support – **8 months**.
- Pincer grasp – **9–10 months**; walks alone – **12–13 months**.
- Copies circle – **3 yrs**; cross – **4 yrs**; triangle – **5 yrs**.
- Erikson toddler stage – **autonomy vs shame & doubt**; school age – **industry vs inferiority**.
- Object permanence – **sensorimotor stage (Piaget)**; conservation – **concrete operational**.
- Toddler play – **parallel**; school-age – **cooperative**.
- Stunting indicates – **chronic malnutrition** (height-for-age); wasting – **acute**.

3 Nutrition, IYCF & Malnutrition

3.1 Nutritional Requirements

Age	Energy	Protein
0–6 months	**~100 kcal/kg/day** (≈ 90–110)	~1.2–1.5 g/kg
6–12 months	~80 kcal/kg/day	~1.5 g/kg
Fluid (maintenance – Holliday-Segar)	**100 ml/kg (first 10 kg) + 50 ml/kg (next 10 kg) + 20 ml/kg (each kg above 20)**	–

- **Newborn fluid** – day 1: **60 ml/kg**, increase by 15–20 ml/kg/day to **150 ml/kg/day by day 7** (term).
- **Exclusive breastfeeding** till **6 months** – no water, honey, ghutti, glucose water (**prelacteal feeds harmful**).
- **Breast milk** – ~**67–70 kcal/100 ml**, protein 1.1 g/dl (lactalbumin – easily digested), **lactose** (main carbohydrate), **whey : casein = 60 : 40** (cow's milk 20 : 80), **low solute load**, iron poor but **50 % absorbed (lactoferrin)**, rich in **IgA**, lysozyme, bifidus factor, DHA; **colostrum (first 3 days)** – high protein, **IgA, vitamin A**, laxative (meconium passage).
- **Disadvantages of cow's milk < 1 yr** – high protein & solute load, **iron-deficiency anaemia (occult GI bleed)**, low iron/ vit C, allergy, infection risk with bottles.

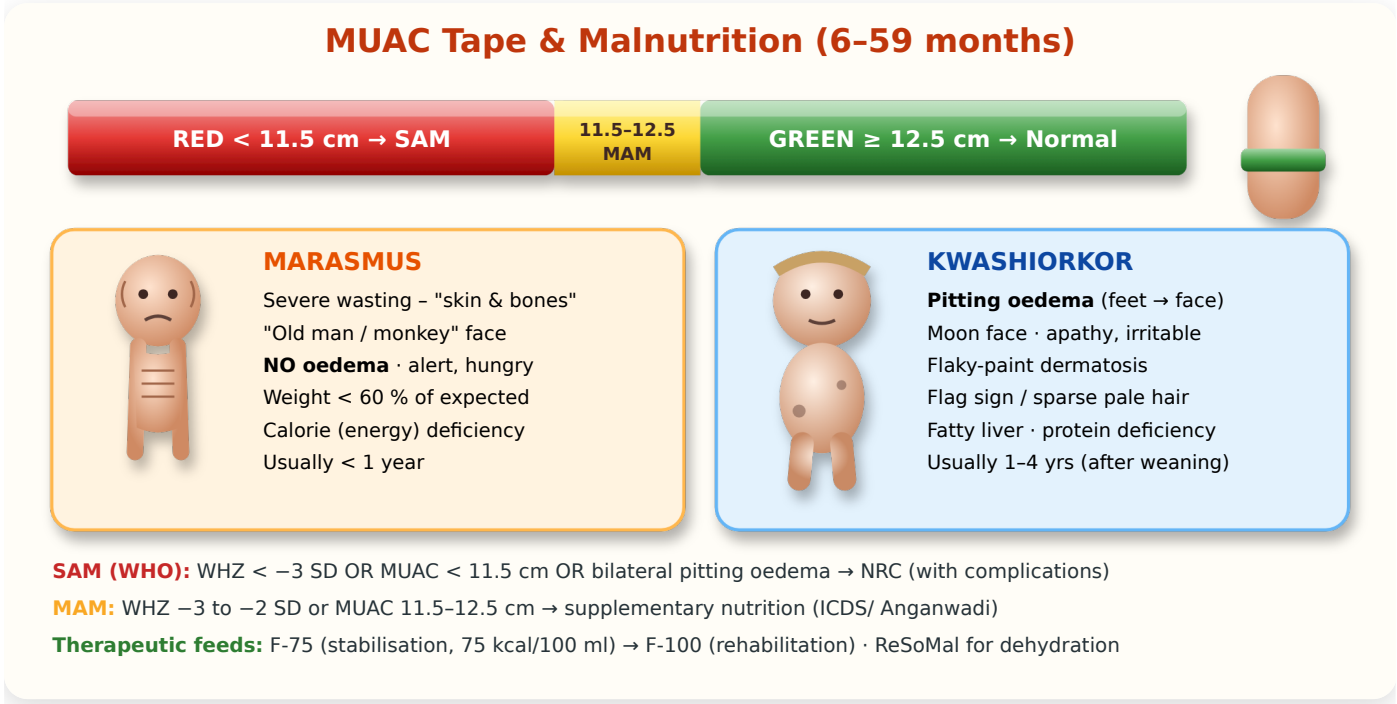
3.2 Infant & Young Child Feeding (IYCF)

- **Early initiation within 1 hour** · **exclusive breastfeeding 0–6 months** · **complementary feeding from 6 months (180 days)** · **continue breastfeeding up to 2 years or beyond**.
- **Complementary feeding** – **semi-solid, mashed**, home-made, **energy-dense** (add a spoon of oil/ ghee), hygienically prepared; start with **2–3 spoons** and increase.

Age	Frequency	Amount per meal	Consistency
6–8 months	**2–3 meals/day** + breastfeeding	**2–3 tablespoons → ½ katori (125 ml)**	Thick mashed/ pureed
9–11 months	**3–4 meals** + 1–2 snacks	**½ katori (250-ml katori)**	Finely chopped, mashed, finger foods
12–24 months	**3–4 meals** + 1–2 snacks	**¾ – 1 katori**	Family foods

- **Responsive feeding**, add **variety (≥ 5 of 8 food groups – minimum dietary diversity)**, avoid **bottle feeding** (use **katori-spoon**/ paladai), avoid junk & sugary drinks.
- **Weaning foods** – khichdi, dalia, mashed banana, suji kheer, **"Sattu"**; **multimix** principle (cereal + pulse + vegetable + fat).
- **Breastfeeding contraindications (rare)** – **galactosaemia** (baby), mother on **anticancer drugs/ radioactive iodine**, untreated active TB (temporary separation; give INH to baby later breastfeed), HIV (India – **exclusive breastfeeding with ART** recommended).

3.3 Protein Energy Malnutrition (PEM)



MUAC tape colours, marasmus vs kwashiorkor & SAM criteria

D **Severe Acute Malnutrition (SAM) – WHO / India (6–59 months)**

Any one: **Weight-for-height < −3 SD** OR **MUAC < 11.5 cm** OR **bilateral pitting oedema** (nutritional). **MAM** – WFH −3 to −2 SD or MUAC **11.5–12.5 cm**. **SAM with complications** (medical complication, failed appetite test, < 6 months) → **NRC (Nutrition Rehabilitation Centre)**; uncomplicated → community management (CMAM).

Classification	Criteria
IAP (weight-for-age % of expected/ 50th centile)	Normal > 80 % · **Grade I 71–80 %** · **II 61–70 %** · **III 51–60 %** · **IV ≤ 50 %** (suffix K if oedema)
Gomez (weight-for-age)	1st degree 75–90 % · 2nd 60–75 % · **3rd < 60 %**
Waterlow	Wasting (weight-for-height) & stunting (height-for-age)
Wellcome Trust	Weight 60–80 % + oedema = **kwashiorkor**; < 60 % no oedema = **marasmus**; < 60 % + oedema = **marasmic kwashiorkor**; 60–80 % no oedema = underweight

	Marasmus	Kwashiorkor
Deficiency	**Calories (energy) + protein** – "starvation"	Mainly **protein** (with adequate/ low calories)
Age	**< 1 yr** (early weaning, bottle feeding)	**1–4 yrs** ("disease of the displaced child" – after next sibling)
Oedema	**Absent**	**Present (pitting) – hallmark**
Weight	**< 60 %**, severe wasting – "**skin & bones**", "**old man/ monkey face**", loose skin folds	60–80 % (masked by oedema), **moon face**

	Marasmus	Kwashiorkor
Appetite / mood	**Good appetite, alert**	**Anorexia, apathy, irritable, miserable**
Skin	Dry, loose	**Flaky-paint dermatosis**, depigmentation
Hair	Thin, dry	**Sparse, easily pluckable, pale – "flag sign"** (alternate bands)
Liver	Normal	**Fatty liver – hepatomegaly**
Serum albumin	Normal/ slightly low	**Very low**
Prognosis	Better	Worse

WHO 10 steps of SAM management (facility)

1. **Hypoglycaemia** – treat/ prevent (< 54 mg/dl → 50 ml 10 % glucose/ sucrose; frequent 2-hrly feeds day & night).
2. **Hypothermia** – keep warm (**KMC, cover, warm room**; rectal < 35.5 °C).
3. **Dehydration** – **ReSoMal** (low sodium, high potassium ORS) **5 ml/kg every 30 min × 2 hrs**, then 5–10 ml/kg/hr; **avoid IV** unless shock (IV only in shock – 15 ml/kg over 1 hr).
4. **Electrolytes** – extra **potassium & magnesium**; **sodium restricted**.
5. **Infection** – **give antibiotics to ALL** (signs often absent).
6. **Micronutrients** – vitamin A, zinc, copper, folic acid; **iron NOT in stabilisation phase** (start in rehabilitation, after 2 weeks).
7. **Cautious feeding** – **F-75** (75 kcal & 0.9 g protein/100 ml) – stabilisation phase (days 1–7).
8. **Catch-up growth** – **F-100** (100 kcal/ 100 ml) – rehabilitation phase.
9. Sensory stimulation & emotional support (**play therapy**).
10. Prepare for **follow-up** after recovery.
    - **Phases**: **stabilisation (1–7 days, steps 1–7)** → **rehabilitation (2–6 weeks)** → follow-up.
    - **Refeeding syndrome** – ↓ phosphate, K, Mg on aggressive feeding → go slow.
    - **Discharge criteria (NRC)** – **15 % weight gain** / WFH ≥ −2 SD, oedema free, eating well.
    - **Appetite test** with RUTF (Ready-to-Use Therapeutic Food) – failed → admit.

3.4 Vitamin Deficiencies

Vitamin	Deficiency disease	Key signs	Treatment/ prevention
A (retinol)	**Xerophthalmia**	**Night blindness (XN) – earliest symptom**; **conjunctival xerosis (X1A) – earliest sign**; **Bitot's spots (X1B)** – foamy triangular patches; corneal xerosis (X2), **keratomalacia (X3) – irreversible blindness**	Treatment: **day 0, 1, 14** (< 6 m 50,000 IU; 6–12 m 1 lakh; > 1 yr 2 lakh IU). Prophylaxis: **9 months – 1 lakh IU, then 2 lakh IU every 6 months till 5 yrs (9 doses)**

Vitamin	Deficiency disease	Key signs	Treatment/ prevention
D	**Rickets** (children), osteomalacia (adults)	**Craniotabes (earliest, ping-pong skull)**, large AF, **frontal bossing**, **rachitic rosary** (costochondral junction), **Harrison's sulcus**, **pigeon chest**, **wrist widening**, **bow legs (genu varum)/ knock knees**, delayed dentition, hypotonia, ↑ **ALP (earliest biochemical)**; X-ray – **cupping & fraying of metaphysis**	**Vitamin D** – daily oral dose or large single **"stoss" dose** (as prescribed) + **calcium**; **sunlight**; prophylaxis **400 IU/day** for infants
C	**Scurvy**	**Bleeding (swollen spongy) gums**, subperiosteal haemorrhage – painful limbs, **"frog position"**, **pseudoparalysis**, scorbutic rosary, perifollicular haemorrhage	Vitamin C 100–200 mg/day; citrus fruits, amla
B1 (thiamine)	**Beriberi** – wet (cardiac), dry (neuritis), **infantile beriberi** (breastfed babies of deficient mothers – aphonia, cardiac failure)	Polished rice eaters	Thiamine
B2 (riboflavin)	Ariboflavinosis	**Angular stomatitis, cheilosis**, magenta tongue	Riboflavin
B3 (niacin)	**Pellagra**	**3 D's – Dermatitis (Casal's necklace), Diarrhoea, Dementia** (+ Death); maize/ jowar eaters	Niacin
B6	–	**Convulsions** in infants, anaemia (INH therapy)	Pyridoxine
B12 / Folate	**Megaloblastic anaemia**	Pallor, glossitis; B12 – neuro signs, knuckle pigmentation	B12, folic acid
K	**Haemorrhagic disease of newborn (VKDB)**	Bleeding day 2–7 (classical)	**Vit K 1 mg IM at birth**

3.5 Minerals & Anaemia

- **Iron-deficiency anaemia – most common nutritional deficiency & most common anaemia in children** (6 months–2 yrs peak – depleted stores, low iron diet, worms). S/S – pallor, irritability, **pica**, koilonychia, poor school performance, **breath-holding spells**. Rx – **elemental iron 3–6 mg/kg/day** (orally, between meals, with vit C; **stains teeth – use straw/ brush teeth**; black stools normal).
- **WHO anaemia cut-offs**: **6–59 months Hb < 11 g/dl**; 5–11 yrs < 11.5; 12–14 yrs < 12.

Anaemia Mukt Bharat – prophylaxis	Dose
6–59 months	**IFA syrup 1 ml (20 mg iron + 100 µg folic acid) biweekly** (ASHA)
5–9 years	**IFA pink tablet (45 mg iron + 400 µg FA) weekly** (school)
10–19 years	**IFA blue tablet (60 mg iron + 500 µg FA) weekly** (WIFS)
Deworming	**Albendazole 400 mg** biannually (**200 mg for 1–2 yrs**)

- **Iodine deficiency** – goitre, **cretinism** (endemic), **deaf-mutism**, mental retardation; **iodised salt – 15 ppm at consumer level (30 ppm at production)**; NIDDCP (1992; earlier NGCP 1962).
- **Zinc deficiency** – growth retardation, **acrodermatitis enteropathica**, diarrhoea; **zinc in diarrhoea** for 14 days.
- **Fluorosis** – dental mottling, skeletal fluorosis (fluoride > 1.5 ppm); **dental caries** (< 0.5 ppm).

3.6 Childhood Obesity

- **BMI-for-age > +2 SD (> 95th centile)** = obesity; > +1 SD (85th) = overweight.
- Causes – sedentary lifestyle, screen time, junk food, genetics; rarely hypothyroidism, Cushing's, Prader-Willi.
- Complications – T2DM, HTN, dyslipidaemia, fatty liver, sleep apnoea, low self-esteem, slipped capital femoral epiphysis.
- Management – **lifestyle modification (family-based)**, ≥ 60 min physical activity daily, **screen time < 2 hrs** (none < 2 yrs), healthy diet.

PYQ style

- Exclusive breastfeeding – **6 months**; complementary feeding starts – **6 months (180 days)**.
- Main carbohydrate in breast milk – **lactose**; protein – **lactalbumin (whey)**.
- Colostrum rich in – **IgA, protein, vitamin A**.
- SAM – **MUAC < 11.5 cm / WFH < −3 SD / bilateral oedema**.
- Hallmark of kwashiorkor – **oedema**; flag sign – **kwashiorkor**; old man face – **marasmus**.
- F-75 used in – **stabilisation phase**; ReSoMal – **dehydration in SAM**.
- Iron in SAM started – **rehabilitation phase** (not in first week).
- Earliest symptom of vitamin A deficiency – **night blindness**; earliest sign – **conjunctival xerosis**.
- Vitamin A prophylaxis first dose – **9 months, 1 lakh IU**; total – **9 doses**.
- Earliest sign of rickets – **craniotabes**; rachitic rosary – **costochondral junction**.
- Frog position, bleeding gums – **scurvy**; 3 D's – **pellagra**.
- IFA syrup 6–59 months – **1 ml biweekly (20 mg iron)**.
- Iodised salt at consumer level – **15 ppm**.

4 Immunization

4.1 Immunity – Basics

Type	Example
Natural active	After infection (measles – lifelong)
Artificial active	**Vaccines** (live/ killed/ toxoid) – takes time, long-lasting
Natural passive	**Maternal IgG through placenta**, IgA in breast milk
Artificial passive	**Immunoglobulins/ antisera** – immediate, short (e.g. **TIG, ARS/ RIG, HBIG, ADS, ASV**)

Vaccine type	Examples
Live attenuated	**BCG, OPV, Measles/ MR/ MMR, Rotavirus, JE (SA-14-14-2), Yellow fever, Varicella, Typhoid oral (Ty21a)**
Killed/ inactivated	**IPV**, Pertussis (whole cell), Rabies, Hepatitis A, Influenza, JE (JENVAC)
Toxoid	**Tetanus, Diphtheria**
Subunit/ recombinant	**Hepatitis B**, HPV, acellular pertussis
Conjugate (polysaccharide + protein)	**Hib**, **PCV**, typhoid conjugate (TCV), meningococcal

- **Herd immunity** – protection of unimmunised due to immunity in majority.
- **Cold chain** – system of storing & transporting vaccines at recommended temperature from **manufacturer to beneficiary**.

4.2 National Immunization Schedule (UIP)

National Immunization Schedule (UIP) – Children

Age	Vaccines	Route / Site
Birth	BCG · OPV-0 · Hepatitis B birth dose (within 24 hrs)	BCG 0.1 ml ID L upper arm (0.05 ml if < 1 month)
6 weeks	OPV-1 · Pentavalent-1 · Rotavirus-1 · fIPV-1 · PCV-1	OPV 2 drops · RVV 5 drops oral Penta/PCV 0.5 ml IM thigh
10 weeks	OPV-2 · Pentavalent-2 · Rotavirus-2	Same as 6 weeks
14 weeks	OPV-3 · Pentavalent-3 · Rotavirus-3 · fIPV-2 · PCV-2	fIPV 0.1 ml ID right upper arm
9–12 months	MR-1 · JE-1* · PCV booster · fIPV-3 · Vitamin A (1 lakh IU)	MR 0.5 ml SC right upper arm Vit A 1 ml (oral)
16–24 months	MR-2 · JE-2* · DPT booster-1 · OPV booster · Vitamin A (2 lakh IU)	DPT 0.5 ml IM thigh Vit A 2 ml every 6 mo → 5 y
5–6 yrs · 10 · 16 yrs	DPT booster-2 (5–6 y) · Td (10 y) · Td (16 y)	Td 0.5 ml IM upper arm

VVM: inner square lighter than outer circle = USE · same/darker = DISCARD
Cold chain: ILR +2 to +8 °C · Deep freezer −15 to −25 °C · *JE only in endemic districts
Vitamin A: total 9 doses (9 months → 5 years) · Pentavalent = DPT + Hep B + Hib

UIP schedule for infants & children with route/ site

Age	Vaccines
Birth	**BCG**, **OPV-0** (within 15 days), **Hepatitis B birth dose (within 24 hrs)**
6 weeks	**OPV-1, Pentavalent-1, RVV-1, fIPV-1, PCV-1**
10 weeks	**OPV-2, Pentavalent-2, RVV-2**
14 weeks	**OPV-3, Pentavalent-3, RVV-3, fIPV-2, PCV-2**
9–12 months	**MR-1, JE-1 (endemic), PCV booster, fIPV-3, Vitamin A (1st dose, 1 lakh IU)**
16–24 months	**MR-2, JE-2, DPT booster-1, OPV booster, Vitamin A (2nd dose, 2 lakh IU)**
5–6 years	**DPT booster-2**
10 years	**Td**
16 years	**Td**
Pregnant women	**Td-1, Td-2 (4 weeks apart) / Td booster**

- **Pentavalent = DPT + Hepatitis B + Hib**.
- **12 vaccine-preventable diseases (UIP)**: TB, polio, diphtheria, pertussis, tetanus, hepatitis B, Hib (pneumonia/ meningitis), measles, rubella, **rotavirus diarrhoea**, **pneumococcal pneumonia**, **JE** (endemic).
- **Vitamin A** – 9 doses: 9 months (1 lakh IU = 1 ml), then 2 lakh IU (2 ml) every 6 months till **5 yrs**.
- **Fully immunized child** – BCG, 3 doses OPV & Penta, MR-1 (+ OPV-0, HepB-0) **before 1 year**.

Vaccine	Dose	Route	Site
BCG	**0.1 ml (0.05 ml if < 1 month)**	**Intradermal**	**Left upper arm (over deltoid insertion)**

Vaccine	Dose	Route	Site
OPV	**2 drops**	Oral	Mouth
Hepatitis B	0.5 ml	IM	**Anterolateral thigh**
Pentavalent / DPT	**0.5 ml**	**IM**	**Anterolateral mid-thigh (left)**
fIPV (fractional IPV)	**0.1 ml**	**Intradermal**	**Right upper arm**
Rotavirus (Rotavac)	**5 drops**	Oral	Mouth
PCV	0.5 ml	IM	**Right anterolateral thigh**
MR	**0.5 ml**	**Subcutaneous**	**Right upper arm**
JE (live SA-14-14-2)	0.5 ml	**SC**	**Left upper arm**
Td	0.5 ml	IM	Upper arm
Vitamin A	1 ml / 2 ml	Oral	–

- **Diluents**: BCG – **normal saline**; MR – **sterile water**; use **only the diluent supplied** with the vaccine, **cool it** (keep in ILR 24 hrs before use) – warm diluent → vaccine damage.
- **BCG reaction** – papule (2–3 wks) → pustule → ulcer → **scar by 6–12 weeks** (no dressing/ ointment); **no scar does NOT need repeat** (as per UIP). Complications – **BCG adenitis** (axillary), abscess (deep injection/ overdose), BCG-osis (immunodeficient).
- **OPV** – **if child vomits within 30 min, repeat dose**; OPV can be given with diarrhoea (dose not counted in older schedules – just give).
- **Diphtheria/ tetanus/ pertussis vaccine reactions** – fever, local pain, persistent cry; **DPT contraindication** – encephalopathy within 7 days of previous dose (give DT/ Td instead); anaphylaxis.

4.3 Cold Chain

Equipment	Temperature	Level
Walk-in cooler (WIC)	+2 to +8 °C	State/ regional
Walk-in freezer (WIF)	−15 to −25 °C	State/ regional
ILR (Ice Lined Refrigerator)	**+2 to +8 °C**	**PHC/ district** – keeps cold **for ~8 hrs without power**
Deep freezer	**−15 to −25 °C**	Ice packs (and OPV at higher levels)
Cold box	+2 to +8 °C	Transport, power failure storage (up to ~5–7 days)
Vaccine carrier	+2 to +8 °C	**Outreach session – with 4 conditioned ice packs**, keep closed, in shade

- **Most heat-sensitive vaccine – OPV** (then MR/ measles); **freeze-sensitive (never freeze)** – **Hepatitis B, Pentavalent, DPT, Td, IPV, PCV** – **freezing destroys them; Hep B is the most freeze-sensitive**.
- **ILR arrangement** – all vaccines **inside the basket** (never directly on the ILR floor); the **bottom of ILR is coldest** → keep **heat-sensitive (OPV, MR, BCG, JE) at the bottom of the basket** and **freeze-sensitive (Hep B, Penta, DPT, Td, IPV, PCV) in the upper part**, away from walls.
- **Shake test** – to check if a freeze-sensitive vaccine was **frozen** (frozen vial sediments faster & forms flakes → discard).

- **Conditioned ice packs** – kept at room temperature till **"sound of water" on shaking** (prevents freezing of vaccines in carrier).
- **VVM (Vaccine Vial Monitor)** – heat-sensitive label: **inner square lighter than outer circle = USE**; **inner square matches or darker = DISCARD** (stages 1–2 usable; 3–4 discard). Indicates **cumulative heat exposure, not freezing**.
- **Temperature log** twice daily; **thermometer/ eVIN (electronic Vaccine Intelligence Network)**.
- **"Hold the cold"**, **"first expiry first out" (FEFO)**.

Open Vial Policy (UIP)

- **Can be reused up to 28 days** (if VVM OK, not expired, aseptic, stored +2 to +8 °C, date & time written): **OPV, DPT, Td, Hep B, Pentavalent, PCV, IPV**.
- **Discard 4 hrs after opening/ reconstitution or end of session (whichever earlier)**: **BCG, MR/ measles, JE** (reconstituted live vaccines) – risk of **toxic shock syndrome** (Staph contamination).
- Vitamin A – opened bottle used within 6–8 weeks (as per guidelines).

4.4 Adverse Events Following Immunization (AEFI)

- **Minor** – fever, local pain, swelling, irritability (**paracetamol, cold compress**).
- **Serious** – death, hospitalisation, disability, cluster; **anaphylaxis** (within minutes – **adrenaline 1:1000, 0.01 ml/kg IM, max 0.5 ml**, anterolateral thigh) – hence **observe for 30 min after vaccination**.
- **Programme errors** – **toxic shock syndrome** (reuse of reconstituted vial > 4–6 hrs) – **fever, vomiting, watery diarrhoea within hours, death in 24–48 hrs**; abscess (unsterile injection), BCG overdose.
- Other: **VAPP (vaccine-associated paralytic polio – OPV)**, **cVDPV**, **intussusception (rotavirus – rare)**, febrile seizures, hypotonic-hyporesponsive episode (DPT), persistent inconsolable crying.
- **AEFI reporting** – immediately to **MO/ DIO**; **AEFI kit** (adrenaline, syringes, IV set, hydrocortisone) at every session.

4.5 General Rules & Contraindications

- **Minimum interval** between doses of same vaccine – **4 weeks**; two live vaccines – same day or **≥ 4 weeks apart**.
- **No maximum interval** – if a dose is missed, **continue schedule, don't restart**.
- **Catch-up**: **Pentavalent up to 1 yr**; after 1 yr give **DPT**; **OPV up to 5 yrs**; **MR up to 5 yrs**; **BCG up to 1 yr**; **Hep B birth dose up to 24 hrs** (not after); **Rotavirus – start before 1 yr**; **PCV – up to 1 yr**; **Vitamin A – up to 5 yrs**.
- **True contraindications** – **anaphylaxis to previous dose**, **live vaccines in immunodeficiency/ immunosuppressive therapy, pregnancy**, DPT – progressive neurological disease/ encephalopathy.
- **NOT contraindications (false)** – **mild fever, cold, cough, mild diarrhoea, malnutrition**, prematurity/ LBW (vaccinate by chronological age), breastfeeding, antibiotics, family history of convulsions, local reaction to previous dose, jaundice.
- **Injection safety** – **AD (auto-disable) syringes**, no recapping, **hub cutter**, red & black bags (BMW), **anterolateral thigh in infants (avoid gluteal – sciatic nerve injury)**.
- **4 key messages to parents** after vaccination – what vaccine & disease it prevents, minor side effects & how to manage, **date of next visit**, keep **MCP card** safe.

Exam traps

- **BCG – intradermal, left arm**; **fIPV – intradermal, right arm**; **MR – subcutaneous, right arm**.
- **Hep B birth dose only within 24 hrs**.
- **VVM shows heat damage, shake test shows freezing**.
- **BCG/ MR/ JE – discard 4 hrs after reconstitution**; others – **28 days** open vial policy.

PYQ style

- BCG dose – **0.1 ml (0.05 ml < 1 month)**, route – **intradermal**.
- Pentavalent contains – **DPT + Hep B + Hib**.
- MR first dose – **9–12 months**, route – **subcutaneous**.
- Most heat-sensitive vaccine – **OPV**; most freeze-sensitive – **Hepatitis B**.
- ILR temperature – **+2 to +8 °C**; deep freezer – **−15 to −25 °C**.
- Test for freezing – **shake test**; heat exposure – **VVM**.
- Reconstituted MR must be used within – **4 hours**.
- Toxic shock syndrome after vaccination – **reuse of reconstituted vial**.
- Diluent for BCG – **normal saline**.
- Not a contraindication – **mild fever/ diarrhoea/ malnutrition**.
- Observation after vaccination – **30 minutes**.

5 Hospitalised Child, Paediatric Procedures & Drug Calculations

5.1 Child's Reaction to Hospitalisation

- **Stressors** – **separation (major stressor in infants & toddlers)**, **loss of control**, **bodily injury & pain**, unfamiliar environment, strangers.
- **Separation anxiety** – peaks **6 months – 30 months** (toddler most affected). **3 stages (Robertson/ Bowlby)**:
  1. **Protest** – cries loudly, clings to parent, refuses others, screams.
  2. **Despair** – hopeless, quiet, withdrawn, **regression**, disinterested in play/ food.
  3. **Detachment (denial)** – seems "adjusted", interacts superficially – **most serious (prolonged separation)**, may lose attachment.
- **Regression** – common in toddler (loses toilet training, bottle again) – **don't scold, accept**, it is temporary.
- **Age-wise**: infant – separation, stranger anxiety; toddler – separation, loss of autonomy/ routine; **preschool – fear of bodily injury/ mutilation, illness as punishment (magical thinking)**; school age – loss of control, fear of death, separation from peers; adolescent – **loss of independence, privacy, body image**, separation from peers.
- **Nursing**: **rooming-in / parents stay (family-centred care)**, maintain home routine, favourite toy (**transitional object**), primary nurse, **play/ therapeutic play**, honest age-appropriate explanation, **give choices** (sense of control), privacy for adolescents, school work/ peers contact.
- **Atraumatic care** – **EMLA cream** (60 min before venipuncture), sucrose 24 % for neonates, distraction, parents present during procedures, **treatment room for painful procedures (bed = safe place)**.

5.2 Paediatric Vital Signs

Age	Heart rate (/min)	Respiratory rate (/min)	Systolic BP (mmHg)
Newborn	**120–160**	**30–60**	60–90
Infant (1–12 m)	100–160	30–50	70–100
Toddler (1–3 y)	90–150	24–40	80–110
Preschool (3–6 y)	80–140	22–34	80–110
School (6–12 y)	70–120	18–30	85–120
Adolescent	60–100	12–20	95–130

- **Minimum (lower limit) SBP (1–10 yrs) = 70 + (2 × age in years)**; < 1 yr – 60 (neonate)/ 70 (infant); **> 10 yrs – 90 mmHg**.
- **Pulse – apical** (stethoscope) in **< 2 yrs**, full 1 minute; radial after 2 yrs.
- **RR – count by observing abdomen** (infants are **abdominal/ diaphragmatic breathers**), **full 1 minute**, when calm (before disturbing procedures).
- **Temperature** – **axillary preferred in newborn/ infants** (safe); **rectal – avoided routinely** (perforation, NEC, diarrhoea; insert ≤ 1.5 cm infants); tympanic > 2 yrs.

- **Cuff width** – **40 % of arm circumference / 2/3 of upper arm length**; bladder encircles 80–100 % of arm. **Narrow cuff → falsely high BP**.
- **Order of assessment** – least invasive first: **RR → HR → temperature → BP**.

5.3 Pain Assessment in Children

Scale	Age
NIPS (Neonatal Infant Pain Scale), **PIPP** (premature), CRIES	Neonates
FLACC (Face, Legs, Activity, Cry, Consolability) – each 0–2, total 10	**2 months – 7 yrs / non-verbal**
Wong-Baker FACES	**≥ 3 yrs**
Oucher scale (photographs + numbers)	3–12 yrs
Poker chip tool ("pieces of hurt")	4–8 yrs
Numeric (0–10)/ VAS	**≥ 7–8 yrs** (who can count/ understand numbers)

- **Non-pharmacological** – **non-nutritive sucking, 24 % sucrose (neonates, 2 min before)**, **breastfeeding during procedure**, **KMC/ skin-to-skin**, swaddling, distraction (bubbles, cartoons), guided imagery, cold/ heat.
- **Drugs** – **paracetamol (15 mg/kg/dose, 4–6 hourly; max 60 mg/kg/day)**, ibuprofen (10 mg/kg/dose, > 3 months), opioids (morphine 0.1 mg/kg); **aspirin avoided (< 16 yrs – Reye's syndrome)**.

5.4 Medication Administration

- **Dose based on body weight (mg/kg)** – most accurate; or **BSA** (chemotherapy). Always **double-check** high-alert drugs (insulin, digoxin, potassium, heparin, opioids).

Rule	Formula
Clark's rule (weight)	**Child dose = (Weight in lb ÷ 150) × adult dose** (kg version: weight kg ÷ 70 × adult dose)
Young's rule (age, 1–12 yrs)	**Child dose = Age ÷ (Age + 12) × adult dose**
Fried's rule (infants < 1 yr)	**Child dose = Age in months ÷ 150 × adult dose**
Cowling's rule	(Age at next birthday ÷ 24) × adult dose
BSA rule (most accurate)	**Child dose = (BSA in m^2 ÷ 1.73) × adult dose**
BSA (Mosteller)	**√ (height cm × weight kg ÷ 3600)**

- **Example**: 6-yr child, adult dose 500 mg → Young's = 6/(6+12) × 500 = **166.7 mg**.
- **Example**: paracetamol 15 mg/kg for 12-kg child = **180 mg**; syrup 250 mg/5 ml → 180 × 5 / 250 = **3.6 ml**.
- **Oral** – use **calibrated dropper/ oral syringe** (not household spoon), place in **side of mouth (cheek pocket)** slowly – prevents aspiration; **hold infant semi-upright**; **don't mix in milk/ essential foods** (food aversion, incomplete dose); crush tablets with small amount of jam/ honey (> 1 yr); never call medicine "candy".

- **IM injection** – **vastus lateralis (anterolateral thigh) – preferred site for infants & children < 3 yrs**; **ventrogluteal** safe for all ages; **deltoid** after 18 months–3 yrs (small volume ≤ 0.5–1 ml); **dorsogluteal avoided until child walks ≥ 1 yr (sciatic nerve)**. Max volume: infant 0.5–1 ml, child 1–2 ml.
- **Ear drops** – **< 3 yrs: pull pinna DOWN & BACK**; **> 3 yrs (and adults): UP & BACK**.
- **Eye drops** – lower conjunctival sac, press inner canthus 1 min.
- **Nasal drops** – head extended over edge of bed/ pillow.
- **Rectal** – suppository inserted with **little finger** in < 3 yrs.

5.5 IV Therapy & Fluids

- **Maintenance fluid (Holliday-Segar)**: **100 ml/kg/day (first 10 kg) + 50 ml/kg/day (11–20 kg) + 20 ml/kg/day (> 20 kg)**. Hourly ("4-2-1 rule"): **4 ml/kg/hr + 2 ml/kg/hr + 1 ml/kg/hr**.
- **Example**: 25-kg child → 1000 + 500 + 100 = **1600 ml/day** (≈ 65 ml/hr).
- **Fluid bolus (shock)** – **10–20 ml/kg NS/ RL** over 5–20 min (reassess after each); SAM/ cardiac – 10–15 ml/kg slowly.
- **Microdrip (paediatric) set: 60 drops = 1 ml → drops/min = ml/hr**. Macro set: 15–20 drops/ml.
- **Use infusion pump/ burette (Soluset, 100–150 ml)** – never hang large bottle for infants (fluid overload).
- **IV site** – **dorsum of hand/ foot**, scalp veins (infants), antecubital; **check site hourly** (infiltration, phlebitis); splint the limb.
- **Intraosseous (IO)** – emergency when IV fails in 90 sec – **proximal tibia (1–2 cm below tibial tuberosity, anteromedial)**.
- **Daily weight** – best indicator of fluid balance; **urine output normal: infant 2 ml/kg/hr, child 1 ml/kg/hr** (< 1 ml/kg/hr = oliguria in child).
- **Diaper weighing** – **1 g wet diaper weight = 1 ml urine**.

5.6 Restraints

Restraint	Use
Mummy (swaddle) restraint	**Short procedures on head/ neck** – venipuncture (scalp/ jugular), gavage, eye exam
Elbow restraint	**Prevent touching face/ head** – **cleft lip repair**, eczema, scalp IV
Jacket restraint	Keep child in bed/ chair, high chair
Clove-hitch	**Single limb** (extremity) – IV site
Papoose board	Commercial, whole body
Crib net / top	Prevent climbing out

- Rules – **order required**, explain to parents, **check circulation every 15–30 min**, **release every 2 hrs** (ROM, skin care), **tie with square/ quick-release knot to bed frame (not side rails)**, document; least restrictive.

5.7 Specimen Collection & Procedures

- **Urine** – **sterile urine bag** (perineum cleaned & dried; check every 15–30 min); **clean-catch**; **suprapubic aspiration – most reliable culture in infants**; catheter specimen.
- **Stool** – from diaper; **24-hr urine** – start with empty bladder, discard first void.
- **Blood** – **heel prick (lateral/ medial plantar surface of heel – avoid posterior curvature/ centre – calcaneus osteomyelitis)**; warm heel; **neonatal screening (TSH, PKU)**.
- **Lumbar puncture** – **sitting (infants) or lateral recumbent with knee-chest (flexed spine)**, **L3–L4 / L4–L5**; post-LP lie flat; LP contraindicated in **↑ ICP (papilloedema), local infection, bleeding disorder**.
- **Nasogastric/ orogastric (gavage) tube** – **neonates prefer orogastric (obligatory nose breathers)**; length – **NEX: nose tip → earlobe → xiphisternum** (+ midpoint to umbilicus in newborn — NEMU); confirm by **aspirate pH < 5.5** / X-ray; feeding by **gravity (not push)**, 15–20 cm above; hold baby/ right side/ prone after feed.
- **Gastrostomy** feeding; **tracheostomy care** (suction ≤ 5 sec, catheter half size of tube).
- **Suctioning** – **mouth before nose**; pressure – neonate 60–80, infant 80–100, child 100–120 mmHg; < 5 sec.
- **Oxygen therapy** – **nasal prongs** (≤ 2 L/min infant), **head box/ hood (5–7 L/min to wash CO_2)**, mask, incubator; **retinopathy of prematurity (ROP)** risk – target SpO_2 **90–95 % in preterms**.
- **Exchange transfusion, phototherapy** – see newborn chapter.
- **Bathing** – sponge bath till cord falls; **tub bath** water **37–38 °C (test with elbow)**; never leave child alone in water.

PYQ style

- Most affected by separation anxiety – **toddler**; first stage – **protest**.
- Most serious stage of separation – **detachment**.
- Preschooler fear in hospital – **body mutilation**.
- Pain scale 2 months–7 yrs – **FLACC**; ≥ 3 yrs – **Wong-Baker FACES**.
- Normal newborn HR – **120–160**; RR – **30–60**.
- Minimum SBP formula – **70 + 2 × age**.
- Preferred IM site in infant – **vastus lateralis**.
- Ear drops in child < 3 yrs – pinna **down & back**.
- Young's rule – **age/(age + 12) × adult dose**; Clark's – **weight (lb)/150 × adult dose**.
- Maintenance fluid for 15-kg child – **1250 ml/day**.
- Microdrip set – **60 drops/ml**.
- Restraint after cleft lip repair – **elbow restraint**.
- Heel prick site – **lateral/ medial aspect of plantar heel**.
- Gavage tube length – **nose → earlobe → xiphisternum**.

6 High-Risk Newborn & Neonatal Disorders

6.1 Classification of Newborns

By	Category
Gestational age	**Preterm < 37 wks** (extremely < 28, very 28–< 32, moderate/ late 32–< 37) · **Term 37–42** · **Post-term ≥ 42 wks**
Birth weight	**LBW < 2500 g** · **VLBW < 1500 g** · **ELBW < 1000 g** · Macrosomia > 4000 g
Weight for gestation	**SGA < 10th centile** · AGA 10–90th · **LGA > 90th centile**

- **High-risk newborn** – preterm, LBW, SGA/ LGA, infant of diabetic mother, birth asphyxia, sepsis, congenital anomalies, jaundice, multiple birth, Rh problems.
- **Gestational age assessment** – **New Ballard Score** (neuromuscular + physical maturity), Dubowitz.

Feature	Preterm	Term
Skin	**Thin, shiny, red/ gelatinous, visible veins**, **lanugo abundant**	Thick, pale, creases, little lanugo
Sole creases	**Few/ only anterior**	Cover whole sole
Breast nodule	**< 5 mm**, areola barely visible	5–10 mm, raised areola
Ear	**Soft, flat, slow recoil**	Firm, instant recoil
Genitalia	**Male – testes undescended, few rugae; female – prominent clitoris & labia minora**	Testes descended, rugae; labia majora cover minora
Posture	**Extended, hypotonic (frog-like)**	**Fully flexed**
Scarf sign	Elbow crosses midline	Elbow doesn't reach midline

Problems of preterm/ LBW

- **Hypothermia** (little brown fat, large surface area), **hypoglycaemia**, **RDS (surfactant deficiency)**, apnoea of prematurity (**caffeine citrate**), **jaundice**, **infections/ sepsis**, feeding problems (poor suck-swallow coordination < 32–34 wks), **NEC**, **IVH**, **PDA**, **ROP**, anaemia, BPD, hypocalcaemia.

6.2 Thermoregulation

- Newborns lose heat by **evaporation (most at birth – wet baby)**, **conduction** (cold surface), **convection** (draughts), **radiation** (cold walls/ windows). **Brown fat → non-shivering thermogenesis** (newborns cannot shiver).
- **Normal axillary temperature 36.5–37.5 °C**.

Hypothermia (WHO)	Temperature
Cold stress (mild)	**36.0–36.4 °C**

Hypothermia (WHO)	Temperature
Moderate	**32.0–35.9 °C**
Severe	**< 32 °C**

- **Warm chain (10 steps)** – warm delivery room (**25–28 °C**), **immediate drying**, skin-to-skin, breastfeeding, **postpone bathing (≥ 24 hrs)/ weighing**, appropriate clothing & bedding, mother & baby together, warm transport, warm resuscitation, training/ awareness.
- **Touch method** – **abdomen warm & feet warm = normal**; abdomen warm, feet cold = cold stress; both cold = hypothermia.
- **Radiant warmer** (servo-control, probe on abdomen/ upper quadrant), **incubator**.
- **Hyperthermia** – > 37.5 °C – undress, remove from heat source, frequent feeds.

Kangaroo Mother Care (KMC)

- **Components** – **continuous skin-to-skin contact** (baby in nappy, cap & socks, between breasts, upright, frog position), **exclusive breastfeeding**, **early discharge with follow-up**.
- **Eligibility** – **stable LBW babies** (especially < 2000 g); can start in **SNCU when stable**; duration – as long as possible (**minimum 1 hr per session**, ideally 24 hrs) until baby reaches ~2500 g/ 40 wks or wriggles out.
- **Benefits** – thermal control, ↑ breastfeeding, weight gain, ↓ infection, ↓ mortality, bonding; **iKMC** (immediate KMC) after birth – WHO 2022.

6.3 Feeding the Preterm/ LBW

Gestation/ weight	Feeding method
< 28 wks / < 1200 g (or sick)	**IV fluids initially → trophic feeds via OG tube**
28–31 wks	**Gavage (OG/ NG) feeding** with expressed breast milk
32–34 wks	**Paladai/ katori-spoon** (± gavage)
≥ 34 wks	**Direct breastfeeding** (suck-swallow-breathe coordination)

- **Expressed breast milk (EBM) – best**; human milk fortifier for VLBW; donor milk banks (**"Comprehensive Lactation Management Centres"**).
- Feeding every **2 hrs (VLBW)**/ 2–3 hrs; check **gastric residual** before feeds; abdominal distension → suspect NEC.
- **Supplements** – **vitamin D 400 IU/day**, **iron 2 mg/kg/day from 2 weeks** (preterm) till 1 yr, multivitamin, calcium/ phosphorus (VLBW).

6.4 Neonatal Jaundice (Hyperbilirubinaemia)

	Physiological	Pathological
Onset	**After 24 hrs** (day 2–3)	**Within first 24 hrs**
Peak	**Day 3–5 (term)**, day 5–7 (preterm)	Any time
Duration	**Disappears by 14 days (term)**, 21 days (preterm)	**> 2 weeks (term)** / 3 weeks (preterm) – prolonged

	Physiological	Pathological
Bilirubin rise	< 5 mg/dl/day	**> 5 mg/dl/day** (> 0.5 mg/dl/hr)
Max level	< 12–15 mg/dl (term)	Exceeds phototherapy range
Conjugated	Unconjugated	**Conjugated > 2 mg/dl (or > 20 %) – always pathological** (cholestasis – pale stools, dark urine)
Palms/ soles yellow	No	Yes (severe)

- **Causes of physiological** – ↑ RBC breakdown (high Hb, short RBC life **70–90 days**), **immature liver (low glucuronyl transferase)**, ↑ enterohepatic circulation.
- **Pathological** – **haemolytic (Rh/ ABO incompatibility, G6PD deficiency)**, sepsis, **hypothyroidism**, breast milk jaundice, biliary atresia (conjugated), cephalhaematoma, polycythaemia.
- **Breastfeeding jaundice** (first week – inadequate intake – feed more) vs **breast milk jaundice** (late, 2nd–3rd week – factors in milk – continue BF).
- **Kramer's rule (dermal zones)** – **face 5 → chest 10 → abdomen 12 → legs 15 → palms & soles > 15 mg/dl** (cephalocaudal progression).
- **Kernicterus (bilirubin encephalopathy)** – unconjugated bilirubin crosses BBB → **basal ganglia** – lethargy, poor suck, hypotonia → **hypertonia, opisthotonus, high-pitched cry, setting-sun eyes, seizures**; sequelae – **choreoathetoid CP, sensorineural deafness, upward gaze palsy, dental enamel dysplasia**.

Phototherapy – nursing

- **Blue light (wavelength 450–460 nm)** converts bilirubin to water-soluble **lumirubin** (structural isomerisation) excreted in bile/ urine.
- **Baby naked** (only diaper/ minimal), **eyes covered (eye shields – retinal damage)**, genitals covered (debated), **distance 30–45 cm** (LED closer as per manufacturer).
- **Change position every 2 hrs**; **continue breastfeeding** (frequent feeds) – **extra fluids 10–20 %** (insensible loss ↑).
- **Monitor temperature 2–4 hourly** (hypo/ hyperthermia), **daily weight**, hydration, **serum bilirubin 12–24 hourly** (skin colour unreliable during phototherapy).
- **Side effects** – **dehydration (↑ insensible water loss)**, **loose green stools**, skin rash, **bronze baby syndrome (if conjugated ↑ – contraindicated in cholestasis)**, hyper/ hypothermia, retinal damage, separation from mother.
- **Exchange transfusion** – removes bilirubin & antibodies; **double volume = 2 × 80 ml/kg = 160 ml/kg** (removes ~85 % RBCs, ~50 % bilirubin); **umbilical vein**; fresh **O-negative/ compatible blood**; 5–10 ml aliquots (push-pull); monitor vitals, Ca^{2+}, glucose; complications – emboli, NEC, infection, hypocalcaemia, electrolyte imbalance, arrhythmia.

6.5 Neonatal Sepsis

- **Early-onset (EOS) < 72 hrs** – maternal (vertical) – **PROM > 18–24 hrs, maternal fever, foul liquor, prolonged labour, multiple PV exams**; organisms – **E. coli, GBS, Klebsiella**.
- **Late-onset (LOS) > 72 hrs** – nosocomial/ community – **Klebsiella, Staph aureus**, Pseudomonas, CONS.

- **S/S** – **refusal to feed (earliest)**, **lethargy**, hypo/ hyperthermia (**hypothermia more common**), poor cry, **apnoea**, jaundice, abdominal distension, vomiting, **sclerema**, seizures, bulging fontanelle (meningitis), shock.
- **Investigations** – **blood culture (gold standard)**, **sepsis screen** (TLC < 5000, ANC low, I/T ratio > 0.2, **CRP > 1 mg/dl**, micro-ESR > 15 mm) – **≥ 2 positive = probable sepsis**; LP.
- **Treatment** – **ampicillin/ cloxacillin + gentamicin** (first line); cefotaxime, piperacillin-tazobactam, vancomycin, meropenem as per sensitivity; supportive – warmth, IV fluids, glucose, O_2.
- **Prevention** – **hand hygiene (most important)**, **6 cleans**, exclusive breastfeeding, **dry cord care**, rooming-in, aseptic procedures, **chlorhexidine 7.1 % cord application in high-NMR/ home delivery areas**.
- **Local infections** – **umbilical sepsis (omphalitis)**, skin pustules, **oral thrush (candida – white curdy patches not wiped off → nystatin/ clotrimazole)**, conjunctivitis (**ophthalmia neonatorum – gonococcal day 2–5, chlamydial day 5–14**).

6.6 Birth Asphyxia & HIE

- **Failure to initiate/ sustain breathing at birth** (WHO); **APGAR ≤ 3 at 1 min** (NNF – gasping/ no breathing).
- **HIE (Hypoxic Ischaemic Encephalopathy) – Sarnat staging**: **Stage I (mild)** – hyperalert, irritable, normal tone, dilated pupils, no seizures; **Stage II (moderate)** – lethargy, hypotonia, **seizures**, miosis; **Stage III (severe)** – **stupor/ coma**, flaccid, absent reflexes, apnoea.
- Management – **NRP**, maintain temperature (**therapeutic hypothermia 33–34 °C for 72 hrs in moderate-severe HIE – within 6 hrs** in NICUs), glucose, fluids (restrict – SIADH), treat seizures (**phenobarbitone 20 mg/kg loading – first-line**), ventilation.

6.7 Respiratory Problems

Condition	Key points
RDS (Hyaline Membrane Disease)	**Preterm**, **surfactant deficiency**; onset **within 6 hrs**, ↑ RR, **expiratory grunt**, retractions, nasal flaring, cyanosis; X-ray – **ground glass, air bronchogram**; Rx – **CPAP**, **surfactant (intratracheal – INSURE/ LISA)**; **prevention – antenatal steroids**; **Silverman-Anderson score** (0–10; ≥ 7 impending failure) / **Downes' score**
TTN (Transient tachypnoea)	**Term, LSCS**, delayed clearance of lung fluid; resolves in **24–72 hrs**; X-ray – fluid in fissures
MAS (Meconium aspiration)	**Post-term, fetal distress**; X-ray – patchy infiltrates, hyperinflation; risk of **PPHN, pneumothorax**; **routine intrapartum/ tracheal suction no longer recommended for vigorous baby**
Apnoea of prematurity	Pause **≥ 20 sec** (or shorter with bradycardia/ cyanosis); < 34 wks; **tactile stimulation, caffeine citrate**

6.8 Other Neonatal Problems

- **Neonatal hypoglycaemia** – **blood glucose < 45 mg/dl** (Indian practice; treat < 40–45); at risk – **IDM, LBW, preterm, SGA, sepsis, hypothermia, asphyxia**; S/S – jitteriness, lethargy, poor feeding, apnoea, seizures (may be asymptomatic); Rx – breastfeed/ EBM if asymptomatic; symptomatic – **10 %**

dextrose 2 ml/kg IV bolus → infusion **6–8 mg/kg/min** (GIR); **avoid 25 % / 50 % dextrose (rebound, vein damage)**.

- **Hypocalcaemia** – < 7 mg/dl (total) – jitteriness, seizures – **10 % calcium gluconate 2 ml/kg slow IV with cardiac monitoring**.
- **NEC (Necrotising enterocolitis)** – **preterm, formula-fed, asphyxia**; **abdominal distension, bilious aspirates/ vomiting, bloody stools**; X-ray – **pneumatosis intestinalis (gas in bowel wall – pathognomonic)**; Rx – **NPO, NG decompression, IV antibiotics, TPN**; prevention – **breast milk**.
- **IVH (intraventricular haemorrhage)** – preterm < 32 wks, first 72 hrs; prevent – avoid rapid fluid/ osmolar changes, minimal handling, head midline.
- **ROP (Retinopathy of prematurity)** – **< 34 wks / < 1750 g**, oxygen toxicity; **screening by day 20–30 of life** (or 4 weeks); target SpO_2 90–95 %; Rx – laser photocoagulation, anti-VEGF.
- **Polycythaemia** – venous Hct > 65 %; partial exchange.
- **Infant of diabetic mother (IDM)** – **macrosomia**, **hypoglycaemia (most common)**, hypocalcaemia, RDS, polycythaemia, jaundice, **sacral agenesis/ caudal regression**, cardiomyopathy, birth injuries.
- **Neonatal seizures** – subtle (most common – lip smacking, eye deviation, cycling), clonic, tonic, myoclonic; causes – **HIE (most common)**, hypoglycaemia, hypocalcaemia, sepsis/ meningitis, IVH; Rx – glucose/ calcium if low, **phenobarbitone** (first line).

6.9 Levels of Newborn Care (India)

Unit	Where	Function
NBCC (Newborn Care Corner)	**Every delivery point** (labour room/ OT)	Resuscitation, warmth, early BF, vitamin K
NBSU (Newborn Stabilisation Unit)	**FRU/ CHC**	Stabilisation (4 beds), referral
SNCU (Special Newborn Care Unit)	**District hospital** (≥ 12 beds)	Level II care – except ventilation & surgery
NICU	Medical colleges/ tertiary	Level III – ventilation, surgery

- **Incubator care** – temperature per weight/ age (**thermo-neutral environment**), humidity, clean daily with soap & water, open portholes only, change every 7 days.
- **Infection control in NICU** – **hand washing (40–60 sec)/ alcohol rub between babies**, gown, minimal handling, no sharing equipment, visitors restricted, cohorting.

PYQ style

- Preterm – **< 37 completed weeks**; LBW – **< 2500 g**; VLBW – **< 1500 g**.
- Gestational age assessment – **New Ballard Score**.
- Major heat loss at birth – **evaporation**; non-shivering thermogenesis – **brown fat**.
- Moderate hypothermia – **32–35.9 °C**; cold stress – **36–36.4 °C**.
- KMC components – **skin-to-skin, exclusive BF, early discharge/ follow-up**.
- Physiological jaundice appears – **after 24 hrs**; jaundice within 24 hrs – **pathological**.
- Kernicterus affects – **basal ganglia**.
- Phototherapy wavelength – **450–460 nm (blue)**; eyes – **covered**.
- Phototherapy side effect – **dehydration, loose stools, bronze baby**.
- Earliest sign of neonatal sepsis – **refusal to feed/ lethargy**.
- Gold standard for sepsis – **blood culture**.
- RDS due to – **surfactant deficiency**; scoring – **Silverman-Anderson**.
- Pneumatosis intestinalis – **NEC**.
- Neonatal hypoglycaemia – **< 45 mg/dl**; Rx – **10 % dextrose 2 ml/kg**.
- SNCU located at – **district hospital**.

7 IMNCI, Diarrhoea & Dehydration

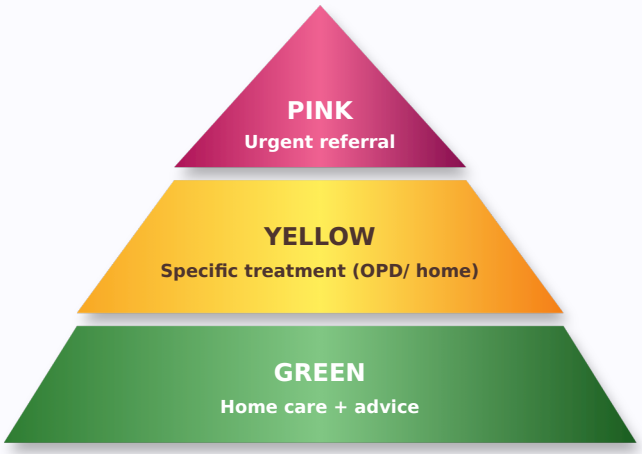
7.1 IMNCI – Integrated Management of Neonatal & Childhood Illness

D IMNCI

WHO-UNICEF **IMCI (1992/1995)** adapted in India as **IMNCI (2003)** – added **newborn (0–7 days) care** and **home visits**. Two age groups: **young infant 0–2 months** and **child 2 months – 5 years**. Focus on **5 major killers** – **pneumonia, diarrhoea, malaria, measles, malnutrition** (+ neonatal problems).

- **Case management process**: **Assess → Classify → Identify treatment → Treat → Counsel the mother → Follow-up**.
- **Components**: improve health worker case-management skills, **health system** (drugs, referral), **family & community practices**.

IMNCI – Colour-coded Classification (2 months – 5 years)



General Danger Signs (any one = PINK)

1. Not able to drink or breastfeed
2. Vomits everything
3. Convulsions (now or in this illness)
4. Lethargic or unconscious

Fast breathing (count 1 full minute)

< 2 months → ≥ 60 / min
2–12 months → ≥ 50 / min
12 months – 5 yrs → ≥ 40 / min

Count when child is calm; look for chest indrawing

Age groups: Young infant 0–2 months · Child 2 months – 5 years · India launched IMNCI in 2003 (IMNCI = IMCI + newborn)
Steps: Assess → Classify (colour) → Identify treatment → Treat → Counsel mother → Follow-up
Main symptoms asked: cough/ difficult breathing · diarrhoea · fever · ear problem → then nutrition & immunization

IMNCI colour coding, general danger signs & fast breathing cut-offs

Colour	Meaning	Action
PINK	Severe classification	**Urgent pre-referral treatment & referral**
YELLOW	Moderate	**Specific treatment at OPD/ home** (antibiotic, ORS, antimalarial)
GREEN	Mild	**Home management & advice**

- **General danger signs (2 m–5 y)**: (1) **not able to drink or breastfeed**, (2) **vomits everything**, (3) **convulsions**, (4) **lethargic or unconscious** (+ convulsing now). Any one → **pink → urgent referral**.
- **Main symptoms assessed**: **cough or difficult breathing, diarrhoea, fever, ear problem** → then **malnutrition & anaemia**, **immunization status**, feeding, other problems.

Cough/ difficult breathing (2 months – 5 yrs)

Signs	Classification	Treatment
Any danger sign **or stridor in calm child** (or SpO_2 < 90 %)	**Severe pneumonia or very severe disease** (pink)	First dose **injectable antibiotic**, refer urgently
Fast breathing and/ or **chest indrawing**	**Pneumonia** (yellow)	**Oral amoxicillin × 5 days** (chest indrawing – as per updated guideline; older Indian chart – refer)
No signs	**Cough or cold – no pneumonia** (green)	Soothe throat (home remedies), follow-up 5 days if not better

- **Fast breathing**: **< 2 months ≥ 60/min**; **2–12 months ≥ 50/min**; **12 months – 5 yrs ≥ 40/min** (count for **full 1 min**, calm child).
- **Chest indrawing** – **lower chest wall goes IN when child breathes IN** (subcostal).
- **Young infant (0–2 m)** – **possible serious bacterial infection (PSBI)** signs: not feeding well, convulsions, **fast breathing ≥ 60**, **severe chest indrawing**, **temp ≥ 37.5 or < 35.5 °C**, movement only when stimulated, umbilicus redness extending to skin, skin pustules many; **jaundice – palms/ soles yellow = severe**.

Fever & ear problem

- **Fever** – classify by malaria risk: high-risk area – **RDT/ blood smear**; **very severe febrile disease (any danger sign or stiff neck)** – pink. **Measles** – rash + cough/ runny nose/ red eyes – vitamin A; complications (clouding cornea, deep mouth ulcers) – pink.
- **Ear** – **tender swelling behind ear = mastoiditis (pink)**; pus < 14 days = **acute ear infection** (amoxicillin 5 days, **dry ear by wicking**); pus ≥ 14 days = **chronic ear infection** (wicking, topical quinolone drops).

7.2 Diarrhoea

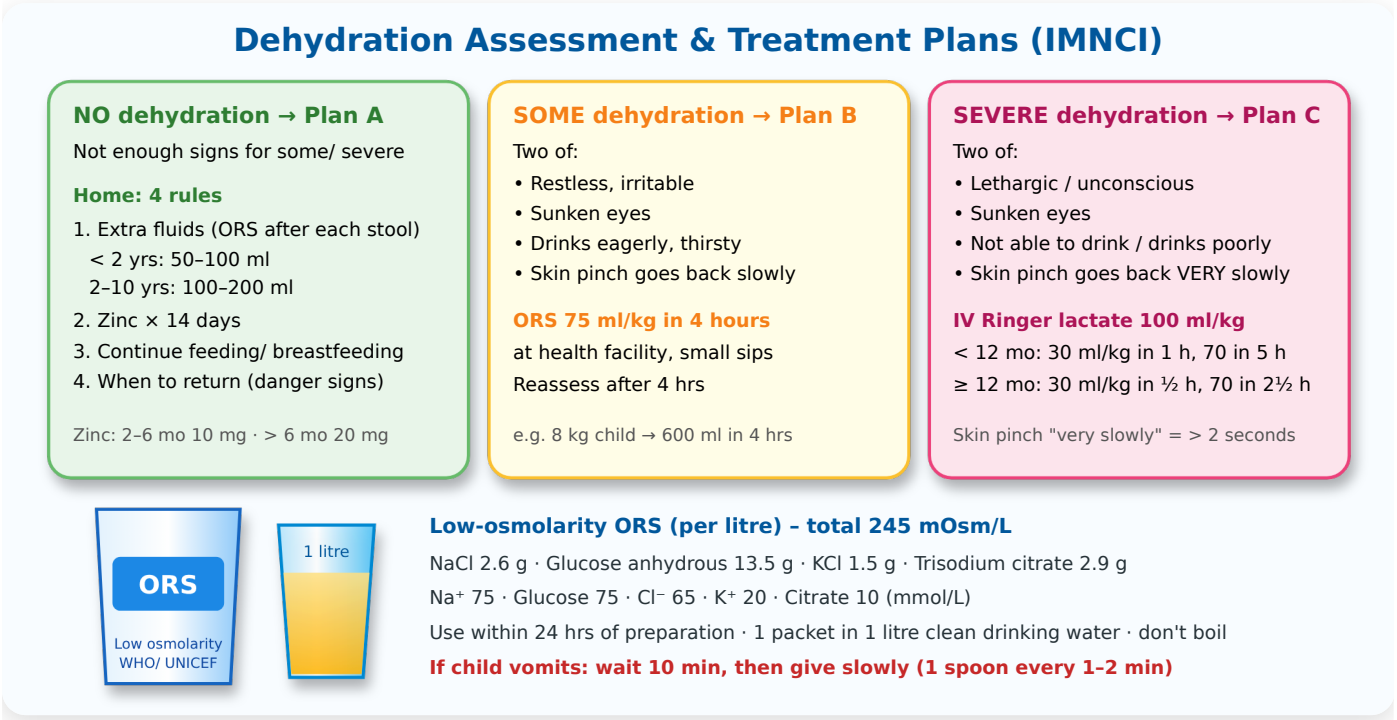
D Diarrhoea (WHO)

Passage of **3 or more loose/ watery stools in 24 hours** (or more frequent than normal for that individual). **Frequent passing of formed stools is not diarrhoea**; **breastfed babies' pasty stools are normal**.

Type	Definition	Key point
Acute watery diarrhoea	< 14 days	Most common – **rotavirus (commonest cause < 2 yrs)**, E. coli (ETEC), cholera – **dehydration** is the danger
Dysentery	**Blood in stool**	**Shigella (commonest)** – **ciprofloxacin 15 mg/kg BD × 3 days** (IMNCI); amoebic – metronidazole
Persistent diarrhoea	**≥ 14 days** (started acutely)	Malnutrition, needs feeding advice; severe persistent (with dehydration) – refer
Chronic diarrhoea	> 30 days / insidious	Malabsorption – coeliac, cystic fibrosis

- **Cholera** – **rice-water stools**, fishy smell, no fever, rapid severe dehydration.

- **Mechanisms** – secretory (toxins – cholera, ETEC), osmotic (lactose intolerance), invasive (Shigella), motility.
- **Complications** – **dehydration (most common cause of death)**, **hypokalaemia**, hypo/ hypernatraemia, **metabolic acidosis (Kussmaul breathing)**, AKI, malnutrition, HUS (E. coli O157, Shigella).



Dehydration classification & Plan A, B, C with ORS composition

Sign	No dehydration	Some dehydration	Severe dehydration
General condition	Well, alert	**Restless, irritable**	**Lethargic or unconscious**
Eyes	Normal	**Sunken**	**Sunken**
Thirst	Drinks normally	**Drinks eagerly, thirsty**	**Not able to drink/ drinks poorly**
Skin pinch	Goes back quickly	**Goes back slowly**	**Goes back very slowly (> 2 sec)**
Fluid deficit	< 5 %	**5–10 %**	**> 10 %**
Plan	**A (home)**	**B (ORS at facility)**	**C (IV)**

- **Classify** with **two or more signs** in a column.
- **Skin pinch** – abdomen, midway between umbilicus & side, pinch longitudinally 1 sec, release; in SAM/ obese – unreliable.
- **Other signs**: depressed AF, dry mouth, no tears, oliguria, weak rapid pulse, cold extremities, delayed **CRT > 3 sec** (shock).

Treatment plans

- **Plan A (no dehydration) – 4 rules of home treatment**:
  1. **Give extra fluids** – ORS after each loose stool: **< 2 yrs: 50–100 ml**; **2–10 yrs: 100–200 ml**; > 10 yrs – as much as wanted; plus home fluids (rice water, lassi, soup, nimbu pani, coconut water) – **avoid sugary drinks/ soft drinks/ glucose**.
  2. **Give zinc** – **2–6 months: 10 mg/day**; **> 6 months: 20 mg/day × 14 days** (dissolved in breast milk/ water) – ↓ duration, severity & recurrence (2–3 months).

3. **Continue feeding** (breastfeeding + usual food) – **never starve**.
4. **When to return** – not able to drink, becomes sicker, blood in stool, fever, drinks poorly.

- **Plan B (some dehydration)** – **ORS 75 ml/kg over 4 hours** at health facility (or by age/weight table), **small frequent sips/ spoon**; **if vomits – wait 10 min, then give more slowly**; continue breastfeeding; **reassess after 4 hrs** → reclassify; give zinc.
- **Plan C (severe dehydration)** – **IV Ringer lactate (or NS) 100 ml/kg**:

Age	First 30 ml/kg in	Then 70 ml/kg in
Infants (< 12 months)	**1 hour**	**5 hours** (total 6 hrs)
Children (12 months – 5 yrs)	**30 minutes**	**2½ hours** (total 3 hrs)

- Repeat first 30 ml/kg once if radial pulse still weak; **reassess every 15–30 min**; start **ORS 5 ml/kg/hr** as soon as child can drink (3–4 hrs infants, 1–2 hrs children); if IV not possible – **NG ORS 20 ml/kg/hr × 6 hrs**; refer if no IV facility within 30 min.
- **Antibiotics – NOT routinely** (only dysentery, cholera with severe dehydration, associated infections). **Antidiarrhoeals/ antimotility drugs (loperamide), antiemetics – not recommended in children**.

ORS

- **Low-osmolarity ORS (WHO 2002) – 245 mOsm/L**: **NaCl 2.6 g, glucose (anhydrous) 13.5 g, KCl 1.5 g, trisodium citrate dihydrate 2.9 g per litre** → Na 75, Cl 65, K 20, citrate 10, glucose 75 mmol/L. (Old ORS 311 mOsm/L – Na 90, glucose 111.)
- **Advantages of low-osmolarity** – ↓ stool output by ~20 %, ↓ vomiting ~30 %, ↓ need for IV by ~33 %.
- Principle – **glucose-coupled sodium transport (SGLT-1)** remains intact in diarrhoea.
- **Preparation** – wash hands, **1 packet in 1 litre clean drinking water** (boiled & cooled), stir; **use within 24 hrs**; don't boil ORS solution, don't add sugar/ salt.
- **Home-made salt-sugar solution (SSS)** – **1 level teaspoon salt (5 g) + 8 level teaspoons sugar (40 g) in 1 litre** (when ORS not available).
- **ReSoMal** – for **SAM** (less sodium, more potassium).
- **ORS + zinc = "ORS-Zinc" co-pack**; **ORS Day 29 July**.

Prevention

- Exclusive breastfeeding, safe water, **hand washing with soap (critical times)**, sanitation (**ODF**), **rotavirus & measles vaccine**, vitamin A, food hygiene.

7.3 Intestinal Worms

Worm	Key points
Roundworm (Ascaris)	Most common; **Loeffler's pneumonitis**, intestinal obstruction, malnutrition
Hookworm (Ancylostoma/ Necator)	**Barefoot walking** (skin penetration); **iron-deficiency anaemia**, ground itch
Pinworm/ threadworm (Enterobius)	**Perianal itching at night**, autoinfection; **cellophane (Scotch) tape test** in morning; treat whole family, cut nails
Tapeworm (Taenia)	Undercooked pork/ beef; **neurocysticercosis (seizures)**
Giardia	Malabsorption, foul greasy stools – metronidazole

- **Albendazole 400 mg single dose** (200 mg for 1–2 yrs); **National Deworming Day 10 Feb & 10 Aug** (1–19 yrs); mebendazole 100 mg BD × 3 days.

PYQ style

- IMNCI launched in India – **2003**; age groups – **0–2 months & 2 months–5 years**.
- Pink in IMNCI – **urgent referral**; green – **home care**.
- General danger sign – **not able to drink, vomits everything, convulsions, lethargic/ unconscious**.
- Fast breathing 2–12 months – **≥ 50/min**; 1–5 yrs – **≥ 40/min**.
- Diarrhoea – **≥ 3 loose stools in 24 hrs**; persistent – **≥ 14 days**.
- Commonest cause of diarrhoea in < 2 yrs – **rotavirus**; dysentery – **Shigella**.
- Some dehydration – **Plan B: ORS 75 ml/kg in 4 hrs**.
- Plan C fluid – **Ringer lactate 100 ml/kg**; infant 30 ml/kg in – **1 hr**.
- Skin pinch very slow – **> 2 seconds**.
- Low-osmolarity ORS – **245 mOsm/L**; sodium – **75 mmol/L**.
- Zinc for child > 6 months – **20 mg × 14 days**.
- Night perianal itching – **pinworm (Enterobius)**; test – **cellophane tape**.
- Hookworm causes – **iron-deficiency anaemia**.

8 Respiratory Disorders in Children

8.1 Why Children Are More Prone

- **Narrow airways** (small oedema → big obstruction – resistance ∝ $1/r^4$), **short, horizontal Eustachian tube** (otitis media), **obligatory nose breathers (up to 3–4 months)**, large tongue, **large occiput**, funnel-shaped larynx – **narrowest at cricoid (< 8 yrs)**, immature immunity, ↑ metabolic rate & O_2 demand, abdominal breathing.
- **ARI (Acute Respiratory Infection)** – most common cause of OPD visits; **pneumonia – leading infectious killer of under-5 children**.

8.2 Upper Respiratory Tract

Condition	Key points & nursing
Common cold (nasopharyngitis)	**Rhinovirus**; nasal saline drops, suction before feeds, fluids, no antibiotics
Pharyngitis/ Tonsillitis	Viral or **Group A β-haemolytic Streptococcus** (**risk of rheumatic fever, AGN**) → **penicillin/ amoxicillin × 10 days**; warm saline gargles
Tonsillectomy	Indications – **recurrent tonsillitis (≥ 7/yr, 5/yr × 2 yrs, 3/yr × 3 yrs)**, obstructive sleep apnoea, peritonsillar abscess (quinsy); **post-op: prone/ side-lying (tonsillar position)**, **ice collar**, **cold fluids, ice cream**; avoid **red/ brown fluids, straws, citrus, hot food**, avoid vigorous coughing/ nose blowing; **frequent swallowing = earliest sign of haemorrhage** (within 24 hrs/ 5–10 days); **avoid aspirin**
Acute otitis media (AOM)	**Short, wide, horizontal Eustachian tube**; **S. pneumoniae, H. influenzae**; ear pulling, fever, irritability; **bottle feeding in supine position** ↑ risk; Rx – amoxicillin; **myringotomy/ grommets** for recurrent; **CSOM** – wicking, keep ear dry
Sinusitis, adenoid hypertrophy	Adenoid facies, mouth breathing, snoring

8.3 Croup & Epiglottitis

	Croup (Laryngotracheobronchitis)	Acute Epiglottitis
Cause	**Parainfluenza virus** (commonest)	**H. influenzae type b** (bacterial)
Age	**6 months – 3 yrs**	**2–6 yrs**
Onset	Gradual (after URI), worse at night	**Sudden, rapid progression**
Features	**Barking (seal-like) cough**, **inspiratory stridor**, hoarseness, low fever	**High fever, toxic, drooling, dysphagia, dysphonia, tripod position**, **no cough**, muffled voice
X-ray	**Steeple sign** (subglottic narrowing, AP view)	**Thumb sign** (swollen epiglottis, lateral view)
Treatment	**Humidified O_2/ cool mist, dexamethasone (0.15–0.6 mg/kg), nebulised adrenaline**	**Airway first – intubation in OT**, IV ceftriaxone; **prevented by Hib vaccine**

! Epiglottitis – NEVER examine the throat

Do not use tongue depressor or throat swab, don't lay child supine, keep child calm in parent's lap (sitting) – can trigger **complete laryngospasm/ obstruction**. Emergency tracheostomy/ intubation set at bedside.

- **Spasmodic croup** – sudden night attacks, no fever. **Diphtheria** – grey pseudomembrane, bull neck (see communicable diseases).
- **Laryngomalacia** – most common cause of **congenital stridor** (floppy larynx, improves by 1–2 yrs).

8.4 Lower Respiratory Tract

Bronchiolitis

- **RSV (Respiratory Syncytial Virus) – commonest cause**; **< 2 yrs (peak 2–6 months)**, winter.
- Coryza → **cough, wheeze, tachypnoea, chest retractions, hyperinflation, fine crepitations**, apnoea in young infants.
- **Management – supportive**: **humidified O_2 (SpO_2 < 90–92 %)**, nasal suction, hydration (NG/ IV), small frequent feeds; **bronchodilators/ steroids/ antibiotics NOT routinely**; hypertonic saline nebulisation (some); **contact isolation** (RSV spreads by hands/ fomites); **palivizumab** (prophylaxis in high risk).

Pneumonia

- **Most common cause – S. pneumoniae** (bacterial); **viral – RSV**; **Staph aureus** (post-measles, pneumatocele, empyema); **H. influenzae**; neonates – GBS, E. coli; **atypical – Mycoplasma** (school age).
- **Types**: lobar, **bronchopneumonia (most common in children)**, interstitial.
- **S/S** – **fever, cough, fast breathing (most sensitive sign)**, **chest indrawing**, nasal flaring, **grunting**, cyanosis, crepitations, bronchial breathing.
- **Investigations** – CXR, CBC, CRP, blood culture, **pulse oximetry**.
- **Treatment** – **amoxicillin (oral) – non-severe**; **severe – injectable ampicillin + gentamicin / ceftriaxone**; O_2 if SpO_2 < 90 %.
- **Nursing** – **semi-Fowler's/ propped-up position**, O_2, **airway clearance (suction, chest physiotherapy, postural drainage)**, tepid sponging/ paracetamol for fever, **hydration** (avoid fluid overload), small frequent feeds (NG if RR > 60–70), rest, monitor RR/ SpO_2, isolation if needed.
- **Complications** – **empyema (Staph)**, pleural effusion, lung abscess, **pneumatocele**, sepsis, respiratory failure.
- **Prevention** – **PCV, Hib, measles, pertussis vaccines**, exclusive BF, nutrition, **reduce indoor air pollution**, **SAANS campaign (2019)**.

Bronchial Asthma

- **Chronic inflammatory airway disease** with **reversible** airway obstruction, bronchial hyperresponsiveness; **most common chronic disease in children**.
- Triggers – **allergens (house dust mite – commonest)**, smoke, cold air, viral infections, exercise, pollen, pets, aspirin.

- **S/S** – **recurrent wheeze (expiratory)**, cough (**nocturnal**), chest tightness, breathlessness; **silent chest = ominous (severe)**.
- **Investigations** – **PEFR**, spirometry (**> 5 yrs**) – **↑ FEV_1 ≥ 12 % after bronchodilator**.
- **Acute attack** – **O_2, nebulised salbutamol (SABA) – first-line (every 20 min × 3)**, **ipratropium**, **systemic steroids (prednisolone 1–2 mg/kg)**, $MgSO_4$ IV (severe), **aminophylline** (rarely).
- **Long-term** – **inhaled corticosteroids (budesonide/ fluticasone) – most effective controller**, LABA (+ICS > 5 yrs), montelukast.
- **Inhaler devices**: **< 4 yrs – MDI + spacer with face mask**; **4–8 yrs – MDI + spacer (mouthpiece)**; > 8 yrs – DPI/ MDI; **rinse mouth after ICS (oral thrush)**.
- **Nursing** – **high Fowler's / tripod**, calm child, avoid triggers, **peak flow diary**, action plan (green/ yellow/ red zones), exercise allowed with pre-treatment.

Foreign Body Aspiration

- **Common 1–3 yrs**; peanuts, beads, coins (coin → oesophagus).
- **Right main bronchus** – more common (**wider, shorter, more vertical**).
- Sudden **choking, coughing, wheeze (unilateral)**, stridor, decreased air entry; X-ray – **obstructive emphysema (check-valve)**/ collapse.
- **Rx – rigid bronchoscopy**. First aid – **infant: 5 back blows + 5 chest thrusts**; **child > 1 yr: abdominal thrusts (Heimlich)**; **no blind finger sweep**.

Cystic Fibrosis

- **Autosomal recessive**, **CFTR gene (chromosome 7)**; thick secretions – lungs, pancreas, intestines.
- **Meconium ileus (earliest manifestation)**, **salty skin ("salty kiss")**, steatorrhoea (bulky, greasy, foul stools), recurrent chest infections (**Pseudomonas, Staph**), FTT.
- **Diagnosis – sweat chloride test > 60 mEq/L (pilocarpine iontophoresis)**.
- **Rx** – **chest physiotherapy (postural drainage) 2–4 times/day**, pancreatic enzymes **with meals**, high-calorie high-protein diet with **extra salt & fat-soluble vitamins (A, D, E, K)**, antibiotics, dornase alfa.

8.5 Tuberculosis in Children

- Usually **primary complex (Ghon focus + hilar lymph node)**; **sputum often unavailable** – **gastric aspirate (early morning, 2 consecutive days)**, induced sputum; **CBNAAT/ GeneXpert**; **Mantoux (TST)** – **0.1 ml (5 TU) PPD intradermal, read at 48–72 hrs; ≥ 10 mm induration = positive** (≥ 5 mm in HIV/ SAM).
- Forms – pulmonary, **TB lymphadenitis (cervical) – most common extrapulmonary**, **TB meningitis** (most serious), miliary TB, spinal (Pott's).
- **Treatment (NTEP)** – **2HRZE + 4HRE** (intensive 2 months + continuation 4 months), **weight-band paediatric FDC (dispersible)**; **TBM/ bone TB – 2HRZE + 10HRE** (12 months) + steroids.
- **TB preventive therapy (TPT)** – **child < 5 yrs household contact** of pulmonary TB (after excluding active TB) – **INH 10 mg/kg × 6 months** (or 3HP for older children).
- **Ni-kshay Poshan Yojana** – **₹ 1000/month (increased from ₹ 500 in 2024)** nutrition support during treatment.
- **BCG** – protects against **severe forms (TBM, miliary)** in children.

PYQ style

- Narrowest part of child's airway (< 8 yrs) – **cricoid cartilage**.
- Commonest cause of croup – **parainfluenza virus**; X-ray – **steeple sign**.
- Epiglottitis organism – **Hib**; X-ray – **thumb sign**; nursing – **don't examine throat**.
- Bronchiolitis organism – **RSV**; age – **< 2 years**.
- Commonest cause of bacterial pneumonia – **S. pneumoniae**.
- Most sensitive sign of pneumonia – **fast breathing**.
- Earliest sign of post-tonsillectomy bleeding – **frequent swallowing**; position – **prone/ side lying**.
- Foreign body commonly lodges in – **right main bronchus**.
- Infant choking – **5 back blows + 5 chest thrusts**.
- Cystic fibrosis test – **sweat chloride**; earliest sign – **meconium ileus**.
- Asthma controller drug – **inhaled corticosteroid**; reliever – **salbutamol**.
- Mantoux positive – **≥ 10 mm** induration at 48–72 hrs.
- Sample for TB in young children – **gastric aspirate**.

9 Cardiovascular Disorders in Children

9.1 Circulation Changes at Birth

- Fetal shunts – **ductus venosus** (umbilical vein → IVC), **foramen ovale** (RA → LA), **ductus arteriosus** (pulmonary artery → aorta).
- At birth – lungs expand → **pulmonary resistance ↓**, **LA pressure > RA → foramen ovale closes (functional within hours)**; ↑ PaO_2 & ↓ prostaglandins → **ductus arteriosus closes functionally in 10–15 hrs (anatomically 2–3 wks)**.
- Remnants – **fossa ovalis, ligamentum arteriosum, ligamentum venosum, ligamentum teres** (umbilical vein), **medial umbilical ligaments** (umbilical arteries).

9.2 Congenital Heart Disease (CHD)

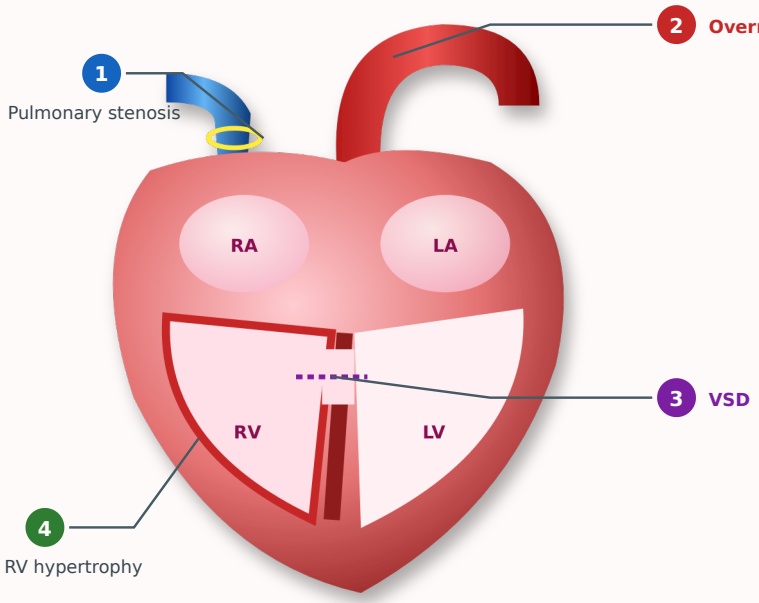
- **Incidence ~8–10 per 1000 live births**. Causes – mostly multifactorial; **maternal rubella (PDA, PS)**, **diabetes (TGA, VSD)**, alcohol, drugs (lithium – **Ebstein's**; warfarin; phenytoin), **chromosomal – Down syndrome (AVSD/ endocardial cushion defect)**, **Turner (coarctation)**, **Marfan**, **DiGeorge (truncus, interrupted arch)**.

	Acyanotic (L → R shunt / obstruction)	Cyanotic (R → L shunt)
Examples	**VSD (most common CHD)**, **ASD**, **PDA**, AVSD; obstructive – **coarctation, aortic stenosis, pulmonary stenosis**	**TOF (most common cyanotic after infancy)**, **TGA (most common cyanotic in newborn)**, tricuspid atresia, TAPVC, truncus arteriosus, Ebstein's, HLHS
Mnemonic	–	**"5 T's"** – TOF, TGA, Tricuspid atresia, TAPVC, Truncus arteriosus
Features	**CHF, recurrent chest infections, FTT**, murmur	**Cyanosis, clubbing, polycythaemia, squatting, hypoxic spells**, brain abscess, stroke

Defect	Key points
VSD	**Most common CHD (~25–30 %)**; **pansystolic murmur, left lower sternal border**; small (muscular) may **close spontaneously**; large → CHF, pulmonary HTN → **Eisenmenger syndrome** (reversal R → L, cyanosis)
ASD (secundum – commonest)	**Wide fixed split S2** (pathognomonic); often asymptomatic childhood; ejection systolic murmur (pulmonary area)
PDA	**Preterms, rubella**; **continuous "machinery" murmur** (left infraclavicular), **bounding pulses, wide pulse pressure**; closure – **indomethacin/ ibuprofen/ paracetamol** (preterm), device/ surgical ligation; **keep open in duct-dependent lesions – Prostaglandin E1 (alprostadil)**
Coarctation of aorta	Narrowing near ductus; **upper limb HTN, weak/ delayed femoral pulses (radio-femoral delay)**, **BP arms > legs**; X-ray – **rib notching (> 8 yrs)**, **"3" sign**; Turner syndrome
TOF	4 defects – **VSD, pulmonary stenosis, overriding aorta, RVH**; **boot-shaped heart (coeur en sabot)**; cyanosis after 3–6 months; **squatting**; **tet spells**

Defect	Key points
TGA	Aorta from RV, PA from LV – **parallel circulation**; severe cyanosis at birth; **"egg on a string"** X-ray; survival needs mixing (PDA/ ASD/ VSD) – **PGE1, balloon atrial septostomy (Rashkind)**, **arterial switch (Jatene)**
TAPVC	**"Snowman / figure of 8"** heart (supracardiac)
Ebstein's anomaly	Downward displaced tricuspid; maternal **lithium**; **"box-shaped" heart**
HLHS	Most common cause of cardiac death in first week

Tetralogy of Fallot (TOF) – Most Common Cyanotic CHD



4 defects – "PROV"

Pulmonary stenosis (RVOT obstruction)
Right ventricular hypertrophy
Overriding of aorta
Ventricular septal defect

Features

Cyanosis, clubbing, polycythaemia
X-ray: boot-shaped heart
Squatting (↑ SVR → less R→L shunt)

Tet (hypercyanotic) spell

Knee-chest position · O_2 · calm child
Morphine · propranolol · IV fluids
Definitive: total surgical correction

Deoxygenated blood RV → overriding aorta through VSD (right-to-left shunt) → central cyanosis

Tetralogy of Fallot – 4 defects, features & management of tet spell

Hypercyanotic (Tet) spell – nursing

- Triggers – crying, feeding, defecation, dehydration, waking up; peak **2–4 months to 2 yrs**, morning.
- S/S – ↑ cyanosis, hyperpnoea, irritability, **disappearance/ softening of murmur**, limpness, convulsions, LOC.
- **Management**: **knee-chest position** (↑ SVR → ↓ R→L shunt), **calm the child (parent's arms)**, **humidified O_2**, **morphine 0.1–0.2 mg/kg SC/ IV** (↓ infundibular spasm, hyperpnoea), **IV fluids, sodium bicarbonate** (acidosis), **propranolol** (prevention & acute), phenylephrine (↑ SVR); avoid dehydration (polycythaemia → stroke).
- Older children **squat** spontaneously after exertion.
- Surgery – **Blalock-Taussig (BT) shunt** (palliative – subclavian to pulmonary artery), **total correction at ~6–12 months**.

Nursing care in CHD

- Assess – colour, **apical pulse 1 min**, RR, **SpO_2 (pre- & post-ductal)**, feeding difficulty (**sweating during feeds**, tires easily), weight gain, edema.
- **Feeding** – **small frequent feeds**, **soft nipple with larger hole**, rest during feeds, **semi-upright position**, high-calorie formula/ fortified EBM, NG if needed (feeding time ≤ 30 min).

- **Rest**, organize care, avoid crying, **prevent infection** (hand hygiene, immunization – including influenza, PCV), **dental hygiene** (infective endocarditis prophylaxis in selected cyanotic CHD before dental procedures).
- **Hydration** in cyanotic CHD (polycythaemia – thrombosis), **iron supplementation** (relative anaemia).
- **Cardiac catheterisation post-care** – **check distal pulses, colour, warmth below the site every 15 min** (first hour), **keep limb straight 4–6 hrs**, pressure dressing, bleeding, vitals, hydration.
- Parent education – **CPR**, signs of CHF, medications (digoxin).

9.3 Congestive Heart Failure (CHF) in Children

- Causes – **infants: CHD (large VSD, PDA, AVSD – most common)**; older: **rheumatic heart disease**, myocarditis, cardiomyopathy, severe anaemia, hypertension (AGN).
- **S/S in infants** – **tachypnoea, tachycardia, feeding difficulty, sweating (forehead) during feeds**, **poor weight gain/ FTT**, **hepatomegaly (most reliable sign of right failure in infants)**, gallop rhythm, wheeze, **periorbital oedema** (peripheral oedema & ↑ JVP uncommon in infants), cardiomegaly.
- **Management**: **propped-up (semi-Fowler's/ infant seat 45°)**, **O_2**, **restrict fluids & sodium**, small frequent high-calorie feeds, **diuretics (frusemide – monitor K^+)**, **ACE inhibitors (captopril/ enalapril)**, **digoxin**, daily weight & I/O, treat cause.

! Digoxin in children

- **Check apical pulse for 1 full minute before giving**: **withhold if HR < 90–110/min (infant)**, **< 70 (older child)**, **< 60 (adolescent/ adult)** – inform doctor.
- Measure dose exactly (**0.1 ml can be a big difference**), give **1 hr before or 2 hrs after feeds**, don't mix in food, **don't repeat dose if vomited**; regular intervals.
- **Toxicity – vomiting (earliest in children), nausea, anorexia, bradycardia, arrhythmias**, visual disturbances; **hypokalaemia ↑ toxicity** (diuretics) – antidote **digoxin-specific Fab (Digibind)**.

9.4 Rheumatic Fever & Rheumatic Heart Disease

- **Autoimmune reaction** 2–4 weeks after **Group A β-haemolytic streptococcal pharyngitis** (**not skin infection**); age **5–15 yrs**; **molecular mimicry** (M protein).
- **Revised Jones criteria (AHA 2015)**: **2 major OR 1 major + 2 minor**, **plus evidence of preceding GAS infection** (↑/ rising **ASO titre**, throat culture, rapid antigen test).

Major (J♥NES)	Minor
J – Joints: **migratory polyarthritis** (large joints; most common, ~75 %, **heals without deformity**)	**Fever** (≥ 38.5 °C)
♥ – **Carditis** (pancarditis; **most serious – only one causing permanent damage**; mitral valve most affected)	**Arthralgia**
N – Subcutaneous **Nodules** (painless, over bony prominences)	**↑ ESR (≥ 60)/ CRP (≥ 3 mg/dl)**
E – **Erythema marginatum** (non-itchy, trunk, evanescent)	**Prolonged PR interval** (ECG)
S – **Sydenham's chorea** (St. Vitus dance – late, may occur alone; emotional lability)	

- **Treatment** – **bed rest** (strict in carditis), **eradicate strep – benzathine penicillin G 1.2 MU IM single dose** (0.6 MU if < 27 kg) or oral penicillin/ amoxicillin × 10 days; **aspirin (anti-inflammatory) 75–100 mg/kg/day** for arthritis; **steroids (prednisolone) for severe carditis**; haloperidol for chorea.
- **Secondary prophylaxis (most important)** – **benzathine penicillin 1.2 MU IM every 21 days (3 weeks)** in India; duration – **no carditis: 5 yrs or till 21 yrs (whichever longer)**; carditis without residual: 10 yrs or till 21; **residual valve disease: 10 yrs or till 40 yrs (sometimes lifelong)**.
- **Primary prevention** – treat streptococcal sore throat with penicillin × 10 days.
- **RHD** – **mitral regurgitation (commonest in children)**, mitral stenosis (adults), aortic regurgitation.
- **Nursing** – rest, **monitor pulse (sleeping pulse ↑ = carditis)**, joint care (warm compress, support with pillows, cradle), safety in chorea (padded side rails), **compliance with penicillin prophylaxis** (test dose, anaphylaxis tray), nutrition, school continuity.

9.5 Other Disorders

- **Infective endocarditis** – Strep viridans (post-dental), Staph aureus; fever, new murmur, splinter haemorrhages, Osler's nodes, **Janeway lesions**, Roth spots; **blood cultures × 3** before antibiotics; IV antibiotics 4–6 weeks.
- **Kawasaki disease (mucocutaneous lymph node syndrome)** – < 5 yrs; **fever ≥ 5 days** + 4 of 5 (**CRASH**): **C**onjunctivitis (bilateral, non-purulent), **R**ash, **A**denopathy (cervical, unilateral), **S**trawberry tongue/ cracked red lips, **H**ands & feet – oedema, erythema, **periungual desquamation** (2nd–3rd week); complication – **coronary artery aneurysm** (most serious; echo); Rx – **IVIG 2 g/kg** single infusion + **aspirin** (high dose → low dose); **one of the few indications of aspirin in children**.
- **Hypertension in children** – BP ≥ 95th centile; **secondary HTN more common in young children (renal – most common)**.
- **Myocarditis** – viral (Coxsackie B).

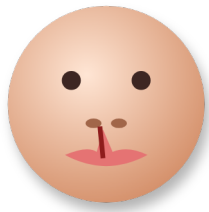
PYQ style

- Most common CHD – **VSD**; most common cyanotic CHD – **TOF**; cyanotic at birth – **TGA**.
- Boot-shaped heart – **TOF**; egg on string – **TGA**; snowman – **TAPVC**.
- Machinery murmur – **PDA**; fixed wide split S2 – **ASD**.
- Radio-femoral delay & rib notching – **coarctation of aorta**.
- PDA closure drug – **indomethacin/ ibuprofen**; keep ductus open – **PGE1**.
- Tet spell position – **knee-chest**; drug – **morphine**.
- Eisenmenger syndrome – **reversal of shunt (R → L)**.
- Most reliable sign of CHF in infants – **hepatomegaly**; feeding sign – **sweating while feeding**.
- Withhold digoxin in infant if apical pulse – **< 90–110/min**.
- Rheumatic fever follows – **streptococcal pharyngitis**; test – **ASO titre**.
- Jones major criterion causing permanent damage – **carditis**.
- Secondary prophylaxis – **benzathine penicillin 1.2 MU every 21 days**.
- Kawasaki complication – **coronary aneurysm**; Rx – **IVIG + aspirin**.

10 Gastrointestinal Disorders in Children

Common Congenital GI Anomalies – Key Signs

Cleft Lip & Palate



Feeding difficulty, aspiration

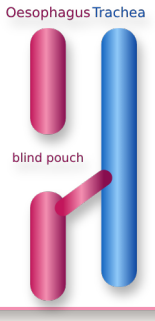
Feed upright, special nipple/ spoon/ ESSR, burp often

Lip repair: ~3 months (Rule of 10s)

Palate: 9–18 months (before speech)

Post-op lip: elbow restraints, Logan bow, supine/ side

TEF – Type C (~85 %)



Upper blind pouch + lower segment joined to trachea

3 C's: Coughing, Choking, Cyanosis on feeding

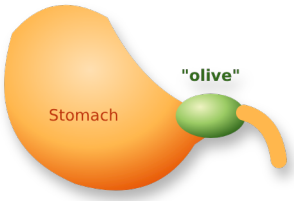
Excess frothy saliva/ drooling

Mother: polyhydramnios

Catheter can't pass to stomach

Nursing: NPO, head-up 30°, continuous pouch suction

Pyloric Stenosis



2–6 weeks, first-born male

Projectile, non-bilious vomiting after feeds

Olive-shaped mass (RUQ)

Visible peristalsis L → R

Hypochloraemic hypokalaemic metabolic alkalosis

Rx: Ramstedt's pyloromyotomy

Intussusception



6 months – 2 yrs (ileocolic)

Sudden colicky pain, draws legs up, vomiting

Red currant jelly stool

Sausage-shaped mass

Dance's sign (empty RIF)

USG: target/ doughnut sign

Rx: air/ hydrostatic enema reduction → surgery

Cleft lip, TEF type C, pyloric stenosis & intussusception – key signs

10.1 Cleft Lip & Cleft Palate

- **Failure of fusion** – lip (maxillary & medial nasal processes, **5–6 wks**), palate (palatal shelves, **8–12 wks**). **Cleft lip ± palate** – more in **males**; isolated cleft palate – females.
- Causes – genetic, **maternal smoking, anticonvulsants (phenytoin, valproate)**, **folic acid deficiency**, alcohol, steroids.
- **Problems** – **feeding difficulty (most immediate)**, aspiration, **recurrent otitis media/ hearing loss** (palate), speech defects, dental problems, cosmetic/ psychological.
- **Feeding** – **upright position (45–60°)**, **ESSR method (Enlarge nipple, Stimulate suck, Swallow, Rest)**, special nipples (cleft palate/ squeeze bottle, **Haberman feeder**), **spoon/ paladai/ dropper**, burp frequently (more air swallowed), small frequent feeds; **breastfeeding possible with cleft lip**; palatal **obturator**.
- **Surgery timing** – **cleft lip (cheiloplasty) ~3 months** – **"Rule of 10s": ≥ 10 weeks age, ≥ 10 lb (4.5 kg) weight, Hb ≥ 10 g/dl, WBC < 10,000**; **cleft palate (palatoplasty) 9–18 months** (before speech develops; many centres 6–12 months).

Post-operative care

Cleft lip repair	Cleft palate repair
Supine / side-lying – NEVER prone (protects suture line)	**Prone / side-lying** (drainage of secretions)

Cleft lip repair	Cleft palate repair
Logan bow (metal bar)/ tape – protects suture line from tension	No straws, spoons, pacifiers, forks in mouth
Elbow restraints – prevent touching	**Elbow restraints**
Clean suture line with **saline/ sterile water after feeds** (remove crusts), apply antibiotic ointment	**Rinse mouth with water after feeding**
Avoid crying & sucking (stress on sutures) – feed with dropper/ syringe on side	**Cup feeding**; soft/ liquid diet; no hard food ~ 3 weeks

- Team care – plastic surgeon, ENT, **speech therapist**, orthodontist, audiologist.

10.2 Oesophageal Atresia & Tracheo-oesophageal Fistula (TEF)

- **Type C (proximal oesophageal atresia + distal TEF) – most common (~85 %)**; Type A – pure atresia (no gas in abdomen); **H-type (E)** – fistula without atresia (recurrent pneumonia, late diagnosis).
- **Antenatal** – **maternal polyhydramnios**.
- **S/S** – **excessive frothy salivation/ drooling (first sign)**, **3 C's – Choking, Coughing, Cyanosis during feeding**, regurgitation, abdominal distension (type C), aspiration pneumonia.
- **Diagnosis** – **inability to pass a stiff red rubber catheter (No. 8–10 Fr) into stomach – coils in pouch** (X-ray shows tube coiled at T2–T4); associated **VACTERL** anomalies (Vertebral, Anorectal, Cardiac, TE, Renal, Limb).
- **Pre-op nursing** – **NPO**, **head end elevated 30–45° (reverse Trendelenburg) / prone** (prevents reflux of gastric acid via fistula), **continuous/ frequent suction of upper pouch (Replogle tube)**, IV fluids, O_2, antibiotics, warmth.
- **Surgery** – primary anastomosis with fistula ligation; gastrostomy if long gap. Post-op – suction (marked length, not beyond anastomosis), chest drain care, **watch anastomotic leak, stricture (dilatation), GERD**.

10.3 Hypertrophic Pyloric Stenosis (HPS)

- **Hypertrophy & hyperplasia of pyloric circular muscle**; **first-born males (4 : 1)**; **age 2–6 weeks (peak 3 wks)**; erythromycin exposure.
- **S/S** – **projectile, non-bilious vomiting** shortly after feeds, **hungry after vomiting ("hungry vomiter")**, **visible gastric peristalsis (left → right)**, **palpable olive-shaped mass (RUQ/ epigastrium) – after vomiting/ during test feed**, dehydration, weight loss, constipation, jaundice.
- **Metabolic** – **hypochloraemic, hypokalaemic metabolic alkalosis** (loss of HCl), paradoxical aciduria.
- **Diagnosis** – **USG (pyloric muscle thickness ≥ 3–4 mm, length ≥ 15–17 mm)** – investigation of choice; barium – **string sign, shoulder sign**.
- **Management** – correct dehydration & electrolytes FIRST (NS + KCl), NPO, NG decompression → **Ramstedt's pyloromyotomy** (it is **not a surgical emergency** – fluid & electrolyte stabilisation comes first).
- **Post-op** – start feeds **4–6 hrs** after surgery (small, clear fluids → breast milk), **upright position, burp**, mild vomiting may occur initially.

10.4 Intussusception

- **Telescoping of one bowel segment (intussusceptum) into adjacent distal (intussuscipiens)**; **ileocolic (most common)**.
- **Age 6 months – 2 yrs (peak 5–9 months)**, boys; **most common cause of intestinal obstruction in this age group**; idiopathic (Peyer's patch hypertrophy after viral infection), lead points (Meckel's, polyp) in older; rotavirus vaccine (rare).
- **Classic triad** – **sudden severe colicky abdominal pain** (child draws legs up, screams, pale, then quiet between attacks), **vomiting**, **red currant jelly stools** (blood + mucus; late), **sausage-shaped mass (RUQ)**, **Dance's sign (empty RIF)**, lethargy.
- **Diagnosis** – **USG – target/ doughnut sign (transverse), pseudokidney sign (longitudinal)**; contrast enema – **claw sign, coiled spring**.
- **Treatment** – **non-operative reduction – air (pneumatic) or hydrostatic (saline/ barium) enema under USG/ fluoroscopy** (if no peritonitis/ perforation/ shock); **passage of normal brown stool = reduction successful – report to doctor**; surgery (manual reduction/ resection) if failed. **Recurrence** ~10 % (mostly in first 72 hrs).

10.5 Hirschsprung Disease (Congenital Aganglionic Megacolon)

- **Absence of ganglion cells (Meissner & Auerbach plexus) in distal bowel** – **rectosigmoid (most common)** → no peristalsis → functional obstruction, proximal dilatation (megacolon). Males; **Down syndrome**.
- **Newborn** – **failure to pass meconium within 24–48 hrs** (**most important early sign**), bilious vomiting, abdominal distension; **"blast/ squirt sign"** on PR exam (explosive gas & stool).
- **Older child** – **chronic constipation since birth, ribbon-like/ pellet foul-smelling stools**, abdominal distension, FTT, palpable faecal mass, **empty rectum** on PR.
- **Complication** – **Hirschsprung-associated enterocolitis (HAEC)** – fever, explosive diarrhoea – life-threatening.
- **Diagnosis** – **rectal suction biopsy (gold standard – absent ganglion cells, ↑ AChE)**; barium enema – **transition zone**; anorectal manometry (absent recto-anal inhibitory reflex).
- **Treatment** – **rectal washouts (NS – never tap water: water intoxication)**; **colostomy** → definitive **pull-through (Swenson, Duhamel, Soave, transanal)**.
- **Nursing** – **NS enemas/ washouts**, stoma care, measure abdominal girth (**at umbilicus, same place**), low-residue diet pre-op, post-op: no rectal temps/ suppositories.

10.6 Anorectal Malformation (Imperforate Anus)

- Absence/ abnormal location of anal opening; **detected at birth – first newborn exam/ first rectal temp (unable to insert)**, **no meconium in 24 hrs**, meconium in urine (rectourethral fistula)/ vagina; abdominal distension.
- Classification – **high (above levator ani – colostomy first)** vs **low (perineal anoplasty)**; **invertogram/ cross-table prone lateral X-ray at 18–24 hrs**.
- Surgery – **PSARP (posterior sagittal anorectoplasty – Pena)**; post-op **anal dilatation** (Hegar dilators) for weeks, stoma care.

10.7 Other GI Conditions

Condition	Key points
Omphalocele (exomphalos)	Abdominal contents **through umbilicus, covered by sac (membrane)**; **associated anomalies common** (Beckwith-Wiedemann, trisomies, cardiac)
Gastroschisis	**Defect lateral (right) to umbilicus, NO covering sac** – bowel exposed; few associated anomalies
Nursing (both)	**Cover with sterile warm moist saline dressing + plastic wrap (bowel bag)**, position on side (prevent kinking), NG decompression, IV fluids, temperature (↑ heat/ fluid loss), **prevent infection**
Umbilical hernia	Common in **preterms, African babies, hypothyroidism**; **most close spontaneously by 3–5 yrs** – **no strapping/ coin application**; surgery if > 4–5 yrs
Inguinal hernia	Patent processus vaginalis; males, preterm; **risk of incarceration – early herniotomy**
Hydrocele	Transilluminates; most resolve by 1–2 yrs
GERD	Regurgitation, poor weight gain, irritability; **thickened feeds, upright position after feeds, small frequent feeds, left-lateral/ prone only when awake & observed** (**sleep supine – SIDS**)
Coeliac disease	**Gluten intolerance (wheat, barley, rye – "BROW": barley, rye, oats, wheat)**; after wheat introduction; **chronic diarrhoea, steatorrhoea, abdominal distension, wasting of buttocks, FTT, anaemia**; **anti-tTG IgA**, **duodenal biopsy (villous atrophy)**; **lifelong gluten-free diet – rice, maize, jowar, bajra allowed**
Biliary atresia	**Obliteration of bile ducts** – **persistent jaundice > 2 weeks, clay/ pale stools, dark urine**, hepatomegaly; **Kasai portoenterostomy before 60 days (8 weeks)** – best outcome; liver transplant
Hepatitis A	Feco-oral; most common viral hepatitis in children; self-limiting; vaccine available
Hepatitis B	Vertical (perinatal) – **90 % chronic if infected at birth → HBIG + Hep B vaccine within 12–24 hrs**
Acute appendicitis	> 5 yrs; periumbilical pain → **RIF (McBurney's point)**, fever, vomiting; **no enema/ laxative/ heat pad** (perforation); perforation common in < 5 yrs
Meckel's diverticulum	**"Rule of 2s"** – 2 % population, 2 ft from ileocaecal valve, 2 inches long, 2 types of ectopic tissue (gastric/ pancreatic), symptomatic by 2 yrs; **painless rectal bleeding**; **technetium-99m (Meckel's) scan**
Lactose intolerance	After gastroenteritis – watery, frothy, acidic stools, perianal excoriation; **lactose-free diet temporarily**
Colic (infantile)	**Rule of 3s** – crying > 3 hrs/day, > 3 days/week, > 3 weeks; 2 wks–4 months; burping, holding, rocking
Constipation	Most functional; fibre, fluids, toilet routine

PYQ style

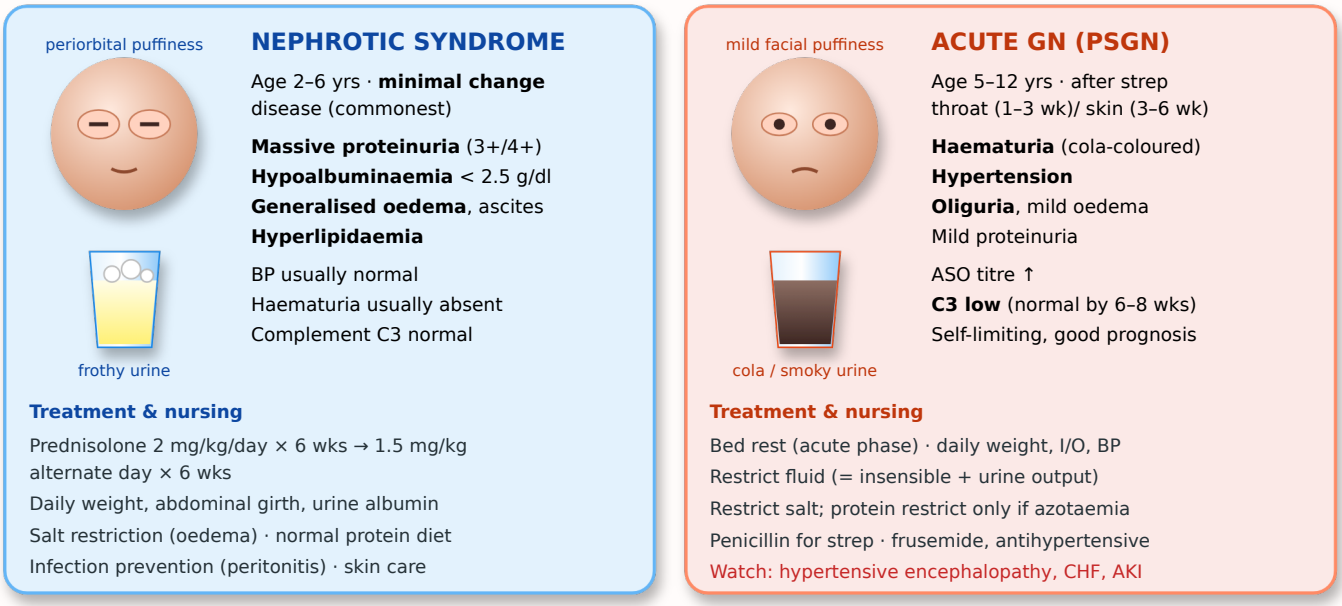
- Cleft lip repair – **~3 months (rule of 10s)**; cleft palate – **9–18 months**.
- Position after cleft lip repair – **supine/ side (not prone)**; after cleft palate – **prone/ side**.
- Device protecting lip suture – **Logan bow**.
- Commonest TEF – **Type C**; first sign – **excess frothy saliva**; 3 C's – **coughing, choking, cyanosis**.
- Pyloric stenosis age – **2–6 weeks**; vomiting – **projectile, non-bilious**; mass – **olive-shaped**.
- Pyloric stenosis metabolic picture – **hypochloraemic hypokalaemic metabolic alkalosis**; surgery – **Ramstedt pyloromyotomy**.
- Red currant jelly stool – **intussusception**; USG – **target/ doughnut sign**.
- Hirschsprung – **failure to pass meconium in 48 hrs**; diagnosis – **rectal biopsy**; washout fluid – **normal saline**.
- Omphalocele – **covered by sac**; gastroschisis – **no sac**.
- Umbilical hernia – **closes spontaneously; no strapping**.
- Coeliac disease – avoid **wheat, barley, rye**.
- Biliary atresia surgery – **Kasai, before 60 days**.
- Rule of 2s – **Meckel's diverticulum**.

11 Genitourinary & Renal Disorders in Children

11.1 Urine Output & Basics

- **First urine** – within **24 hrs** of birth (most within 12 hrs); **delayed > 48 hrs** – investigate.
- **Normal urine output**: **infant 2 ml/kg/hr**, **child 1 ml/kg/hr**, adolescent 0.5 ml/kg/hr; **oliguria < 1 ml/kg/hr (child)**, < 0.5 ml/kg/hr (adult).
- **Bladder capacity (ml) = (age in yrs + 2) × 30** (approx).

Nephrotic Syndrome vs Acute Glomerulonephritis (AGN)



Nephrotic syndrome vs acute glomerulonephritis – comparison & nursing

11.2 Nephrotic Syndrome

D Nephrotic syndrome – tetrad

Heavy proteinuria (**> 40 mg/m²/hr** or **spot urine protein : creatinine > 2** or **3+/ 4+ dipstick**) + **hypoalbuminaemia (< 2.5 g/dl)** + **generalised oedema** + **hyperlipidaemia** (cholesterol > 200 mg/dl).

- **Minimal change disease (MCNS) – ~80–90 % in children, steroid-sensitive**; age **2–6 yrs**, boys.
- **Pathophysiology** – ↑ glomerular permeability → protein loss → ↓ oncotic pressure → oedema; liver ↑ lipoprotein synthesis → hyperlipidaemia; ↑ clotting factors → **hypercoagulable**.
- **S/S** – **periorbital (puffy eyes, morning) → generalised oedema (anasarca)**, ascites, scrotal/ labial oedema, **weight gain**, **frothy urine**, ↓ urine output, pale, irritable, anorexia, diarrhoea (gut oedema); **BP usually normal**, **no gross haematuria**, normal C3.
- **Complications** – **infections (most common – spontaneous bacterial peritonitis – pneumococcus, cellulitis)**, **thrombosis** (renal vein), hypovolaemia/ shock, AKI, malnutrition, steroid side effects.

Treatment (ISPN/ IAP)

- **Initial episode** – **prednisolone 2 mg/kg/day (60 mg/m²) single morning dose × 6 weeks**, then **1.5 mg/kg (40 mg/m²) alternate days × 6 weeks** (total 12 weeks).
- **Remission** – urine protein **nil/ trace for 3 consecutive days**; **relapse** – 3+/4+ for 3 consecutive days.
- **Steroid-dependent/ frequent relapsers** – levamisole, cyclophosphamide, MMF, tacrolimus, rituximab; **steroid-resistant** → renal biopsy (FSGS).
- **Diuretics** (frusemide) with care; **albumin infusion (20 %)** + frusemide for severe oedema.

Nursing care

- **Daily weight (same time, same scale, same clothes)**, **abdominal girth** (at umbilicus), strict I/O, **urine albumin daily (dipstick – first morning sample)**.
- **Diet** – **no added salt during oedema** (salt restriction), **normal protein for age (1.5–2 g/kg) – high protein NOT recommended**, adequate calories; fluid restriction only if severe oedema/ hyponatraemia.
- **Skin care** – oedematous skin breaks easily: **change position 2-hourly**, support scrotum, separate skin surfaces (cotton), loose clothes, **avoid IM injections in oedematous areas**.
- **Infection prevention** – hand hygiene, avoid crowds/ sick contacts, **pneumococcal & varicella vaccination (live vaccines deferred during high-dose steroids)**, recognise peritonitis (fever, abdominal pain).
- **Steroid side effects** – **cushingoid face, weight gain, ↑ appetite, hypertension, hyperglycaemia, behaviour change, peptic ulcer, infection, growth suppression, cataract, osteoporosis**; give **with food in the morning**, **never stop abruptly (adrenal crisis)**.
- **Parent teaching** – home urine testing with dipstick, relapse recognition, medication diary, body-image support.

11.3 Acute Post-Streptococcal Glomerulonephritis (APSGN/ AGN)

- **Immune-complex** disease after **Group A β-haemolytic Streptococcus** – **throat (1–3 weeks)** or **skin – impetigo/ pyoderma (3–6 weeks)**; age **5–12 yrs**.
- **Nephritic syndrome** – **haematuria (cola/ smoky/ tea-coloured urine) – hallmark**, **hypertension**, **oliguria**, **oedema (periorbital, mild)**, mild proteinuria, **azotaemia**.
- **Investigations** – urine: **RBCs & RBC casts**; **↑ ASO titre** (throat), anti-DNase B (skin), **↓ C3 complement (returns to normal in 6–8 weeks)**, ↑ urea/ creatinine.
- **Complications** – **hypertensive encephalopathy** (headache, vomiting, blurred vision, seizures), **pulmonary oedema/ CHF**, **AKI**, hyperkalaemia.
- **Treatment** – mostly **supportive**; **penicillin × 10 days** (eradicate strep – doesn't change course); **fluid restriction = insensible loss (400 ml/m²/day) + previous day's urine output**; **salt restriction**; protein restriction only if azotaemia; **frusemide**, antihypertensives (amlodipine, nifedipine); dialysis if needed.
- **Nursing** – **bed rest during acute phase** (oedema, HTN, gross haematuria), **BP monitoring 4-hourly**, **daily weight, strict I/O**, urine colour, neuro checks (encephalopathy), low-salt diet, diversion activities; **prognosis excellent (> 95 % recover)**.

Feature	Nephrotic	Nephritic (AGN)
Age	**2–6 yrs**	**5–12 yrs**

Feature	Nephrotic	Nephritic (AGN)
Onset	Insidious	Acute, post-strep
Oedema	**Massive, generalised**	Mild, periorbital
Proteinuria	**Massive (3+/4+)**	Mild (1+/2+)
Haematuria	Absent/ microscopic	**Gross – cola coloured**
BP	Normal	**Raised**
Serum albumin	**Very low**	Normal/ slightly low
Cholesterol	**↑↑**	Normal
C3	Normal	**Low**
Treatment	**Steroids**	**Supportive, fluid/ salt restriction, penicillin**

11.4 Urinary Tract Infection (UTI)

- **E. coli (most common, ~80 %)**, Klebsiella, Proteus.
- **More common in girls (short urethra)** except **first year (boys – uncircumcised, anomalies)**.
- **Risk factors** – **VUR (vesico-ureteric reflux)**, posterior urethral valves (boys), constipation, poor hygiene (**wiping back → front**), bubble baths, tight clothes, holding urine, catheter.
- **S/S** – **infants: fever, poor feeding, vomiting, irritability, FTT, jaundice (non-specific)**; older: **dysuria, frequency, urgency, foul urine, enuresis (new), suprapubic pain**; pyelonephritis – high fever, flank pain.
- **Diagnosis** – **urine culture (gold standard)** – **suprapubic aspiration (any growth significant)**, catheter (≥ 50,000 CFU/ml), **clean-catch midstream (≥ 1,00,000 CFU/ml)**; urine routine (pus cells > 5/hpf, nitrite +ve).
- **Treatment** – oral **amoxicillin-clavulanate/ cefixime/ cotrimoxazole/ nitrofurantoin** 7–14 days; IV for infants < 3 months/ toxic.
- Imaging after first febrile UTI – **USG KUB**, **MCU (micturating cystourethrogram) – for VUR**, **DMSA scan – renal scarring**.
- **Prevention/ teaching** – **wipe front to back**, **adequate fluids**, regular voiding (every 3–4 hrs), **avoid bubble baths & tight nylon underwear (use cotton)**, treat constipation, complete antibiotic course.
- **Posterior urethral valves (PUV)** – most common cause of **lower urinary tract obstruction in boys**; poor stream, dribbling, palpable bladder, antenatal hydronephrosis → **valve ablation**.

11.5 Wilms Tumour (Nephroblastoma)

- **Most common renal malignancy of childhood**; age **2–5 yrs**; **WT1 gene (chromosome 11)**; associations – **WAGR, Beckwith-Wiedemann**, aniridia, hemihypertrophy.
- **S/S** – **painless, firm, smooth abdominal mass, usually unilateral, does not cross midline** (often noticed by mother while bathing/ dressing), haematuria, hypertension (renin), fever.
- (vs **neuroblastoma** – crosses midline, irregular, adrenal, < 2 yrs, ↑ urinary VMA, "raccoon eyes")
- **Treatment** – **nephrectomy + chemotherapy (vincristine, actinomycin D) ± radiotherapy**; **excellent prognosis (~90 %)**.

! Wilms tumour – DO NOT palpate the abdomen

Place sign **"Do not palpate abdomen"** on bed – palpation can **rupture the capsule & spread tumour cells (seeding)**; careful bathing/ handling; surgery within 24–48 hrs of diagnosis.

11.6 Genital Anomalies

Condition	Key points
Hypospadias	**Urethral opening on ventral (under) surface** of penis; ± **chordee** (ventral curvature); **NO circumcision** (foreskin used for repair); repair at **6–18 months** (before toilet training/ body awareness); post-op – **catheter/ stent care**, double diaper, prevent kinking, fluids, analgesia
Epispadias	Opening on **dorsal (upper)** surface; with **bladder exstrophy**
Undescended testis (cryptorchidism)	Common in **preterms**; most descend by **3–6 months**; **orchidopexy at 6–12 months (before 18 months)** – risk of **infertility, malignancy (seminoma), torsion**, inguinal hernia; **retractile testis** – normal (cremasteric reflex)
Phimosis	Non-retractable foreskin – **physiological till 3–5 yrs**; **don't forcibly retract**; pathological – steroid cream/ circumcision
Paraphimosis	Retracted foreskin can't return – **emergency** (cold compress, manual reduction)
Testicular torsion	**Sudden severe scrotal pain**, high-riding testis, **absent cremasteric reflex** – **surgical emergency (within 6 hrs to save testis)**
Hydronephrosis	PUJ obstruction (commonest antenatal finding)
Ambiguous genitalia	**Congenital adrenal hyperplasia (21-hydroxylase – most common)** – salt-wasting crisis; don't assign sex hastily

11.7 Enuresis

- **Involuntary passage of urine in a child ≥ 5 yrs** (at least twice a week × 3 months). **Nocturnal (most common)**, diurnal; **primary** (never dry) – most common, family history; **secondary** (after ≥ 6 months dry) – stress, UTI, diabetes, new sibling.
- **Management** – **reassure, don't punish/ shame**, **star chart (positive reinforcement)**, void before bed, **limit fluids 1–2 hrs before bedtime**, avoid caffeine, wake to void; **enuresis alarm (most effective long-term)**, **desmopressin (DDAVP)** at bedtime (short-term – camps), imipramine (rarely); bladder training.
- **Encopresis** – faecal soiling **after 4 yrs**; usually **chronic constipation with overflow**.

11.8 Renal Failure (brief)

- **AKI** – prerenal (dehydration – most common in children), renal (**HUS – common cause in young children**, AGN), postrenal (PUV); **hyperkalaemia** – most dangerous (ECG – tall T); Rx – fluid balance, **calcium gluconate, salbutamol nebulisation, insulin-glucose, sodium bicarbonate, kayexalate**, dialysis (**peritoneal dialysis** preferred in small children).
- **CKD** – congenital anomalies (CAKUT – commonest cause in children); growth failure, renal osteodystrophy, anaemia (erythropoietin).

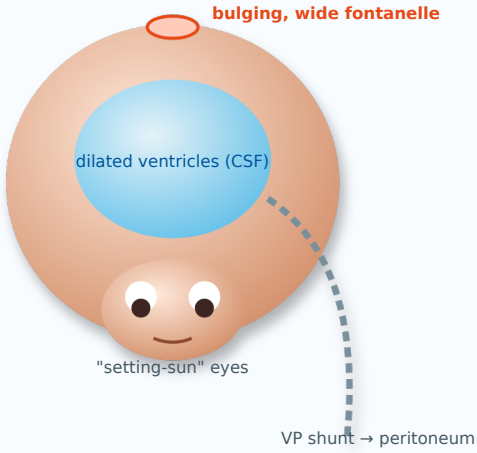
- **HUS** – triad: **microangiopathic haemolytic anaemia, thrombocytopenia, AKI**; after **E. coli O157:H7/ Shigella** dysentery.

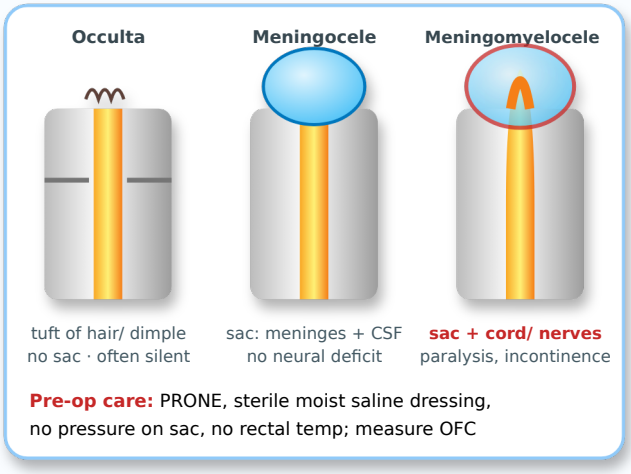
PYQ style

- Nephrotic syndrome commonest type – **minimal change disease**; age – **2–6 yrs**.
- Hallmark of nephrotic – **massive proteinuria + hypoalbuminaemia + oedema + hyperlipidaemia**.
- Earliest oedema in nephrotic – **periorbital (morning)**.
- Prednisolone dose – **2 mg/kg/day × 6 wks → 1.5 mg/kg alternate day × 6 wks**.
- Best indicator of fluid status – **daily weight**.
- Commonest complication of nephrotic – **infection (peritonitis)**.
- AGN follows – **streptococcal throat/ skin infection**; urine – **cola coloured**; C3 – **low**.
- Fluid allowed in AGN – **insensible loss + previous day urine output**.
- Commonest organism in UTI – **E. coli**; gold standard – **urine culture**.
- Teach girls to wipe – **front to back**.
- Wilms tumour nursing – **do not palpate abdomen**.
- Hypospadias – **ventral opening; avoid circumcision**.
- Orchidopexy age – **6–12 months**.
- Enuresis diagnosed after – **5 years**; most effective therapy – **enuresis alarm**.
- HUS triad – **haemolytic anaemia, thrombocytopenia, AKI**.

12 Neurological Disorders in Children

Hydrocephalus & Neural Tube Defects (Spina Bifida)





Hydrocephalus

↑ Head circumference (serial OFC daily) · Macewen's (cracked-pot) sign
Separated sutures · shiny scalp, dilated veins · vomiting
VP shunt post-op: flat/ unoperated side, raise head gradually; watch shunt block (↑ ICP signs) & infection · support large head while lifting

Prevention & associations

Folic acid 400 µg preconception (4–5 mg if previous NTD)
Maternal serum AFP ↑ · valproate in pregnancy ↑ risk
Meningomyelocele + hydrocephalus (Arnold-Chiari II)
Latex allergy precautions (repeated catheterisation)

Hydrocephalus signs, VP shunt & spina bifida types with pre-op care

12.1 Hydrocephalus

D Hydrocephalus

Excess accumulation of **CSF in the ventricles** due to **obstruction, impaired absorption or (rarely) over-production** → ventricular dilatation & ↑ ICP. Normal CSF ~ **500 ml/day produced** (choroid plexus), volume ~ 50 ml infant / 150 ml adult.

- **Types** – **Non-communicating (obstructive) – most common in children** (**aqueductal stenosis – commonest cause**, Arnold-Chiari II, Dandy-Walker, tumour); **communicating** – impaired absorption at arachnoid villi (post-meningitis, haemorrhage).
- **S/S in infants (sutures open)** – **rapid increase in head circumference (earliest & most important)**, **bulging, tense anterior fontanelle**, **separated sutures**, **Macewen's sign ("cracked-pot" sound on percussion)**, **"setting-sun" eyes** (sclera visible above iris), **dilated scalp veins, shiny thin scalp**, **high-pitched (shrill) cry**, irritability, poor feeding, vomiting, frontal bossing, **head lag**, spasticity.
- **Older children (sutures closed)** – classic ↑ ICP – **headache (morning), projectile vomiting, papilloedema**, diplopia, ataxia, behaviour change; **Cushing's triad** (late) – ↑ BP (wide pulse pressure), ↓ pulse, irregular respiration.
- **Diagnosis** – **serial OFC**, USG (through AF), **CT/ MRI**, transillumination (hydranencephaly).
- **Treatment** – **VP (ventriculoperitoneal) shunt – most common** (one-way valve; peritoneal end allows growth), **ETV (endoscopic third ventriculostomy)** for obstructive; acetazolamide (temporary).

Nursing care

- **Pre-op** – **measure OFC daily** (same site, same time), neuro checks, monitor ↑ ICP, **support large head** while lifting/ feeding (**prevent neck strain**), **change position 2-hrly; skin care on head (sheepskin/ water pillow – pressure sores)**, small frequent feeds (vomiting).
- **Post-op (VP shunt)** – **position flat (supine) initially** – avoids rapid CSF drainage (subdural haematoma) – raise head gradually per order; **lie on unoperated side** (avoid pressure on valve/ shunt site); check **fontanelle (sunken = overdrainage, bulging = blockage)**, OFC, neuro & vitals; watch abdominal distension (peritonitis, ileus); **pump the shunt only if ordered**.
- **Complications** – **shunt infection (most common early, within 2 months – fever, redness along tract)**, **shunt malfunction/ blockage (most common overall – signs of ↑ ICP)**, over-drainage, subdural haematoma, **need for revision with growth**.
- **Parent teaching** – signs of ↑ ICP & infection, development stimulation, regular follow-up.

12.2 Neural Tube Defects (NTDs)

- Failure of **neural tube closure by day 28** (anterior neuropore day 25, posterior day 27–28).
- **Anencephaly** (absent brain/ skull – incompatible with life), **encephalocele** (occipital commonest), **spina bifida**.

Type	Features
Spina bifida occulta	Vertebral arch defect only, **no sac**; **tuft of hair, dimple, lipoma, haemangioma** over lumbosacral area; usually asymptomatic
Meningocele	Sac contains **meninges + CSF** only; **no neurological deficit** (usually); transilluminates
Meningomyelocele (myelomeningocele) – most common & severe	Sac contains **meninges, CSF, spinal cord/ nerve roots**; **lumbosacral** commonest; **flaccid paralysis of lower limbs, loss of sensation, neurogenic bladder (dribbling) & bowel**, club feet, hip dislocation; **associated hydrocephalus (~80–90 %, Arnold-Chiari II)**

- **Prevention** – **folic acid 400 µg (0.4 mg)/day from ≥ 1 month before conception to 12 wks**; **4–5 mg/day if previous NTD/ on antiepileptics**; fortified foods.
- **Antenatal diagnosis** – **↑ maternal serum AFP & amniotic fluid AFP**, acetylcholinesterase; **USG – lemon sign, banana sign**.
- **Pre-op nursing (meningomyelocele)**:
- **Prone position** (or side-lying), **hips slightly flexed & abducted** (pad between legs), head turned to side – **no pressure on sac**.
- **Sterile, moist (warm normal saline) non-adherent dressing** over sac, changed every 2–4 hrs; **plastic drape/ barrier** below sac to **prevent faecal contamination**.
- **Inspect sac for leakage, tears, infection**; **measure OFC daily** (hydrocephalus); **no rectal temperature**; no diaper over sac.
- Feed in prone/ side; touch & cuddle without pressure; **latex precautions (high latex allergy risk)**.
- **Surgery** – closure **within 24–72 hrs** (prevents infection – meningitis).
- **Long-term** – **clean intermittent catheterisation (CIC)**, bowel training, mobility aids/ orthoses, skin care (insensate), UTI prevention.

12.3 Meningitis

- **Inflammation of meninges** (pia & arachnoid).
- **Organisms**: **neonates – E. coli, Group B Streptococcus, Listeria, Klebsiella**; **2 months–5 yrs – S. pneumoniae, Hib, N. meningitidis**; **> 5 yrs – S. pneumoniae, N. meningitidis**; TB meningitis (gradual).
- **S/S** – **fever, headache, vomiting, irritability, photophobia, seizures, altered sensorium**; **neck stiffness (nuchal rigidity)**, **Kernig's sign** (pain on knee extension with hip flexed), **Brudzinski's sign** (neck flexion → hip & knee flexion); **infants – bulging fontanelle (most important), high-pitched cry, poor feeding, lethargy** (neck stiffness may be absent); **meningococcal – petechial/ purpuric rash, Waterhouse-Friderichsen syndrome** (adrenal haemorrhage – shock).
- **Diagnosis – Lumbar puncture (CSF)**:

CSF	Normal	Bacterial	Viral	Tubercular
Appearance	Clear	**Turbid/ purulent**	Clear	Clear/ **cobweb clot**
Cells	< 5/mm^3	**↑ ↑ neutrophils**	↑ lymphocytes	↑ lymphocytes
Protein	15–45 mg/dl	**↑ ↑**	Normal/ ↑	**↑ ↑ ↑**
Glucose	~ 2/3 of blood	**↓ ↓**	Normal	**↓**

- **Treatment** – **ceftriaxone/ cefotaxime ± vancomycin**; neonates – ampicillin + cefotaxime/ gentamicin; **dexamethasone** (Hib – reduces deafness, give before/ with first antibiotic dose); anticonvulsants; fluids (SIADH watch).
- **Nursing** – **droplet isolation – for 24 hrs after start of antibiotics** (meningococcal, Hib); **quiet, dark room** (photophobia), minimal stimulation; **head end elevated 30°**, **side-lying (no neck flexion)**; **seizure precautions**; neuro observations (GCS), **OFC daily in infants**; fever control; I/O, daily weight (**SIADH**); hearing test after recovery.
- **Complications** – **sensorineural deafness (most common sequela)**, hydrocephalus, subdural effusion (Hib/ pneumococcus in infants), seizures, cerebral palsy, mental retardation, SIADH.
- **Chemoprophylaxis for contacts** – **rifampicin** (meningococcal, Hib), ciprofloxacin/ ceftriaxone.
- **Prevention** – **Hib, PCV, BCG (TBM)**, meningococcal vaccines.

12.4 Encephalitis

- Inflammation of brain parenchyma; **JE (Japanese encephalitis – Culex mosquito, pigs reservoir, ardeid birds)** – endemic districts → JE vaccine; HSV, measles, **AES (Acute Encephalitis Syndrome)** – fever + altered sensorium/ seizures.
- Supportive – airway, ICP control, seizures, fever; **acyclovir for HSV**.

12.5 Seizures

Febrile seizures

- **Seizure with fever (≥ 38 °C) without CNS infection**, **age 6 months – 5 yrs (peak 18 months)**; **most common seizure in children**; family history common.
- **Simple (~80 %)** – **generalised, < 15 min, once in 24 hrs**, no post-ictal deficit; **complex** – focal, > 15 min, recurrent within 24 hrs.

- **Recurrence ~30 %**; risk of epilepsy only slightly ↑ (complex).
- **Management** – **protect from injury, lateral (recovery) position, nothing in mouth**, time the seizure, O_2; if **> 5 min** – **midazolam (buccal/ intranasal 0.2–0.3 mg/kg) or diazepam rectal (0.5 mg/kg)/ IV (0.3 mg/kg)/ lorazepam IV 0.1 mg/kg**; **antipyretic (paracetamol) & tepid sponging for comfort** (antipyretics don't prevent recurrence); find & treat fever source; **LP if meningitis suspected (esp. < 12 months)**.
- **No long-term anticonvulsant prophylaxis** for simple febrile seizures; intermittent **clobazam** during fever (selected cases).
- **Parent teaching** – first aid, **don't put anything in mouth, don't restrain, don't give water during seizure**, when to go to hospital.

Epilepsy

- **≥ 2 unprovoked seizures > 24 hrs apart**.
- **Generalised** – **tonic-clonic (grand mal)**, **absence (petit mal)** – brief staring spells **(5–10 sec), many per day, 3 Hz spike-wave EEG**, provoked by **hyperventilation**, school performance ↓ → **ethosuximide/ valproate**; myoclonic, atonic (drop attacks).
- **Focal** – simple/ complex partial.
- **Infantile spasms (West syndrome)** – **3–12 months**, **"jack-knife/ salaam" spasms in clusters**, developmental regression, **hypsarrhythmia (EEG)**; Rx – **ACTH, vigabatrin (tuberous sclerosis)**, prednisolone.
- **Drugs** – **valproate** (broad spectrum; hepatotoxicity, weight gain, tremor), **carbamazepine** (focal; rash – SJS, hyponatraemia), **phenytoin** (**gum hypertrophy – oral hygiene**, hirsutism, ataxia, nystagmus, **IV in NS only – precipitates with dextrose**), **phenobarbitone** (neonates; sedation, hyperactivity), levetiracetam, lamotrigine.
- **Nursing/ teaching** – **regular medication, never stop abruptly** (status), drug levels, seizure diary, **safety – no swimming/ cycling alone, showers not tub baths, helmet**, school informed, **medical ID card**, adequate sleep, avoid triggers (flashing lights, sleep deprivation).

Status epilepticus

- Seizure **≥ 5 minutes** or recurrent seizures without regaining consciousness (operational); **medical emergency**.
- **ABC, O_2, glucose check, IV access → 0–5 min: benzodiazepine (lorazepam 0.1 mg/kg IV / midazolam / diazepam)**, repeat once after 5–10 min → **phenytoin/ fosphenytoin 20 mg/kg** or **valproate 20–40 mg/kg** or **levetiracetam 40–60 mg/kg** → **phenobarbitone 20 mg/kg** → refractory: midazolam infusion/ thiopentone (ICU).

12.6 Cerebral Palsy (CP)

D Cerebral palsy

Non-progressive disorder of **movement & posture** due to **injury to the developing brain** (prenatal, perinatal, postnatal up to ~2 yrs). Lesion is static but clinical picture may change with growth.

- **Causes** – **prematurity (most important risk factor – periventricular leukomalacia)**, **birth asphyxia/ HIE**, kernicterus (**dyskinetic/ athetoid CP**), infections (TORCH, meningitis), LBW, multiple births, head trauma.

- **Types**: **Spastic (most common ~70–80 %) – UMN – ↑ tone, brisk reflexes, scissoring gait, clasp-knife** (quadriplegia, **diplegia – preterm**, hemiplegia); **dyskinetic/ athetoid** (involuntary writhing – **basal ganglia – kernicterus**); **ataxic** (cerebellum – wide-based gait, intention tremor); **mixed**.
- **Early signs** – **persistence of primitive reflexes (Moro, ATNR beyond 6 months)**, delayed milestones, **abnormal tone**, **hand preference before 12–18 months** (hemiplegia), poor head control, feeding difficulty, **scissoring of legs**.
- **Associated problems** – **intellectual disability, seizures (epilepsy)**, visual (strabismus), hearing & speech defects, feeding/ swallowing (aspiration), drooling, constipation, contractures, dental problems.
- **Management (multidisciplinary – no cure)** – **physiotherapy (most important)**, occupational & speech therapy, **orthoses**, **baclofen, diazepam, botulinum toxin (spasticity)**, surgery (tendon release), anticonvulsants, special education.
- **Nursing** – **safety (fall/ injury)**, **feeding – upright, small bites, chin tuck, high-calorie**, aspiration prevention, **ROM exercises & positioning (prevent contractures)**, skin care, communication aids, promote independence (ADLs), **family support & respite**, school inclusion.

12.7 Other Neurological Conditions

Condition	Key points
Head injury	Falls (commonest); GCS (modified paediatric for < 2 yrs); **"shaken baby syndrome"** – retinal haemorrhages, subdural – abuse; monitor neuro obs, vomiting, ↑ ICP
Reye's syndrome	**Aspirin during viral illness (influenza, varicella) → acute encephalopathy + fatty liver (hepatic dysfunction)**; vomiting, lethargy → coma; ↑ ammonia; **avoid aspirin < 16 yrs**
Guillain-Barré syndrome (GBS)	Post-infectious **ascending, symmetrical flaccid paralysis**, areflexia, **albumino-cytological dissociation (CSF)**; watch **respiratory failure**; IVIG/ plasmapheresis; **AFP surveillance < 15 yrs** (GBS most common non-polio cause)
Duchenne muscular dystrophy	**X-linked recessive (boys)**, dystrophin absent; onset 3–5 yrs; **Gowers' sign** (climbs up own legs), **calf pseudohypertrophy**, waddling gait, ↑ **CPK**; wheelchair by ~12 yrs; death – cardio-respiratory
Breath-holding spells	6 months–5 yrs; cyanotic (crying → apnoea → cyanosis → LOC) or pallid; **benign; iron deficiency association**; reassure, iron
Autism, ADHD	See behavioural disorders chapter
Brain tumours	**Most common solid tumours of childhood**; **infratentorial (posterior fossa) more common** – medulloblastoma, cerebellar astrocytoma; morning headache, vomiting, ataxia

PYQ style

- Earliest sign of hydrocephalus in infant – **rapidly increasing head circumference**.
- Setting-sun eyes & cracked-pot sign – **hydrocephalus**.
- Commonest cause of obstructive hydrocephalus – **aqueductal stenosis**.
- VP shunt – most common complication – **malfunction/ blockage**; early – **infection**.
- Position after VP shunt – **flat, on unoperated side**.
- Meningomyelocele pre-op position – **prone**; dressing – **sterile moist saline**.
- NTD prevention – **folic acid 400 µg preconception**; marker – **↑ AFP**.
- Meningitis signs – **Kernig's, Brudzinski's, neck rigidity**; infant – **bulging fontanelle**.
- CSF in bacterial meningitis – **↑ protein, ↓ glucose, ↑ neutrophils**.
- Isolation in meningococcal meningitis – **droplet, 24 hrs after antibiotics**.
- Commonest sequela of meningitis – **deafness**.
- Febrile seizure age – **6 months – 5 years**; simple – **< 15 min, generalised, once in 24 hrs**.
- Status epilepticus – **≥ 5 min**; first drug – **benzodiazepine (lorazepam/ midazolam)**.
- Phenytoin side effect – **gum hypertrophy**; dilute only in **normal saline**.
- Absence seizure drug – **ethosuximide/ valproate**; infantile spasms – **ACTH/ vigabatrin**.
- Commonest type of CP – **spastic**; kernicterus causes – **athetoid CP**.
- Reye's syndrome – **aspirin + viral infection**.
- Gowers' sign – **Duchenne muscular dystrophy**.

13 Haematological & Oncological Disorders

13.1 Normal Values

Parameter	Value
Hb – newborn	**14–20 g/dl** (cord ~16.5)
Hb – physiological nadir	**~9–11 g/dl at 2–3 months (term)** – "physiological anaemia of infancy" (preterm earlier & lower)
Hb – 6 months–5 yrs	≥ 11 g/dl
HbF at birth	**~70–80 %** → < 2 % by 1 yr
Platelets	1.5–4.5 lakh/mm^3
Blood volume	**Newborn ~80–85 ml/kg**; child ~75 ml/kg

13.2 Anaemias

- **Iron-deficiency anaemia – most common** (see nutrition chapter) – **microcytic hypochromic**, ↓ ferritin; Rx – oral iron 3–6 mg/kg/day × 3 months after Hb normalises.
- **Megaloblastic** – B12/ folate (goat milk – folate deficiency).
- **Aplastic anaemia** – pancytopenia; BMT.

Thalassaemia

- **Autosomal recessive**; ↓/ absent **β-globin chain** synthesis → ineffective erythropoiesis, haemolysis. **Common in Gujarat (Kutchi, Sindhi, Lohana, Bhanushali communities)**, Punjabis, Bengalis; **carrier rate in India ~3–4 %**.
- **β-thalassaemia major (Cooley's anaemia)** – symptoms at **6 months** (HbF → HbA switch); **severe pallor, jaundice, hepatosplenomegaly (massive spleen)**, FTT, **"chipmunk/ thalassaemic facies"** (frontal bossing, maxillary prominence – marrow expansion), **"hair-on-end" skull X-ray**.
- **Investigations** – **microcytic hypochromic, target cells**, **Hb electrophoresis/ HPLC – ↑ HbF, ↓/ absent HbA**; trait – **↑ HbA2 (> 3.5 %)**; **NESTROFT** (screening test for carriers).
- **Treatment** – **regular packed RBC transfusion every 2–4 weeks – keep pre-transfusion Hb 9–10.5 g/dl** (hypertransfusion); **iron chelation – desferrioxamine (SC infusion over 8–12 hrs, 5–7 nights/wk)**, **deferasirox/ deferiprone (oral)** – start when **serum ferritin > 1000 ng/ml** (or after 10–20 transfusions); **folic acid**; **avoid iron supplements & iron-rich foods (tea with meals ↓ absorption)**; **splenectomy** (hypersplenism, > 5 yrs; pneumococcal, Hib, meningococcal vaccines + penicillin prophylaxis); **bone marrow (HSC) transplant – only cure**.
- **Complications** – **iron overload (haemosiderosis)** – heart failure (**most common cause of death**), liver cirrhosis, diabetes, hypothyroidism, delayed puberty, bronze skin; transfusion infections (HBV, HCV, HIV).
- **Prevention** – **carrier screening, premarital/ antenatal screening & counselling**, **prenatal diagnosis (CVS 10–12 wks)**.

Sickle Cell Disease

- **Autosomal recessive**; **HbS** (valine replaces glutamic acid at 6th position of β-chain); **tribal belts of Gujarat** (south Gujarat – Dangs, Valsad, Surat, Tapi), MP, Maharashtra, Odisha, Chhattisgarh. **National Sickle Cell Anaemia Elimination Mission (2023, by 2047)**.
- Sickling triggered by **hypoxia, dehydration, acidosis, cold, infection, fever, exertion, high altitude**.
- **Crises**: **vaso-occlusive (painful) – most common**; **dactylitis/ hand-foot syndrome (first presentation in infants 6 m–2 yrs)**; **acute chest syndrome (leading cause of death)**; **splenic sequestration** (young children – sudden splenomegaly + shock); **aplastic crisis (parvovirus B19)**; haemolytic crisis; stroke; priapism.
- **Autosplenectomy** (by 5 yrs) → **infection risk (pneumococcus, Salmonella osteomyelitis)**.
- **Diagnosis** – **sickling test, solubility test, Hb electrophoresis/ HPLC (HbS)**.
- **Management of crisis** – **HYDRATION (IV/ oral 1.5 × maintenance) – most important**, **analgesia (paracetamol/ NSAIDs → opioids – morphine)**, **O_2 if hypoxic**, **warmth (warm compress – NOT cold)**, rest, treat infection, transfusion (severe).
- **Long-term** – **hydroxyurea (↑ HbF)** – mainstay; **penicillin prophylaxis till 5 yrs**, pneumococcal & all vaccines, **folic acid**, avoid triggers (dehydration, cold, high altitude, strenuous exercise), regular follow-up.

G6PD Deficiency

- **X-linked recessive (boys)**; **haemolysis on oxidant stress** – drugs (**primaquine, dapsone, sulphonamides, nitrofurantoin, high-dose aspirin**), **fava beans (favism)**, naphthalene (moth balls), infections; neonatal jaundice.

13.3 Bleeding Disorders

	Haemophilia A	Haemophilia B (Christmas disease)
Factor	**VIII** (most common – ~80–85 %)	**IX**
Inheritance	**X-linked recessive – males affected, females carriers**	Same
Lab	**↑ aPTT**, **normal PT, platelets & bleeding time**	Same

- **S/S** – **haemarthrosis (most common – knees, elbows, ankles)** → joint deformity, prolonged bleeding after circumcision/ dental extraction/ injury, **intramuscular haematomas**, **intracranial haemorrhage (most serious)**, easy bruising as toddler starts walking.
- **Treatment** – **factor replacement (factor VIII/ IX concentrate) – prophylaxis or on-demand**; **cryoprecipitate/ FFP** (if concentrate unavailable); **desmopressin (DDAVP)** – mild haemophilia A; **tranexamic acid** (mucosal bleeding); emicizumab.
- **Nursing** – **RICE for joint bleed (Rest, Ice, Compression, Elevation)** + immobilise, factor replacement **ASAP**; **avoid IM injections** (SC/ oral/ IV only), **avoid rectal temperature**, **no aspirin/ NSAIDs** (paracetamol for pain); **pressure on venipuncture sites ≥ 5 min**; soft toothbrush; **safe environment – padded cot, helmet, knee/ elbow pads**; **swimming, cycling (with helmet) allowed – avoid contact sports**; **medical alert ID**; genetic counselling; physiotherapy after bleed.
- **Von Willebrand disease** – **most common inherited bleeding disorder** overall; **autosomal dominant**; mucosal bleeding (epistaxis, menorrhagia); ↑ BT.

Immune Thrombocytopenic Purpura (ITP)

- **Autoimmune platelet destruction**, often **1–4 weeks after viral infection**; age 2–5 yrs.
- **Sudden petechiae, purpura, bruises, epistaxis** in otherwise **well child**, **no hepatosplenomegaly/ lymphadenopathy**; **platelets < 1 lakh (often < 20,000)**, normal Hb & WBC.
- **Mostly self-limiting (~80 % resolve within 6 months)**; observation if no bleeding; **IVIG, steroids (prednisolone), anti-D** if significant bleeding; **platelet transfusion only in life-threatening bleed**.
- Nursing – **avoid trauma/ contact sports, IM injections, aspirin/ NSAIDs**; soft toothbrush; watch for ICH (headache).
- **Henoch-Schönlein purpura (IgA vasculitis)** – **palpable purpura on buttocks & legs (normal platelets)**, arthritis, abdominal pain, nephritis.

13.4 Childhood Cancers

- **Most common childhood cancer – leukaemia (ALL)**; then **brain tumours (most common solid tumour)**, lymphomas, neuroblastoma, Wilms, retinoblastoma, bone tumours.

Acute Lymphoblastic Leukaemia (ALL)

- **~75–80 % of childhood leukaemia**; **peak 2–5 yrs**; **Down syndrome ↑ risk** (also AML), radiation, chemicals.
- **Bone marrow failure** – **anaemia** (pallor, fatigue), **thrombocytopenia** (petechiae, bruising, bleeding), **neutropenia** (fever, infections); **infiltration** – **bone pain/ limp** (refuses to walk), **hepatosplenomegaly, lymphadenopathy**, CNS (headache, vomiting), testicular swelling, mediastinal mass (T-cell).
- **Diagnosis – bone marrow aspiration/ biopsy (> 20–25 % blasts) – confirmatory**; CBC – blasts, CSF for CNS involvement.
- **Treatment phases** – **induction** (vincristine, prednisolone/ dexamethasone, L-asparaginase, ± daunorubicin) → **remission** → **consolidation/ CNS prophylaxis (intrathecal methotrexate)** → **delayed intensification** → **maintenance (6-mercaptopurine daily + methotrexate weekly) – 2–3 yrs total**.
- **Prognosis** – **~85–90 % cure** (good: age 1–10 yrs, WBC < 50,000, girls, rapid response); **CNS & testes – sanctuary sites (relapse)**.
- **Tumour lysis syndrome** – at start of chemo – **hyperuricaemia, hyperkalaemia, hyperphosphataemia, hypocalcaemia** → AKI; prevention – **hyperhydration, allopurinol/ rasburicase**, monitor electrolytes & urine output.

Retinoblastoma

- **Most common intraocular malignancy of childhood**; **< 5 yrs (mostly < 2 yrs)**; **RB1 gene – chromosome 13q14**; **hereditary (bilateral) ~40 %**.
- **Leukocoria (white pupillary reflex / "cat's eye reflex") – most common sign** (noticed in photos), **strabismus (second)**, red painful eye, glaucoma, proptosis.
- Rx – **enucleation** (large), chemotherapy, laser, cryotherapy, radiotherapy; **genetic counselling**; screening of siblings; **second malignancy – osteosarcoma**.
- Post-enucleation care – pressure dressing, prosthesis (~6 weeks), eye socket cleaning.

Other tumours

Tumour	Key points
Neuroblastoma	Adrenal medulla/ sympathetic chain; **< 2 yrs**; **irregular abdominal mass crossing midline**, **"raccoon eyes"** (periorbital ecchymosis), **↑ urinary VMA/ HVA**, opsoclonus-myoclonus; may regress spontaneously in infants (stage 4S)
Wilms tumour	Kidney, 2–5 yrs, smooth, **doesn't cross midline – don't palpate**
Hodgkin lymphoma	Adolescents; **painless cervical lymphadenopathy**, **Reed-Sternberg cells**, B symptoms (fever, night sweats, weight loss)
Osteosarcoma	Adolescents, **metaphysis of long bones (around knee)**, **sunburst appearance, Codman's triangle**
Ewing sarcoma	Diaphysis, **"onion-peel"** periosteal reaction, fever mimics osteomyelitis
Medulloblastoma	Most common malignant brain tumour, posterior fossa (cerebellum), ataxia

Nursing care of child on chemotherapy

- **Infection prevention (neutropenia – ANC < 500/mm^3)** – **protective (reverse) isolation**, **strict hand hygiene**, no fresh flowers/ plants, **no raw/ uncooked food**, avoid crowds & sick people, **no rectal temperature/ suppositories**, report **fever ≥ 38 °C immediately (febrile neutropenia – emergency: IV broad-spectrum antibiotics within 1 hr)**; **no live vaccines** (siblings – inactivated polio, avoid OPV contact).
- **Bleeding (thrombocytopenia)** – soft toothbrush, avoid IM, NSAIDs, contact sports, nose picking.
- **Nausea/ vomiting** – **antiemetics (ondansetron) before chemo**, small frequent bland meals.
- **Mucositis** – soft bland diet, **saline/ soda-bicarbonate mouthwash**, oral hygiene, topical anaesthetic, nystatin/ clotrimazole for thrush.
- **Alopecia** – reassure (temporary; regrows in 3–6 months), cap/ scarf, body-image support.
- **Specific drug toxicities** – **vincristine – peripheral neuropathy, constipation, jaw pain** (fatal if intrathecal); **doxorubicin/ daunorubicin – cardiotoxicity, red urine**; **cyclophosphamide – haemorrhagic cystitis (hydrate, MESNA)**; **L-asparaginase – anaphylaxis, pancreatitis**; **methotrexate – mucositis (leucovorin rescue)**; **steroids – moon face, ↑ appetite, mood changes, hyperglycaemia**; **cisplatin – nephro/ ototoxicity**.
- **Extravasation** of vesicants – stop infusion, aspirate, cold (most)/ warm (vinca alkaloids) compress, antidote.
- **Safe handling** – **chemo gloves, gown, biosafety cabinet, cytotoxic waste (yellow bag/ purple label)**; pregnant staff avoid.
- **Psychosocial** – honest age-appropriate information, play/ school continuity, sibling support, **palliative care & pain management (WHO analgesic ladder)**, bereavement support.
- **Child's concept of death** – **< 3 yrs: no concept (separation)**; **3–5 yrs: temporary, reversible (sleep)**; **5–9 yrs: irreversible but may personify (bogeyman), not universal**; **> 9–10 yrs: irreversible, universal, inevitable (adult concept)**.

PYQ style

- Physiological anaemia of infancy – **2–3 months**.
- Thalassaemia major presents at – **6 months**; X-ray – **hair-on-end**; ↑ **HbF**.
- Iron chelator (SC) – **desferrioxamine**; oral – **deferasirox/ deferiprone**.
- Thalassaemia carrier screening test – **NESTROFT**; trait – **↑ HbA2**.
- First sign of sickle cell in infants – **hand-foot syndrome (dactylitis)**.
- Most important in sickle cell crisis – **hydration**; long-term drug – **hydroxyurea**.
- Haemophilia A deficiency – **factor VIII**; inheritance – **X-linked recessive**.
- Most common bleeding in haemophilia – **haemarthrosis**; avoid – **IM injection, aspirin**.
- Most common inherited bleeding disorder – **von Willebrand disease**.
- ITP – **low platelets in a well child after viral infection**.
- Most common childhood cancer – **ALL**; confirmatory test – **bone marrow aspiration**.
- Sanctuary sites in ALL – **CNS & testes**.
- Leukocoria – **retinoblastoma (chromosome 13)**.
- Raccoon eyes & ↑ VMA – **neuroblastoma**.
- Febrile neutropenia – **antibiotics within 1 hr**.
- Haemorrhagic cystitis – **cyclophosphamide** (prevent – MESNA, hydration).
- Child understands death as irreversible & universal – **after 9–10 yrs**.

14 Endocrine, Metabolic & Genetic Disorders

14.1 Congenital Hypothyroidism (Cretinism)

- **Most common preventable cause of intellectual disability**; ~1 in 2000–4000 births (higher in India); **thyroid dysgenesis (most common)**, dyshormonogenesis, **iodine deficiency (endemic cretinism)**.
- **Newborn screening – TSH (heel prick) at 48–72 hrs** (cord blood TSH also used); confirm with serum T4/ TSH.
- **S/S (often normal at birth)** – **prolonged physiological jaundice (earliest)**, **large posterior fontanelle**, **macroglossia (large protruding tongue)**, **umbilical hernia**, **constipation**, **hoarse cry**, **lethargy, sleepy, poor feeding**, hypotonia, dry coarse skin, **hypothermia**, puffy face, **delayed milestones & dentition**, short stature (if untreated).
- **Treatment** – **levothyroxine 10–15 µg/kg/day orally** – start **ASAP (within 2 weeks of birth)** for normal IQ; **lifelong**; crush tablet in small amount of breast milk/ water (**not with soy, iron, calcium**); monitor T4/ TSH regularly.
- **Juvenile (acquired) hypothyroidism** – Hashimoto's thyroiditis (commonest); growth retardation, delayed puberty, ↓ school performance.

14.2 Type 1 Diabetes Mellitus

- **Autoimmune destruction of β-cells → absolute insulin deficiency**; peak **5–7 yrs & puberty**; most common type in children (T2DM ↑ with obesity).
- **S/S** – **polyuria, polydipsia, polyphagia, weight loss**, **secondary enuresis (bedwetting – often first clue)**, fatigue, **DKA (first presentation in ~25–40 %)**.
- **Diagnosis** – FBS ≥ 126, RBS ≥ 200 mg/dl with symptoms, HbA1c ≥ 6.5 %.
- **Treatment** – **insulin (lifelong)** – **basal-bolus regimen** (long-acting glargine/ detemir + rapid-acting before meals) or split-mixed; **pump**; **target HbA1c < 7 %**.
- **Insulin injection teaching** – **rotate sites** (abdomen, thigh, buttock, upper arm) – lipohypertrophy; **SC at 45–90°**; **draw clear (regular) before cloudy (NPH)**; store unopened in fridge, in-use at room temp ≤ 28 days; **SMBG** (self-monitoring 4–6 times/day).
- **Diet** – balanced, carbohydrate counting, regular meals & snacks, no restriction of calories for growth; **exercise** – snack before, monitor glucose; **sick-day rules – never omit insulin, check ketones, fluids**.
- **Honeymoon period** – transient remission (↓ insulin need) weeks–months after diagnosis.

Diabetic Ketoacidosis (DKA)

- **Glucose > 200 mg/dl + venous pH < 7.3 or HCO_3 < 15 mmol/L + ketonaemia/ ketonuria**.
- **S/S** – dehydration, **Kussmaul breathing (deep, rapid)**, **fruity (acetone) breath**, abdominal pain, vomiting, drowsiness → coma.
- **Management (ISPAD)**:
- **Fluids FIRST – 0.9 % NS 10–20 ml/kg over 1 hr** (shock) → deficit replacement **slowly over 48 hrs** (**avoid rapid fluids – cerebral oedema**).

- **Insulin infusion 0.05–0.1 U/kg/hr – start 1 hr after fluids** (**no bolus**); add **dextrose (5–10 %)** when glucose falls to **250–300 mg/dl**; reduce glucose by **≤ 100 mg/dl/hr**.
- **Potassium** – add after urine output & K^+ < 5.5 (**40 mEq/L**) – insulin drives K^+ into cells (**hypokalaemia**).
- **Bicarbonate – not routinely**.
- **Monitor** – hourly glucose, vitals, **neuro observations (GCS hourly)**, I/O; electrolytes, pH 2–4 hourly.
- **Cerebral oedema – most common cause of death in paediatric DKA** – headache, ↓ HR, ↑ BP, irritability, altered sensorium (4–12 hrs into treatment) → **mannitol 0.5–1 g/kg / 3 % hypertonic saline**, head-end elevated, reduce fluids.

Hypoglycaemia in diabetic child

- **< 70 mg/dl** – sweating, tremor, hunger, pallor, irritability, behaviour change, confusion, seizures.
- **Conscious → "rule of 15"**: **15 g fast-acting sugar** (glucose tablets, juice ½ cup, 3 tsp sugar) → recheck in **15 min** → repeat if < 70 → then complex carbohydrate snack.
- **Unconscious → glucagon IM/ SC (0.5 mg < 25 kg; 1 mg > 25 kg) / IV 10 % dextrose 2 ml/kg**; **nothing by mouth**.

14.3 Other Endocrine Disorders

Condition	Key points
Growth hormone deficiency	**Short stature with normal proportions**, delayed bone age, chubby, hypoglycaemia (infant), micropenis; **GH stimulation test**; **recombinant GH SC at bedtime**
Short stature causes	**Familial/ constitutional (commonest – normal growth velocity)**, malnutrition, chronic disease, hypothyroidism, GHD, Turner syndrome, skeletal dysplasia (achondroplasia – disproportionate)
Precocious puberty	**Girls < 8 yrs, boys < 9 yrs**; central (GnRH dependent – idiopathic in girls, CNS tumour in boys) – **GnRH analogue (leuprolide)**
Delayed puberty	Girls no breast development by **13 yrs**, boys no testicular enlargement by **14 yrs**
Congenital adrenal hyperplasia (CAH)	**21-hydroxylase deficiency (90 %)** – autosomal recessive; **ambiguous genitalia in girls**; **salt-wasting crisis (1–3 weeks) – vomiting, dehydration, hyponatraemia, hyperkalaemia, shock**; **↑ 17-OH progesterone**; **hydrocortisone + fludrocortisone lifelong**, stress doses
Diabetes insipidus	ADH deficiency – **polyuria (dilute urine), polydipsia, hypernatraemia**; **desmopressin**
SIADH	↑ ADH – hyponatraemia, fluid restriction (meningitis, pneumonia)
Cushing's syndrome	Mostly iatrogenic (steroids) – moon face, buffalo hump, growth suppression

14.4 Inborn Errors of Metabolism

Disorder	Key points
Phenylketonuria (PKU)	**Phenylalanine hydroxylase deficiency** (autosomal recessive) → phenylalanine ↑; **musty/ mousy odour**, **fair skin, blue eyes, blond hair**, eczema, seizures, **intellectual disability**; screening – **Guthrie test** (after 24–48 hrs of protein feeds); **low-phenylalanine diet (special formula)** – **avoid high-protein foods & aspartame (artificial sweetener)**; started in first weeks, continued lifelong (especially before & during pregnancy)
Galactosaemia	**GALT deficiency**; **vomiting, jaundice, hepatomegaly, cataract (oil-drop), E. coli sepsis** after milk feeds; **lactose/ galactose-free diet (soy formula)**; **breastfeeding contraindicated**
Maple syrup urine disease	Branched-chain amino acids; sweet (maple syrup) odour urine
Glycogen storage disease	Hypoglycaemia, hepatomegaly
Lysosomal storage (Gaucher, Tay-Sachs – cherry-red spot)	Enzyme replacement (some)

- **Newborn screening (India – RBSK/ state programmes)** – **congenital hypothyroidism, CAH, G6PD deficiency, galactosaemia, PKU**, hearing, CCHD (pulse oximetry), sickle cell (selected areas).

14.5 Chromosomal & Genetic Disorders

Down Syndrome (Trisomy 21)

- **Most common chromosomal disorder & most common genetic cause of intellectual disability**; ~1 in 700–800 births; **↑ with maternal age > 35 yrs**.
- **Types** – **non-disjunction (~95 % – maternal age related)**, **translocation (~4 % – hereditary, any age; t(14;21))**, mosaicism (~1 %).
- **Features** – **hypotonia (most common neonatal sign – "floppy baby")**, **flat facial profile/ occiput (brachycephaly)**, **upslanting palpebral fissures, epicanthic folds**, **Brushfield spots** (iris), small low-set ears, **protruding tongue**, short neck with excess skin, **single transverse palmar (simian) crease**, **sandal gap** (wide 1st–2nd toe space), short stature, clinodactyly (5th finger).
- **Associated** – **CHD (~40–50 %) – AVSD/ endocardial cushion defect (most characteristic)**, VSD; **duodenal atresia ("double bubble")**, Hirschsprung; **hypothyroidism**; **leukaemia (ALL/ AML ↑ 10–20 times)**; atlanto-axial instability; hearing/ vision problems; early Alzheimer's; infertility (males).
- **Screening/ diagnosis** – **1st trimester combined test (NT ↑ on USG + PAPP-A ↓, free β-hCG ↑)**, quadruple test (AFP ↓, uE3 ↓, hCG ↑, inhibin A ↑), **NIPT**; **confirm – karyotype (CVS/ amniocentesis)**.
- **Nursing** – early intervention & **stimulation programmes**, feeding (hypotonia – support head, protruding tongue), constipation, prevent infections (respiratory), **thyroid check yearly**, cardiac evaluation (echo), parent support groups, special education, **genetic counselling**.

Disorder	Karyotype	Key features
Edwards syndrome	**Trisomy 18**	**Clenched fists with overlapping fingers, rocker-bottom feet**, micrognathia; most die in 1st year
Patau syndrome	**Trisomy 13**	**Cleft lip/ palate, polydactyly, holoprosencephaly**, microphthalmia, scalp defects
Turner syndrome	**45, XO** (female)	**Short stature, webbed neck, shield chest (widely spaced nipples)**, **coarctation of aorta**, lymphoedema of hands/ feet at birth, **streak ovaries – primary amenorrhoea, infertility**; GH & oestrogen therapy
Klinefelter syndrome	**47, XXY** (male)	**Tall, gynaecomastia, small testes, infertility (azoospermia)**, ↓ testosterone; most common cause of male hypogonadism
Fragile X syndrome	X-linked (CGG repeat)	**Most common inherited cause of intellectual disability**; long face, **large ears, macroorchidism**, autism features
Cri-du-chat	**5p deletion**	**Cat-like cry**, microcephaly
Achondroplasia	Autosomal dominant (FGFR3)	Short-limbed dwarfism, normal intelligence

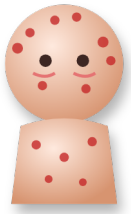
- **Inheritance patterns** – **autosomal dominant** (Huntington's, Marfan, achondroplasia, NF-1); **autosomal recessive** (**thalassaemia, sickle cell, cystic fibrosis, PKU, galactosaemia, CAH, albinism**); **X-linked recessive** (**haemophilia A & B, G6PD, DMD, colour blindness**); X-linked dominant (vitamin D-resistant rickets).
- **Autosomal recessive – both parents carriers → each pregnancy: 25 % affected, 50 % carrier, 25 % normal**.
- **Genetic counselling** – **non-directive**, pedigree, risk estimation, prenatal diagnosis options, support; advise against consanguineous marriage where a recessive disorder runs in the family.

PYQ style

- Most common preventable cause of intellectual disability – **congenital hypothyroidism**.
- Newborn screening for hypothyroidism – **TSH heel prick at 48–72 hrs**.
- Earliest sign of congenital hypothyroidism – **prolonged jaundice**; other – **macroglossia, umbilical hernia**.
- Levothyroxine dose – **10–15 µg/kg/day**, start within **2 weeks**.
- Common first sign of T1DM in a toilet-trained child – **secondary enuresis**.
- DKA – first step – **fluids (NS)**; insulin infusion – **0.05–0.1 U/kg/hr**.
- Most common cause of death in paediatric DKA – **cerebral oedema**.
- Insulin mixing – **clear (regular) before cloudy (NPH)**.
- Hypoglycaemia – **rule of 15**; unconscious – **glucagon**.
- Mousy odour & Guthrie test – **PKU**; avoid – **aspartame, high protein**.
- Galactosaemia – **breastfeeding contraindicated**.
- CAH – **21-hydroxylase deficiency**, ambiguous genitalia, salt-wasting.
- Down syndrome – **trisomy 21**; cardiac defect – **AVSD (endocardial cushion)**; sign – **simian crease, sandal gap**.
- Double bubble sign – **duodenal atresia (Down)**.
- Rocker-bottom feet – **Edwards (trisomy 18)**; polydactyly & cleft – **Patau (13)**.
- Webbed neck, coarctation – **Turner (45 XO)**; gynaecomastia, tall – **Klinefelter (47 XXY)**.
- Most common inherited cause of intellectual disability – **Fragile X**.

15 Communicable Diseases in Children

Childhood Rash (Exanthematous) Fevers – Identify Quickly



MEASLES

Rubeola virus (paramyxo)

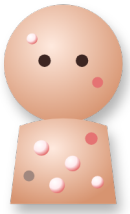
Koplik's spots – buccal mucosa (prodrome)

3 C's: cough, coryza, conjunctivitis + fever

Rash day 4: behind ears → face → trunk (downward)

Vitamin A (2 doses)

Complication: pneumonia (commonest), SSPE late



CHICKENPOX

Varicella-zoster virus

Centripetal, pleomorphic rash in crops

Macule → papule → vesicle ("dew drop on rose petal")

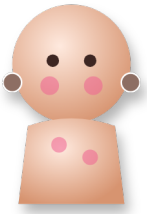
→ crust; itchy

Infectious till all crusted

Acyclovir · calamine

No aspirin (Reye's)

Cut nails, prevent scratching



RUBELLA

"German / 3-day measles"

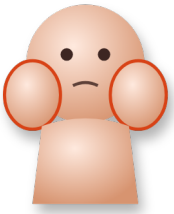
Post-auricular & sub-occipital nodes

Mild fever, pink rash face → body, 3 days

Forchheimer spots (palate)

Danger in pregnancy: congenital rubella – cataract, deafness, PDA

Prevent: MR vaccine



MUMPS

Paramyxovirus

Parotid swelling – ear lobe pushed up/ out

Pain on chewing, sour foods ↑ pain

Complications: orchitis (after puberty), aseptic meningitis, pancreatitis

Soft diet, warm/ cold compress, oral hygiene

Measles, chickenpox, rubella & mumps – identifying features

15.1 Viral Exanthems

Disease	Agent	Incubation	Key features	Infectivity/ isolation
Measles (Rubeola)	Paramyxovirus (morbillivirus)	**10–14 days**	**Prodrome – fever, 3 C's (cough, coryza, conjunctivitis), Koplik's spots (pathognomonic – white spots on red base, buccal mucosa opposite lower molars, before rash)**; **maculopapular rash on day 4 – behind ears/ hairline → face → trunk → limbs (centrifugal, downward)**, fades with **brown staining & branny desquamation**	**4 days before to 4 days after rash – airborne isolation**
Rubella (German/ 3-day measles)	Togavirus	14–21 days	Mild fever, **post-auricular & sub-occipital lymphadenopathy**, pink rash (face → body, 3 days), **Forchheimer spots** (soft palate); arthralgia	7 days before to 7 days after rash; **keep away from pregnant women**
Chickenpox (Varicella)	**Varicella-zoster virus**	**14–16 days (10–21)**	**Centripetal rash (trunk > limbs), pleomorphic – all stages at same time (macule → papule → vesicle → pustule → crust) in crops**, **"dew drop on rose petal"**, **intensely itchy**, involves scalp & mucosa	**1–2 days before rash till all lesions crusted** (~5–7 days) – airborne + contact

Disease	Agent	Incubation	Key features	Infectivity/ isolation
Mumps	Paramyxovirus	**16–18 days**	**Parotitis (bilateral ~70 %)** – earache, pain on chewing/ sour food; ear lobe pushed up & out	7 days before to 9 days after swelling (droplet)
Roseola infantum (exanthem subitum)	**HHV-6**	5–15 days	**6 months–2 yrs, high fever 3–4 days → rash appears when fever subsides**; febrile seizures	–
Erythema infectiosum (fifth disease)	**Parvovirus B19**	–	**"Slapped cheek"** appearance, lacy rash; aplastic crisis in sickle cell; hydrops fetalis	–
Hand, foot & mouth disease	**Coxsackie A16, Enterovirus 71**	3–5 days	Vesicles on **palms, soles, mouth**, buttocks; < 5 yrs	–
Scarlet fever (bacterial)	**Group A Strep**	2–4 days	**Sandpaper rash**, **strawberry tongue**, **Pastia's lines**, circumoral pallor; penicillin	–

Measles – details

- **Complications** – **pneumonia (most common cause of death)**, **diarrhoea**, otitis media, **malnutrition/ vitamin A deficiency → corneal ulcer/ blindness**, encephalitis, **SSPE** (subacute sclerosing panencephalitis – 7–10 yrs later), **flare-up of TB** (immunosuppression – Mantoux may become negative).
- **Treatment** – **supportive + vitamin A (2 doses on 2 consecutive days: < 6 m 50,000 IU; 6–11 m 1 lakh IU; ≥ 12 m 2 lakh IU; third dose after 2–4 wks if eye signs)**, eye care (clean with saline), fluids, nutrition, isolation.
- **Prevention** – **MR vaccine at 9–12 & 16–24 months**; **measles vaccine within 72 hrs of exposure** (post-exposure); **immunoglobulin within 6 days**; **Measles-Rubella elimination target** (India).

Chickenpox – nursing

- **Calamine lotion, antihistamines**, **cut nails short / mittens** (prevent scratching & secondary infection), cool baths with baking soda/ oatmeal, loose cotton clothes, **paracetamol for fever – NEVER aspirin (Reye's syndrome)**, oral hygiene, **isolate till all lesions crusted**.
- **Acyclovir** – adolescents, immunocompromised, severe; **VZIG** – high-risk exposed (newborn of mother with rash 5 days before–2 days after delivery, immunocompromised).
- **Complications** – **secondary bacterial skin infection (most common)**, pneumonia, encephalitis (cerebellar ataxia), Reye's; **congenital varicella syndrome** (1st/ 2nd trimester – limb hypoplasia, scarring); **herpes zoster (shingles)** later – reactivation.

Mumps – complications & care

- **Orchitis (post-pubertal males – sterility rare)**, **aseptic meningitis (most common extra-salivary)**, pancreatitis, oophoritis, **sensorineural deafness (unilateral)**, encephalitis.
- **Soft/ liquid diet, avoid sour/ acidic foods**, warm/ cold compress, oral hygiene, scrotal support & ice packs in orchitis, analgesics.

15.2 Vaccine-Preventable Bacterial Diseases

Disease	Key points
Diphtheria	**Corynebacterium diphtheriae** (exotoxin); **2–5 days** IP; **greyish-white adherent pseudomembrane** (bleeds on removal) on tonsils/ pharynx, **"bull neck"** (cervical adenitis + oedema), low fever, toxic; **laryngeal → stridor, airway obstruction**; **complications – myocarditis (most common cause of death, 2nd week), palatal/ neuritic palsy**; Dx – throat swab (**Albert's stain – Chinese letter pattern, Loeffler's serum slope**); **Rx – Anti-Diphtheritic Serum (ADS) immediately (after test dose)** + **penicillin/ erythromycin**; **strict isolation till 2 negative cultures**; **bed rest 2–3 weeks** (myocarditis); contacts – erythromycin + booster; **Schick test** (susceptibility)
Pertussis (whooping cough)	**Bordetella pertussis**; **7–14 days** IP; **3 stages – catarrhal (1–2 wks, most infectious), paroxysmal (2–6 wks – bouts of cough ending in inspiratory "whoop", vomiting, face red/ cyanosed, tongue protruded), convalescent ("100-day cough")**; infants – apnoea instead of whoop; **marked lymphocytosis**; complications – **pneumonia (death), subconjunctival haemorrhage, hernia, rectal prolapse, convulsions, encephalopathy**; **Rx – azithromycin/ erythromycin** (reduces infectivity); **droplet isolation 5 days after antibiotics**; small frequent feeds (**re-feed after vomiting**), suction, O_2, quiet environment
Tetanus neonatorum	**Clostridium tetani** (spores – **unclean cord cutting/ dressing – cow dung, ash**); **onset day 3–10 ("8th-day disease")**; **first sign – inability to suck (trismus)**, excessive crying, **risus sardonicus**, **spasms/ rigidity, opisthotonus**, triggered by touch/ sound/ light; **CFR very high (~80 %)**; Rx – **TIG (human tetanus immunoglobulin 500 IU IM)**, **diazepam** for spasms, metronidazole/ penicillin, NG feeding, airway; **quiet, dark room, minimal handling**; **prevention – Td to mother (2 doses/ booster) + clean delivery (6 cleans)**; **India eliminated maternal & neonatal tetanus (MNTE) – May 2015** (< 1 case per 1000 live births in every district)
Tetanus (child)	Wound contamination; trismus (lockjaw); TIG + vaccine + wound care
Poliomyelitis	**Poliovirus (types 1, 2, 3; feco-oral)**; **anterior horn cells → acute flaccid paralysis – asymmetric, proximal > distal, legs**, **no sensory loss**, fever at onset; most infections asymptomatic (~90 %+); **India polio-free since 27 March 2014** (WHO SEAR); **AFP surveillance – any child < 15 yrs with acute flaccid paralysis (or any age with suspected polio) → 2 stool samples 24 hrs apart within 14 days of onset (reverse cold chain)**; nursing – **bed rest in acute phase (no IM injections – provocation paralysis)**, **positioning – footboard (foot drop)**, **hot packs**, physiotherapy after acute phase, respiratory support (bulbar)
Typhoid (enteric fever)	**Salmonella Typhi** (feco-oral); **step-ladder fever**, relative bradycardia, **coated tongue**, **rose spots (trunk)**, splenomegaly, abdominal pain, constipation/ diarrhoea; complications – **intestinal perforation & haemorrhage (3rd week)**; **blood culture – 1st week (best)**, **Widal (after 1st week, ≥ 1:160 O)**, stool culture (3rd week); Rx – **ceftriaxone/ cefixime/ azithromycin**; **typhoid conjugate vaccine (TCV) ≥ 6 months**; carriers – gallbladder

15.3 Vector-Borne Diseases

Dengue

- **Dengue virus (4 serotypes) – Aedes aegypti (day-biting, breeds in clean stored water)**; IP 4–7 days.
- **Phases** – **febrile (2–7 days – high fever, retro-orbital pain, myalgia "breakbone fever", rash, positive tourniquet test)** → **critical (around defervescence, 24–48 hrs – plasma leakage)** → recovery (convalescent rash – "white islands in sea of red", bradycardia, fluid reabsorption).

- **Warning signs** – **abdominal pain/ tenderness, persistent vomiting, clinical fluid accumulation (ascites, effusion), mucosal bleeding, lethargy/ restlessness, liver enlargement > 2 cm, ↑ Hct with rapid ↓ platelets**.
- **Severe dengue** – shock (**dengue shock syndrome – narrow pulse pressure ≤ 20 mmHg**), severe bleeding, organ impairment.
- **Diagnosis** – **NS1 antigen (day 1–5)**, **IgM ELISA (after day 5)**; CBC – **thrombocytopenia, ↑ haematocrit (haemoconcentration)**, leucopenia.
- **Management** – **paracetamol for fever (NO aspirin/ NSAIDs/ steroids)**, **oral fluids (ORS, juices)**, tepid sponging, **monitor Hct & platelets**, IV crystalloids for warning signs/ shock (5–10–20 ml/kg as per protocol), **platelet transfusion only if bleeding** (or < 10,000 with risk), watch fluid overload in recovery phase.
- **Prevention** – **source reduction (empty water containers weekly – "dry day")**, mosquito nets/ repellents, full-sleeve clothes; **National Dengue Day – 16 May**.

Malaria

- Plasmodium vivax **(most common in India)**, P. falciparum (**severe/ cerebral malaria – most dangerous**); **female Anopheles (night-biting)**.
- **Paroxysm** – cold stage (chills) → hot stage → sweating stage; anaemia, splenomegaly; **children – severe anaemia, hypoglycaemia, convulsions, cerebral malaria**.
- **Diagnosis** – **RDT (rapid diagnostic test – bivalent)**, **peripheral blood smear (gold standard – thick for detection, thin for species)**.
- **Treatment (NVBDCP/ NCVBDC)** – **vivax: chloroquine 3 days (25 mg/kg total) + primaquine 0.25 mg/kg × 14 days** (radical cure; **not < 1 yr / pregnancy / G6PD deficiency**); **falciparum: ACT (artesunate + sulfadoxine-pyrimethamine; AL in north-east) + primaquine single dose (0.75 mg/kg) on day 2**; **severe malaria – IV/ IM artesunate**.
- **Prevention** – **LLINs (long-lasting insecticidal nets)**, IRS, source reduction; **World Malaria Day – 25 April**.

Others

- **Japanese encephalitis** – Culex (rice fields), pigs amplify; vaccine in endemic districts.
- **Chikungunya** – Aedes, severe arthralgia.
- **Kala-azar** – sandfly, Leishmania donovani; Bihar/ UP; hepatosplenomegaly, darkening skin.
- **Scrub typhus** – mite (chigger) – **eschar**; doxycycline/ azithromycin; AES cause.
- **Leptospirosis** – rat urine, flood water; jaundice, renal failure.
- **Rabies** – dog bite; **wash wound with soap & running water 15 min (most important first aid)**; **ARV (intradermal 0.1 ml at 2 sites on days 0, 3, 7, 28 – Updated Thai Red Cross)** / IM; **RIG (rabies immunoglobulin) for category III** – infiltrated into wound; **hydrophobia, aerophobia**; 100 % fatal once symptoms appear.

15.4 HIV/ AIDS in Children

- **Mother-to-child transmission (MTCT) – most common route (~90 %)** – pregnancy, **labour & delivery (most)**, breastfeeding.
- **PPTCT (Prevention of Parent-To-Child Transmission)** – **universal HIV screening of pregnant women**, **lifelong ART for mother (TLD – tenofovir + lamivudine + dolutegravir)** irrespective of CD4, **infant prophylaxis – nevirapine syrup × 6 weeks (low risk); high risk – nevirapine +**

zidovudine × 12 weeks, **exclusive breastfeeding for 6 months (mother on ART)**, **cotrimoxazole prophylaxis from 6 weeks** till HIV excluded.

- **Infant diagnosis** – **HIV DNA PCR (EID – Early Infant Diagnosis via DBS) at 6 weeks, 6 months, 12 months**; **antibody test valid only after 18 months** (maternal antibodies persist).
- **S/S** – FTT, persistent diarrhoea, **oral thrush (recurrent)**, recurrent pneumonia (**PCP – Pneumocystis jirovecii** in infants), lymphadenopathy, hepatosplenomegaly, parotid enlargement, developmental delay, TB.
- **Immunization** – all UIP vaccines; **BCG avoided in symptomatic HIV**; live vaccines with caution if severe immunosuppression.
- **Nursing** – **standard precautions**, ART adherence (> 95 %), nutrition, infection prevention, **disclosure (age-appropriate)**, stigma reduction, confidentiality.

15.5 Skin Infections & Infestations (brief)

Condition	Key points
Scabies	Sarcoptes scabiei; **intense nocturnal itching**, burrows in **web spaces, wrists, axillae, genitals** (infants – palms, soles, face/ scalp); **permethrin 5 % cream** – apply **neck down (whole body incl. scalp in infants)**, leave **8–12 hrs**, repeat after 1 week; **treat all family members simultaneously**; wash clothes/ bedding in hot water/ sun-dry
Pediculosis capitis (head lice)	Nits on hair shafts; **permethrin 1 %**, fine-toothed comb
Impetigo	**Staph aureus/ Strep**; **honey-coloured crusts** around nose/ mouth; contagious; **mupirocin**, oral antibiotics; risk of **AGN** (strep)
Oral thrush	Candida – white patches not easily wiped off (unlike milk curds); **nystatin/ clotrimazole**
Diaper (napkin) dermatitis	Ammonia/ irritant; **keep dry, frequent changes, zinc oxide barrier**, air exposure; candidal – satellite lesions → clotrimazole
Atopic eczema	Itchy, flexural (older child), cheeks (infant); emollients, avoid soap, topical steroids; **elbow restraints/ mittens**

15.6 Isolation & Incubation – Quick Table

Disease	Incubation	Isolation
Measles	10–14 days	Airborne – till 4 days after rash
Chickenpox	14–16 days	Airborne + contact – till all crusted
Mumps	16–18 days	Droplet – till 9 days after swelling (5 days CDC)
Rubella	14–21 days	Droplet – 7 days after rash
Diphtheria	2–5 days	Droplet – till 2 negative cultures
Pertussis	7–14 days	Droplet – 5 days after antibiotic start
Meningococcal	3–4 days	Droplet – 24 hrs after antibiotics
Hepatitis A	15–45 days (~30)	Contact/ enteric
Typhoid	10–14 days	Enteric precautions

Disease	Incubation	Isolation
Tetanus	3–21 days (neonatal 3–10)	No isolation (not communicable)

PYQ style

- Pathognomonic sign of measles – **Koplik's spots**; rash starts – **behind ears**.
- Most common cause of death in measles – **pneumonia**; late complication – **SSPE**.
- Measles – give **vitamin A**; infectivity – **4 days before to 4 days after rash**.
- Chickenpox rash – **centripetal, pleomorphic, dew drop on rose petal**; avoid – **aspirin**.
- Post-auricular lymphadenopathy – **rubella**; danger – **pregnancy (congenital rubella)**.
- Mumps commonest complication in adult males – **orchitis**; extra-salivary in child – **aseptic meningitis**.
- Rash appearing after fever subsides – **roseola (HHV-6)**; slapped cheek – **parvovirus B19**.
- Strawberry tongue – **scarlet fever/ Kawasaki**.
- Bull neck & pseudomembrane – **diphtheria**; death due to – **myocarditis**; Rx – **ADS**.
- Whoop – **pertussis**; most infectious stage – **catarrhal**; blood – **lymphocytosis**.
- Neonatal tetanus first sign – **inability to suck**; onset – **day 3–10**.
- India MNTE – **2015**; polio-free – **27 March 2014**.
- AFP surveillance age – **< 15 years**; stool samples – **2 samples, 24 hrs apart, within 14 days**.
- Dengue vector – **Aedes aegypti**; avoid – **aspirin/ NSAIDs**; early test – **NS1 antigen**.
- Most common malaria in India – **vivax**; dangerous – **falciparum**; radical cure – **primaquine 14 days**.
- HIV infant diagnosis – **DNA PCR at 6 weeks**; prophylaxis – **nevirapine**.
- Scabies treatment – **permethrin 5 %, whole family**.
- Honey-coloured crusts – **impetigo**.

16 Musculoskeletal Disorders, Fractures & Burns

16.1 Congenital Talipes Equinovarus (CTEV – Clubfoot)

- **Most common congenital foot deformity**; **boys 2 : 1**, bilateral ~50 %; associated with oligohydramnios, spina bifida, arthrogryposis.
- **Components (CAVE)** – **C**avus (high arch), forefoot **A**dduction, hindfoot **V**arus (inversion), **E**quinus (plantar flexion) – foot points **down & in**.
- **Positional** (correctable passively) vs **structural/ true** (rigid).
- **Treatment – Ponseti method (gold standard)** – start **within 1–2 weeks of birth**; **serial manipulation & long-leg plaster casts changed weekly (5–7 casts)** → **percutaneous Achilles tenotomy (~80–90 %)** → **foot abduction brace (Denis Browne splint/ Steenbeek) – 23 hrs/day × 3 months, then nights & naps till 4–5 yrs** (prevents relapse – **non-compliance = most common cause of relapse**).
- Surgery (PMSTR) for resistant/ late cases.
- **Cast care** – elevate, check **5 P's/ neurovascular status (toes – colour, warmth, capillary refill, swelling, movement)**, keep cast dry & clean (cover with plastic during diaper change), **petal edges**, don't insert anything inside, report foul smell/ crying (pressure sore).

16.2 Developmental Dysplasia of Hip (DDH)

- Abnormal development of hip – **subluxation → dislocation**; **risk – females (6:1), breech, first-born, family history, oligohydramnios**, **tight swaddling with legs extended**.
- **Tests (newborn – up to 3 months)**: **Barlow test – dislocates** (adduct & push posteriorly – "clunk out"); **Ortolani test – relocates** (abduct & lift – "clunk in").
- **Later signs** – **limited hip abduction (most reliable after 3 months)**, **asymmetric thigh & gluteal folds**, **Galeazzi/ Allis sign (unequal knee heights)**, **Trendelenburg sign & waddling gait (walking child)**, telescoping.
- **Diagnosis – USG (< 4–6 months)**; X-ray after 4–6 months (ossific nucleus).
- **Treatment** – **0–6 months: Pavlik harness (hips flexed & abducted – "frog position")** worn **23 hrs/day for 6–12 weeks**; 6–18 months: closed reduction + **spica cast**; older: open reduction/ osteotomy.
- **Pavlik harness care** – don't remove unless allowed, check skin (sponge bath), **diaper under the straps**, light knee socks/ vest under straps, don't adjust straps.
- **Hip spica cast care** – **turn every 2 hrs**, **petal & protect perineal area with plastic** (soiling), check circulation/ skin, **never use abduction bar to lift**, pillows for support; drying with exposed cast (24–48 hrs plaster).

16.3 Fractures in Children

- **Bones are more porous & flexible**, thick **periosteum**, **growth plates (epiphyseal)** → **heal faster**, **remodel**; **non-union rare**.

- **Greenstick fracture** – **most common in children** (one cortex breaks, other bends); **buckle/ torus** (compression – distal radius); **plastic deformation**; **Salter-Harris** (growth plate – type II most common; type V crush – growth arrest).
- **Supracondylar fracture of humerus** – **most common elbow fracture** (fall on outstretched hand, 5–7 yrs); **brachial artery injury → Volkmann's ischaemic contracture** (compartment syndrome – **pain on passive finger extension – earliest sign**), median/ radial nerve injury, **cubitus varus (gunstock deformity)**.
- **Clavicle fracture** – commonest birth fracture.
- **Fracture femur < 2 yrs (< 12–15 kg) – Bryant's (gallows) traction**: both legs suspended vertically, **hips flexed 90°, knees extended**, **buttocks slightly off the bed (just clearing – hand-breadth)** – body weight = counter-traction; check circulation (**risk of vascular compromise**).
- **Older children – Russell's traction**, skeletal traction, **hip spica**, elastic nails.
- **Traction care** – **weights hang freely** (never rest on floor/ bed), **don't remove weights**, ropes in pulleys, proper alignment, counter-traction, **neurovascular checks hourly (first 24 hrs)**, pin-site care (skeletal), skin care, diversion/ play, prevent complications of immobility (constipation, pressure sores).
- **Non-accidental injury (child abuse)** – **spiral fractures in non-ambulant infant**, **posterior rib fractures**, multiple fractures of different ages, **metaphyseal "corner/ bucket-handle" fractures**, story inconsistent with injury.
- **Osteogenesis imperfecta** – **brittle bones, blue sclera**, deafness – handle gently (differential of abuse).

16.4 Infections & Other Conditions

Condition	Key points
Acute osteomyelitis	**Staph aureus (most common)**; **metaphysis of long bones (femur, tibia)** – haematogenous; high fever, **severe localised pain, refuses to move limb (pseudoparalysis)**, tenderness; **X-ray changes after 10–14 days**; **MRI** earliest; **IV antibiotics 4–6 weeks** (cloxacillin/ cefazolin), drainage; limb immobilisation; **sequestrum/ involucrum** in chronic
Septic arthritis	**Hip (infants) – emergency**; limb held flexed, abducted, externally rotated; urgent aspiration/ drainage
Perthes disease (Legg-Calvé-Perthes)	**Avascular necrosis of femoral head**, boys 4–8 yrs, painless limp/ knee pain
SCFE (slipped capital femoral epiphysis)	Obese adolescent boys; knee/ hip pain, limp
Scoliosis	**Lateral curvature of spine**; idiopathic adolescent – **girls**; **screening – Adam's forward bend test (rib hump)**; Cobb angle; **Milwaukee/ Boston brace (23 hrs/day)** for 25–40°; **surgery (spinal fusion – Harrington rods) > 40–50°**; post-op log rolling
Juvenile idiopathic arthritis (JIA)	Arthritis **> 6 weeks, onset < 16 yrs**; **systemic (Still's)** – quotidian fever, salmon rash; **iridocyclitis (uveitis) – regular eye exams (oligoarticular, ANA +)**; NSAIDs, methotrexate; **warm baths/ ROM exercises in the morning**, splints
Duchenne MD	See neuro chapter (Gowers' sign)

Condition	Key points
Torticollis	Sternomastoid tumour – head tilted to same side, chin to opposite; stretching exercises
Rickets deformities	Bow legs, knock knees – vit D

16.5 Burns in Children

- **Scalds (hot liquids) – most common burns in children < 5 yrs**; flame (older), electrical, chemical; **child abuse** – immersion burns ("glove/ stocking" pattern), cigarette burns.
- **Children more vulnerable** – **thin skin (deeper burns at lower temperature)**, **larger body surface area : weight ratio (more fluid & heat loss)**, immature immunity, **head relatively larger**.

Depth	Features
Superficial (1st degree)	Epidermis; red, painful, no blisters (sunburn); heals 3–7 days
Partial thickness – superficial (2nd degree)	**Blisters, very painful, moist, blanches**; heals 10–21 days
Partial – deep	Dermis deeper; mottled, less painful; heals > 3 wks/ graft
Full thickness (3rd degree)	**Leathery, white/ charred, painless (nerve endings destroyed)**, no blanching; **needs grafting**
4th degree	Muscle/ bone

Estimating TBSA in children

- **Lund & Browder chart – most accurate for children** (adjusts for age).
- **Modified "Rule of Nines" for children (infant)**: **head & neck 18 %**, **each upper limb 9 %**, **anterior trunk 18 %**, **posterior trunk 18 %**, **each lower limb 14 %**, perineum 1 % (vs adult: head 9 %, leg 18 %). **For each year after 1, subtract 1 % from head and add 0.5 % to each leg** (till adult values at ~10 yrs).
- **Palm method** – **child's palm (with fingers) ≈ 1 % TBSA** (scattered burns).
- **Do NOT include superficial (1st degree) burns** in TBSA calculation.

Fluid resuscitation

- **IV fluids if burns > 10 % TBSA in children** (> 15 % adults).
- **Parkland formula – 4 ml × weight (kg) × % TBSA of Ringer lactate** in first 24 hrs – **half in the first 8 hrs (from time of burn)**, remaining half in next 16 hrs. **Children – add maintenance fluid with dextrose (5 % dextrose in ½ NS/ RL)** – **hypoglycaemia risk (low glycogen)**.
- **Example**: 20-kg child, 30 % burns → 4 × 20 × 30 = **2400 ml** RL → **1200 ml in first 8 hrs**, 1200 in next 16 hrs + maintenance (1500 ml/day).
- **Best guide – urine output 1 ml/kg/hr** (child), 0.5 ml/kg/hr adult; **Foley catheter**.

Nursing care

- **First aid** – stop burning (**stop, drop & roll**), remove clothing/ jewellery (not stuck), **cool with running tap water (15–25 °C) for 20 minutes within 3 hrs** – **NO ice, toothpaste, oil, butter, ghee, turmeric**; cover with **clean cling film/ sheet**; **keep child warm (hypothermia)**.
- **Airway** – inhalational injury (singed nasal hair, soot, hoarseness, facial burns) → early intubation.

- **Pain** – IV morphine (before dressing), distraction.
- **Wound care** – **silver sulfadiazine 1 % (not on face; not < 2 months)**, **closed (occlusive) vs exposed method**, **aseptic technique, reverse barrier isolation**, debridement, **skin grafting**; **tetanus prophylaxis**.
- **Nutrition** – **high-calorie, high-protein diet** (hypermetabolism – **Curreri formula**), NG feeding early, vitamins & zinc.
- **Complications** – **hypovolaemic shock (first 48 hrs)**, **infection/ sepsis (most common cause of death after 48–72 hrs – Pseudomonas, Staph)**, **Curling's ulcer** (stress ulcer – H2 blockers/ PPI), hypothermia, AKI, **contractures** (**position of anti-deformity – neck extended, shoulders abducted, elbows & knees extended**, splints), **hypertrophic scars/ keloid (pressure garments 23 hrs/day for 12–18 months)**, body-image problems.
- **Escharotomy** – circumferential full-thickness burns (limb/ chest).
- **Electrical burns** – entry/ exit wounds, **cardiac arrhythmias (ECG monitoring)**, myoglobinuria (↑ fluids).
- **Chemical burns** – **irrigate with copious water (20–30 min)**.
- **Prevention** – **keep hot liquids/ kettles out of reach, handles turned inward**, test bath water (**≤ 38 °C; geyser at ≤ 49 °C**), matches/ lighters away, kerosene stove safety, **socket covers**, fire-retardant clothing.

PYQ style

- Clubfoot treatment of choice – **Ponseti method (serial casting)**; brace – **Denis Browne/ foot abduction brace**.
- CTEV components – **cavus, adduction, varus, equinus**.
- DDH tests – **Ortolani (relocates), Barlow (dislocates)**; treatment < 6 months – **Pavlik harness**.
- DDH more common in – **females, breech**.
- Most common fracture in children – **greenstick**.
- Supracondylar fracture complication – **Volkmann's ischaemic contracture**.
- Bryant's traction – **< 2 yrs, femur fracture, buttocks just off bed**.
- Acute osteomyelitis organism – **Staph aureus**; site – **metaphysis**.
- Scoliosis screening – **Adam's forward bend test**; brace – **Milwaukee**.
- JIA eye complication – **uveitis/ iridocyclitis**.
- Most accurate burn chart in children – **Lund & Browder**.
- Infant rule of nines – **head 18 %, each leg 14 %**.
- Child's palm – **1 % TBSA**.
- Parkland formula – **4 ml × kg × % TBSA, half in first 8 hrs**.
- Best guide of fluid resuscitation – **urine output 1 ml/kg/hr (child)**.
- Burn first aid – **cool running water 20 min, no ice/ toothpaste**.
- Most common cause of death after 72 hrs in burns – **sepsis**.

17 Behavioural & Developmental Disorders, Child Abuse, Accidents & Poisoning

17.1 Common Behaviour Problems

Problem	Age	Key points & management
Thumb sucking	Normal **< 4 yrs**	Ignore in toddler; after 4–5 yrs – **dental malocclusion**; **positive reinforcement, distraction**; no punishment/ bitter substances forcibly
Nail biting	School age	Anxiety – address stress, keep nails short
Temper tantrums	**1–3 yrs (toddler)** – peak 2 yrs ("terrible twos")	**Ignore the behaviour (if safe), stay calm, be consistent**, **time-out (1 min per year of age)**, don't give in, praise good behaviour
Breath-holding spells	6 months – 5 yrs	Triggered by anger/ pain; benign; reassure; iron therapy; no punishment
Pica	> 2 yrs	Eating non-food (mud, chalk, paint) – **iron deficiency, lead poisoning, worms**; investigate & treat
Enuresis	**≥ 5 yrs**	See GU chapter – alarm, star chart, desmopressin
Encopresis	**≥ 4 yrs**	Faecal soiling – constipation with overflow; bowel training
Sleep problems	Night terrors (**non-REM, child doesn't remember**), nightmares (REM, remembers), sleep walking (somnambulism)	Reassure, regular routine, safety
Stuttering (stammering)	2–5 yrs (developmental)	Don't interrupt/ criticise; speech therapy if persists
Sibling rivalry	New baby	Prepare child, involve in care, spend special time
School phobia (refusal)	School age	Separation anxiety – **gradual return to school**
Lying, stealing	Preschool (fantasy) normal	Older – identify cause, consistent discipline
Juvenile delinquency	< 18 yrs	Antisocial acts – **Juvenile Justice Act 2015**, observation homes, counselling
Substance abuse	Adolescents	Solvent sniffing (whitener, glue), tobacco, alcohol

17.2 Neurodevelopmental Disorders

ADHD (Attention Deficit Hyperactivity Disorder)

- Onset **before 12 yrs**, symptoms in **≥ 2 settings** (home & school) for **≥ 6 months**; **boys > girls**.
- **Triad – inattention, hyperactivity, impulsivity** (can't sit still, talks excessively, interrupts, careless mistakes, loses things).
- Rx – **behaviour therapy (first-line in < 6 yrs)**, **parent training**, classroom modification (front seat, short tasks, structured routine), **methylphenidate (stimulant – drug of choice)** – side effects: **anorexia, weight loss, insomnia (give in morning), growth suppression**; atomoxetine.

Autism Spectrum Disorder (ASD)

- **Onset in early childhood (< 3 yrs)**; boys 4 : 1.
- **Deficits in social communication & interaction** – **poor eye contact**, doesn't respond to name, **no social smile/ pointing (joint attention)**, no pretend play, prefers to be alone, **delayed/ abnormal speech (echolalia)**; **restricted, repetitive behaviours** – **hand flapping, rocking, spinning**, insistence on sameness (routine), lining up toys, unusual sensory responses.
- **Screening – M-CHAT (16–30 months)**; **INCLEN/ ISAA** (India).
- **Management** – **early intensive behavioural intervention (ABA – applied behaviour analysis)**, speech & occupational therapy (sensory integration), special education; **no cure**; risperidone for irritability; **structured, predictable environment**; **vaccines (MMR) do NOT cause autism**.

Intellectual Disability (ID – Mental Retardation)

- **IQ < 70** + deficits in adaptive functioning, **onset during developmental period (< 18 yrs)**.
- **IQ = (Mental age ÷ Chronological age) × 100**.

Severity	IQ	Remarks
Mild	**50–69**	**~85 %**; **"educable"**, achieve ~6th grade, independent living with support
Moderate	**35–49**	**"Trainable"**, self-care, semi-skilled work under supervision
Severe	**20–34**	Basic self-care with training, constant supervision
Profound	**< 20**	Total care, often with physical disabilities

- **Causes** – **Down syndrome (most common genetic)**, **Fragile X (most common inherited)**, **congenital hypothyroidism (most common preventable)**, birth asphyxia, prematurity, infections (TORCH, meningitis), malnutrition, iodine deficiency, fetal alcohol syndrome, lead poisoning, PKU.
- **Management** – early intervention & stimulation, special education, vocational training, **parental counselling**, **NIOS, disability certificate (RPwD Act 2016 – 21 disabilities)**, institutions – **NIEPID (Secunderabad)**.

Learning Disability

- **Dyslexia (reading – most common)**, **dysgraphia (writing)**, **dyscalculia (mathematics)**; **normal intelligence**; remedial education.

17.3 Child Abuse & Neglect

- **Types** – **physical abuse**, **sexual abuse**, **emotional/ psychological abuse**, **neglect (most common type)**, Munchausen syndrome by proxy (fabricated illness by caregiver).
- **Indicators of physical abuse**:
- **History inconsistent with injury/ developmental stage**, **delay in seeking care**, changing stories.
- Bruises of **different ages/ colours** (red-purple fresh → blue → green → yellow → brown), **patterned bruises (belt, hand, bite marks)**, bruises in non-ambulant infant, on **back, buttocks, ears, cheeks, genitals** (non-bony areas).
- **Burns – cigarette (round), immersion (glove/ stocking), iron-shaped**.
- **Fractures – spiral (twisting), posterior rib, metaphyseal corner, multiple at different healing stages, skull**.
- **Shaken baby syndrome (abusive head trauma)** – **< 1 yr**; **subdural haematoma + retinal haemorrhages** (± rib fractures), irritability, vomiting, seizures, no external injury.
- **Sexual abuse** – genital injury/ bleeding, STI, pregnancy, **age-inappropriate sexual behaviour/ knowledge**, regression, fear of particular person.
- **Neglect** – FTT, poor hygiene, missed immunisations, untreated illness, unsupervised child.
- **Behavioural signs** – **"frozen watchfulness"**, withdrawn, fearful of parents/ adults, aggressive, school problems.

Nurse's role

- **Recognise, document (exact words of child in quotes, body map, photographs), report**; **treat injuries**; **ensure child safety (hospital admission if needed)**; **don't confront or accuse parents**; **don't interrogate repeatedly**; use **non-leading open questions**; inform **CWC (Child Welfare Committee)/ police/ Special Juvenile Police Unit, CHILDLINE 1098**.
- **POCSO Act 2012** – **mandatory reporting of sexual offences against < 18 yrs by any person, including doctors & nurses** – **failure to report is punishable**; child-friendly procedures (statement at child's place, female police officer, no aggressive questioning, in-camera trial, special courts); **POCSO amendment 2019 – death penalty for aggravated penetrative sexual assault**.
- **Medical examination of a female child – by a woman doctor**, in presence of parent/ trusted person.
- **Good touch – bad touch** education in schools.

17.4 Accidents & Injuries

- **Accidents – leading cause of death in children > 1 yr** (globally); types by age: **infant – falls, suffocation, aspiration, burns**; **toddler – falls, poisoning, drowning, burns (scalds)** – **most accident-prone (curious, mobile, no sense of danger)**; preschool – road traffic (pedestrian), drowning; school – **RTA, sports, bicycle**; adolescent – **RTA, drowning, violence, suicide**.
- **Prevention** – **"supervision is the best prevention"**; childproofing (**socket covers, stair gates, window guards, cabinet locks**), medicines & chemicals locked & out of reach in **original containers (not in soft-drink bottles)**, **child-resistant caps**, no small objects (< 3.2 cm – coins, beads, button batteries, peanuts) for < 3–4 yrs, **never leave child alone near water (buckets, tubs, ponds)**, car seats (rear-facing infants), helmets, **sleep on back (SIDS)**.
- **Drowning** – toddlers (buckets, tanks); rescue breaths first, CPR, prevent hypothermia, observe 24 hrs (secondary drowning).

- **SIDS (Sudden Infant Death Syndrome)** – **2–4 months**, during sleep; prevention – **"Back to Sleep" (supine)**, firm mattress, no soft bedding/ pillows/ toys, **no smoking**, **room sharing without bed sharing**, avoid overheating, **breastfeeding**, pacifier at sleep.

17.5 Poisoning

- **Most common age – 1–3 yrs (toddler, accidental)**; adolescents – intentional.
- **Common agents in India** – **kerosene (most common in children)**, **household products** (detergents, bleach, toilet cleaners), **medicines (iron, paracetamol)**, **organophosphates (insecticides)**, rodenticides, plant (dhatura), **snake/ scorpion bite**.

Poison	Key points	Specific treatment/ antidote
Kerosene/ hydrocarbons	**Aspiration pneumonitis (main danger)** – cough, choking, tachypnoea, fever	**NO induced vomiting, NO gastric lavage**; O_2, observe 6–8 hrs, CXR; **no antibiotics/ steroids routinely**
Paracetamol	Hepatotoxicity (toxic dose **> 150 mg/kg**); asymptomatic first 24 hrs	**N-acetylcysteine (NAC)** – best within 8 hrs; levels at 4 hrs (Rumack-Matthew nomogram)
Iron	**> 60 mg/kg elemental – serious**; vomiting, GI bleeding → apparent recovery → shock, hepatic failure; radiopaque tablets	**Desferrioxamine IV** (vin-rosé coloured urine); **activated charcoal does NOT bind iron**
Organophosphates	**DUMBELS** (Diarrhoea, Urination, Miosis, Bronchorrhoea/ bradycardia, Emesis, Lacrimation, Salivation), fasciculations	**Atropine** (till secretions dry/ HR > 80) + **pralidoxime (PAM)**; remove clothes, wash skin
Datura (anticholinergic)	**"Dry as bone, red as beet, hot as hare, blind as bat, mad as hatter"** – dilated pupils, flushing, delirium	Physostigmine (rarely), supportive
Lead	Pica, old paint; **anaemia (basophilic stippling), abdominal colic, encephalopathy, lead lines**	**Chelation – succimer (DMSA), $CaNa_2EDTA$, BAL**
Corrosives (acids/ alkalis)	Burns of mouth/ oesophagus, drooling, stridor	**NO emesis, NO lavage, NO neutralisation**; dilute with small sips of water/ milk only if able to swallow; endoscopy (12–24 hrs)
Benzodiazepines	Sedation	**Flumazenil** (caution)
Opioids	**Pin-point pupils, respiratory depression, coma**	**Naloxone**
Snake bite (neurotoxic – cobra/ krait; haemotoxic – viper)	Ptosis, respiratory paralysis / bleeding (**20-min whole blood clotting test – 20WBCT**)	**Polyvalent ASV**; **immobilise limb, reassure, no tourniquet, no cutting/ sucking**, remove rings; neostigmine + atropine (cobra)
Scorpion sting	Autonomic storm, **pulmonary oedema**, myocarditis	**Prazosin** (drug of choice), scorpion antivenom

- **General management** – **ABC first**, decontamination: **activated charcoal 1 g/kg** (within 1 hr; **not for kerosene, corrosives, iron, alcohols, lithium**), **gastric lavage (rarely – within 1 hr, life-threatening, protected airway; contraindicated in corrosives & hydrocarbons)**; **induced emesis (ipecac) – no longer recommended**; enhance elimination; **specific antidote**; supportive care; **psychosocial assessment & prevention education**.
- **Poison control centre** – AIIMS Delhi (National Poison Information Centre).

17.6 Adolescent Health

- **Puberty** – **girls 8–13 yrs** (thelarche – first sign; menarche ~2–2.5 yrs later, after peak height velocity), **boys 9–14 yrs** (**testicular enlargement – first sign**); **Tanner staging (SMR 1–5)**.
- **Problems** – **anaemia (most common nutritional)**, acne, **body image issues**, eating disorders (**anorexia nervosa – girls, BMI < 17.5, amenorrhoea, fear of weight gain; bulimia**), **substance abuse**, **teenage pregnancy**, STIs/ HIV, **depression & suicide**, **RTAs**, internet addiction, academic stress, **menstrual problems (dysmenorrhoea)**, PCOS.
- **Services** – **RKSK – AFHCs ("Ujjwal" clinics)**, **peer educators (Saathiya)**, **WIFS (weekly IFA)**, **Menstrual Hygiene Scheme** (sanitary napkins), **Kishori Shakti Yojana/ SABLA (adolescent girls)**, school health & wellness programme (Ayushman Bharat), **HPV vaccine (9–14 yrs girls)**.
- **HEEADSSS assessment** – **H**ome, **E**ducation/ employment, **E**ating, **A**ctivities, **D**rugs, **S**exuality, **S**uicide/ depression, **S**afety – **confidential, non-judgmental**, interview alone.

PYQ style

- Temper tantrums peak – **toddler (2 yrs)**; management – **ignore, consistent, time-out**.
- Time-out duration – **1 minute per year of age**.
- Pica associated with – **iron deficiency, lead poisoning**.
- ADHD drug – **methylphenidate**; side effect – **anorexia, insomnia**.
- Autism features – **poor eye contact, echolalia, repetitive behaviour**; screening – **M-CHAT**.
- Mild intellectual disability IQ – **50–69 (educable)**; moderate – **35–49 (trainable)**.
- IQ formula – **MA/CA × 100**.
- Most common type of child abuse – **neglect**.
- Shaken baby syndrome – **subdural haematoma + retinal haemorrhages**.
- POCSO Act – **2012**, mandatory reporting, child **< 18 yrs**.
- Most accident-prone age – **toddler**; commonest childhood poisoning – **kerosene**.
- Kerosene poisoning – **don't induce vomiting** (aspiration pneumonitis).
- Paracetamol antidote – **N-acetylcysteine**; iron – **desferrioxamine**; OP – **atropine + PAM**.
- Activated charcoal dose – **1 g/kg**; not useful in – **iron, kerosene, corrosives**.
- SIDS prevention – **supine sleeping**.
- First sign of puberty – girls **thelarche**; boys **testicular enlargement**.

18 Paediatric Emergencies & Life Support

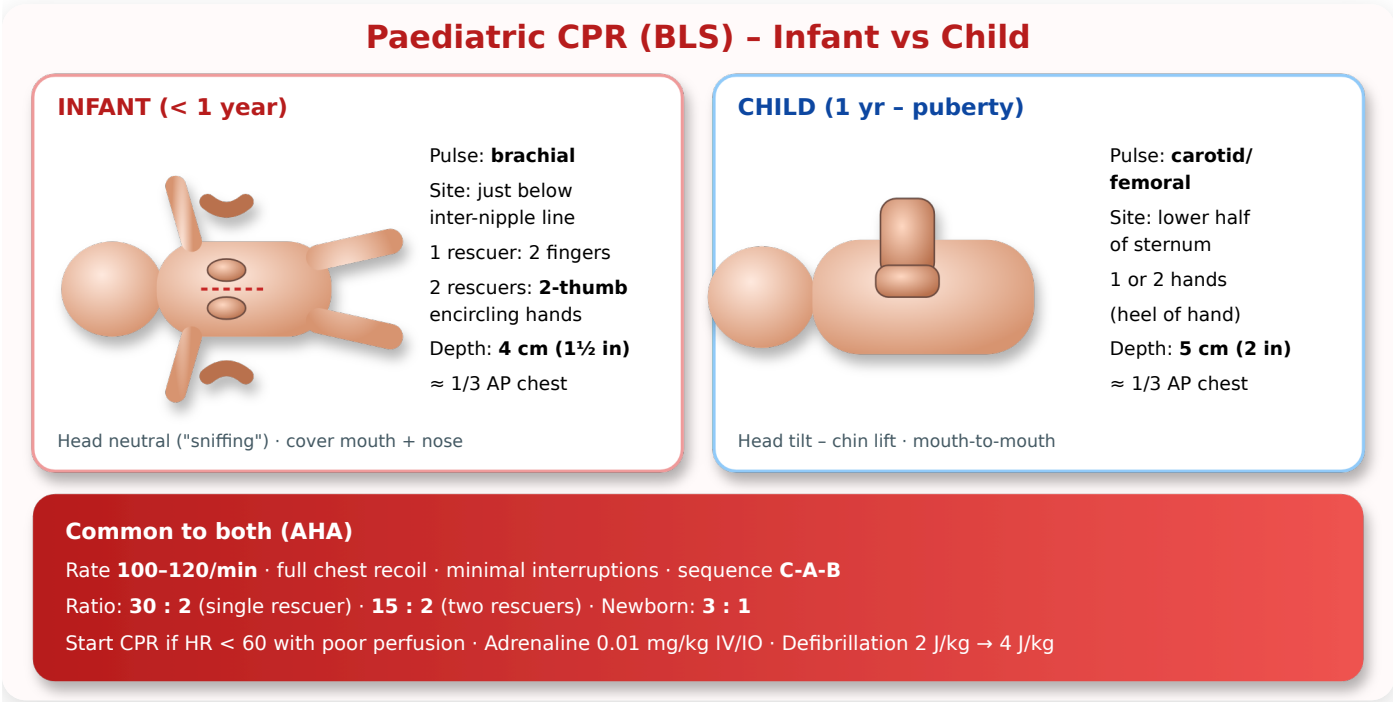
18.1 ETAT – Emergency Triage Assessment & Treatment (WHO)

- **Triage** = sorting patients by urgency on arrival; done in **< 1 minute**.

Category	Signs	Action
E – Emergency (red)	**Obstructed/ absent breathing, severe respiratory distress, central cyanosis, shock (cold hands + CRT > 3 sec + weak fast pulse), coma (AVPU – P/U), convulsing now, severe dehydration (lethargy + sunken eyes + very slow skin pinch)**	**Immediate treatment**
P – Priority (yellow)	**"3 TPR + MOB"** – **T**iny baby (< 2 months), **T**emperature very high, **T**rauma; **P**allor (severe), **P**oisoning, **P**ain (severe); **R**espiratory distress, **R**estless/ irritable/ lethargic, **R**eferral (urgent); **M**alnutrition (visible severe wasting), **O**edema of both feet, **B**urns (major)	Assess & treat **next/ quickly**
Q – Queue / Non-urgent (green)	None of the above	Wait in queue

- **ABCD** – **A**irway, **B**reathing, **C**irculation, **C**oma, **C**onvulsion, **D**ehydration (severe).
- **AVPU scale** – **A**lert, responds to **V**oice, responds to **P**ain, **U**nresponsive (**P or U = emergency**, ≈ GCS ≤ 8).

18.2 Paediatric Basic Life Support (CPR)



Infant vs child CPR – hand position, depth, ratios & key numbers

- **Sequence – C-A-B** (Compressions, Airway, Breathing) (AHA); for drowning/ asphyxial arrest, early breaths are especially important. **Lone rescuer, unwitnessed collapse – 2 minutes (5 cycles) of CPR**

before leaving to call for help; witnessed sudden collapse – call first (shockable).

- **Check responsiveness & breathing (≤ 10 sec)**, pulse check – **infant: brachial**, **child: carotid/ femoral**.

	Newborn	Infant (< 1 yr)	Child (1 yr – puberty)
Compression method	**2-thumb encircling**	**2 fingers (1 rescuer)** / **2-thumb encircling (2 rescuers)**	**1 or 2 hands (heel)**
Site	Lower third of sternum	**Just below inter-nipple line** (lower half of sternum)	**Lower half of sternum**
Depth	1/3 AP diameter	**~4 cm (1.5 inch) – 1/3 AP**	**~5 cm (2 inch) – 1/3 AP**
Rate	**90 compressions + 30 breaths/min**	**100–120/min**	**100–120/min**
Ratio	**3 : 1**	**30 : 2 (single), 15 : 2 (two rescuers)**	**30 : 2 (single), 15 : 2 (two rescuers)**
Start compressions if	**HR < 60 after 30 sec effective PPV**	No pulse / **HR < 60 with poor perfusion**	No pulse / HR < 60 with poor perfusion

- **Airway** – **infant – neutral (sniffing) position** (don't hyperextend – large occiput, soft trachea; **towel under shoulders**); child – **head tilt–chin lift**; trauma – **jaw thrust**.
- **Breaths** – **infant: mouth-to-mouth-and-nose**; each breath over 1 sec, **just enough to make chest rise**; with advanced airway – **1 breath every 2–3 sec (20–30/min)** (AHA 2020).
- **Rescue breathing only (pulse present, inadequate breathing)** – **1 breath every 2–3 sec**.
- **AED/ defibrillation** – **paediatric pads/ attenuator for < 8 yrs**; **2 J/kg first shock → 4 J/kg** (subsequent ≥ 4 J/kg, max 10 J/kg); shockable rhythms – VF/ pulseless VT (less common in children – **most paediatric arrests are asphyxial → bradycardia → asystole**).
- **Most common cause of cardiac arrest in children – respiratory failure/ hypoxia** (not primary cardiac).

Choking (FBAO)

Infant (< 1 yr)	Child (> 1 yr)
5 back blows (head down, prone on forearm, between shoulder blades) **+ 5 chest thrusts** (2 fingers, CPR site) – repeat	**Abdominal thrusts (Heimlich manoeuvre)** – repeat until object out/ unresponsive
No abdominal thrusts (liver injury)	Coughing effectively – **encourage coughing, don't interfere**
If unresponsive → **CPR; look in mouth before breaths, remove only if visible** – **no blind finger sweep**	Same

18.3 Shock in Children

- **Hypovolaemic (most common – diarrhoea/ dehydration, haemorrhage, burns)**, **septic (distributive)**, cardiogenic, obstructive, anaphylactic.
- **Compensated shock** – **tachycardia (earliest sign)**, **CRT > 3 sec (> 2 sec)**, cold peripheries, weak peripheral pulses, ↓ urine output, irritability; **BP maintained**.
- **Hypotension is a LATE sign in children** (decompensated – pre-arrest).

- **Fluid bolus** – **20 ml/kg NS/ RL over 5–20 min**, reassess (liver, crackles, HR, CRT); repeat up to **40–60 ml/kg** in first hour (septic, no overload signs); **cautious 10 ml/kg in SAM (15 ml/kg over 1 hr – WHO), heart disease, severe anaemia (blood)**, DKA.
- **Septic shock** – fluids + **antibiotics within 1 hr**, **adrenaline/ noradrenaline** (vasoactive) if fluid-refractory, correct hypoglycaemia & hypocalcaemia.
- **Vascular access** – peripheral IV; **intraosseous (proximal tibia) if IV fails in 90 sec/ 2 attempts**.

18.4 Emergency Drug Doses (Paediatric)

Drug	Dose
Adrenaline (cardiac arrest)	**0.01 mg/kg IV/ IO = 0.1 ml/kg of 1 : 10,000** (max 1 mg), every 3–5 min; ET 0.1 mg/kg (1:1000)
Adrenaline (anaphylaxis)	**0.01 mg/kg IM of 1 : 1000 (0.01 ml/kg) – max 0.5 mg**, anterolateral thigh; repeat 5–15 min
Atropine	0.02 mg/kg IV (bradycardia from vagal/ AV block)
Amiodarone	5 mg/kg IV (VF/ pVT)
10 % dextrose (hypoglycaemia)	**2–5 ml/kg IV** (neonate 2 ml/kg)
10 % calcium gluconate	**0.5–1 ml/kg slow IV** (max 20 ml) – hypocalcaemia/ hyperkalaemia
Sodium bicarbonate	1 mEq/kg (prolonged arrest/ hyperkalaemia – not routine)
Midazolam (seizure)	0.2–0.3 mg/kg buccal/ intranasal; **0.1–0.2 mg/kg IV**
Lorazepam	**0.1 mg/kg IV** (max 4 mg)
Diazepam	0.3 mg/kg IV / **0.5 mg/kg rectal**
Phenytoin/ fosphenytoin	**20 mg/kg (PE)** IV (in NS, ≤ 1 mg/kg/min)
Phenobarbitone	20 mg/kg IV
Mannitol 20 %	**0.5–1 g/kg (2.5–5 ml/kg)** IV over 20–30 min – ↑ ICP
Naloxone	0.1 mg/kg (opioid)
Salbutamol nebulisation	**2.5 mg (< 20 kg) / 5 mg (> 20 kg)**
Hydrocortisone	4–5 mg/kg IV (asthma, anaphylaxis, adrenal crisis)
Paracetamol	**15 mg/kg/dose** 4–6 hourly
Ibuprofen	10 mg/kg/dose 6–8 hourly (> 3 months)
ET tube size (uncuffed)	**(Age/4) + 4** mm; cuffed (age/4) + 3.5; depth (cm) = **3 × ET size**

- **Broselow tape** – length-based weight & dose estimation in emergencies.
- **Weight estimation (APLS)** – 1–12 months: (0.5 × age in months) + 4; 1–5 yrs: (2 × age) + 8; 6–12 yrs: (3 × age) + 7.

18.5 Respiratory Distress & Oxygen Therapy

- **Signs** – **tachypnoea (earliest)**, **nasal flaring**, **chest retractions (subcostal, intercostal, suprasternal)**, **grunting (expiratory – keeps alveoli open)**, head bobbing, stridor/ wheeze,

cyanosis, ↓ SpO_2 (< 90–92 %), altered sensorium, **silent chest/ bradycardia/ apnoea = impending respiratory failure**.

- **Position** – **position of comfort** (sitting/ in parent's lap), **don't force supine**.
- **O_2 delivery** –

Device	Flow	FiO_2
Nasal prongs/ cannula	**0.5–1 L/min infant, 1–2 L/min (≤ 4) child**	~30–40 %
Nasal catheter	0.5–1 L/min	–
Head box (oxygen hood)	**5–7 L/min** (washes CO_2)	40–80 %
Simple face mask	5–10 L/min	40–60 %
Non-rebreathing mask	**10–15 L/min**	60–90 %
CPAP (bubble CPAP)	5–7 cm H_2O	Neonatal RDS

- **Humidify O_2** (dries mucosa), monitor SpO_2 (target **≥ 94 %** in emergency; **90–95 % preterm**), O_2 is a drug – prescribed, avoid hyperoxia in preterm (ROP).

18.6 Raised ICP & Unconscious Child

- **Signs** – infants: **bulging fontanelle, ↑ OFC, high-pitched cry, setting-sun eyes, irritability, poor feeding**; older: **headache, projectile vomiting, papilloedema, diplopia**, ↓ consciousness; **late – Cushing's triad (↑ BP, ↓ HR, irregular respiration), pupillary changes, decerebrate/ decorticate posturing**.
- **Nursing** – **head end elevated 15–30°**, **head in midline** (venous drainage), avoid neck flexion/ hip flexion, **minimal handling & stimulation**, prevent coughing/ straining (stool softeners), suction ≤ 10–15 sec (pre-oxygenate), **avoid hypotonic fluids** (restrict to 2/3 maintenance), maintain normothermia & normoglycaemia, **mannitol/ 3 % saline**, seizure control, neuro observations (GCS, pupils) hourly.
- **Paediatric GCS (< 2 yrs / pre-verbal)** – Eye (4), **Verbal: 5 coos/ babbles, 4 irritable cry, 3 cries to pain, 2 moans to pain, 1 none**, Motor (6 – spontaneous movement); **GCS ≤ 8 = severe – protect airway (intubation)**.
- **Care of unconscious child** – airway (lateral/ recovery position, suction), **eye care (saline/ lubricant, tape lids)**, **oral care**, **2-hrly position change**, skin care, **NG feeding**, bladder (catheter/ diaper) & bowel care, ROM exercises, talk to child (hearing last to go), family support.

18.7 Other Emergencies

- **Anaphylaxis** – sudden urticaria, angioedema, stridor, wheeze, hypotension after food/ drug/ sting → **IM adrenaline (first & most important)**, call help, supine with legs raised (or sitting if breathless), O_2, IV fluids 20 ml/kg, then antihistamine & hydrocortisone (adjuncts), observe ≥ 6–24 hrs (biphasic).
- **Acute severe asthma** – O_2, **salbutamol nebs (back-to-back)**, ipratropium, steroids, **IV $MgSO_4$ 50 mg/kg**.
- **Hypoglycaemia** – < 54 mg/dl (or < 45 neonate) – **10 % dextrose 2–5 ml/kg IV** (avoid 25 %/ 50 % in small children), then infusion & feeds.
- **Hyperthermia/ high fever** – > 40 °C – remove excess clothes, **tepid sponging (lukewarm water, not cold/ alcohol)** – after antipyretic, fluids; heat stroke – active cooling.

- **Hypothermia** – rewarm (KMC, warm blankets, radiant warmer), check glucose.
- **Status epilepticus** – see neuro chapter (benzodiazepine first).
- **Head injury, poisoning, burns, drowning** – see respective chapters.

PYQ style

- ETAT emergency signs – **ABCD: airway/ breathing, circulation, coma/ convulsion, dehydration (severe)**.
- AVPU – **P or U = emergency**.
- Paediatric CPR sequence – **C-A-B**; rate – **100–120/min**.
- Infant compression – **2 fingers/ 2-thumb, just below nipple line, 4 cm**.
- Child compression depth – **5 cm**; two-rescuer ratio – **15 : 2**.
- Pulse check in infant – **brachial**.
- Infant choking – **5 back blows + 5 chest thrusts**; child – **abdominal thrusts**.
- Most common cause of cardiac arrest in children – **hypoxia/ respiratory failure**.
- Earliest sign of shock in child – **tachycardia**; late sign – **hypotension**.
- Fluid bolus in shock – **20 ml/kg NS/ RL**.
- Adrenaline in arrest – **0.01 mg/kg (0.1 ml/kg of 1:10,000)**; anaphylaxis – **0.01 mg/kg IM of 1:1000**.
- Defibrillation – **2 J/kg then 4 J/kg**.
- ET tube size – **(age/4) + 4**.
- IO site – **proximal tibia**.
- Head box O_2 flow – **5–7 L/min**.
- ↑ ICP position – **head elevated 30°, midline**.
- First drug in anaphylaxis – **IM adrenaline**.

19 Quick Revision – Master Tables & 150 One-Liners

19.1 Master Table – Numbers

Topic	Value
Neonate / infant / toddler / preschool / school	**0–28 d / < 1 y / 1–3 y / 3–6 y / 6–12 y**
Adolescent (WHO)	**10–19 yrs**
Birth weight / length / OFC	**~3 kg / 50 cm / 33–35 cm**
Weight doubles / triples / quadruples	**5 months / 1 yr / 2 yrs**
Length at 1 yr / doubles	**75 cm / 4 yrs**
OFC at 1 yr	**45–47 cm**
Head = chest circumference	**1 yr**
AF / PF closure	**12–18 months / 6–8 weeks**
Weight formula 1–6 yrs	**2 × age + 8**
Height formula 2–12 yrs	**6 × age + 77**
First tooth / all milk teeth	**6 months / 2½ yrs**
No. of milk teeth	**20** (age in months − 6)
Social smile / neck holding	**6–8 wks / 3 months**
Sits without support / walks alone	**8 months / 12–13 months**
Pincer grasp	**9–10 months**
Copies circle / cross / triangle	**3 / 4 / 5 yrs**
Newborn HR / RR	**120–160 / 30–60**
Min SBP	**70 + 2 × age**
Fast breathing (< 2 m / 2–12 m / 1–5 y)	**≥ 60 / ≥ 50 / ≥ 40**
Normal urine output (infant / child)	**2 / 1 ml/kg/hr**
Maintenance fluid	**100 – 50 – 20 ml/kg/day**
Fluid bolus	**20 ml/kg**
Plan B ORS	**75 ml/kg in 4 hrs**
Plan C RL	**100 ml/kg** (infant 1 + 5 hrs; child ½ + 2½ hrs)
ORS osmolarity / Na	**245 mOsm/L / 75 mmol/L**
Zinc (2–6 m / > 6 m)	**10 / 20 mg × 14 days**
SAM MUAC	**< 11.5 cm**
F-75 / F-100	**75 / 100 kcal per 100 ml**
Vit A prophylaxis	**1 lakh IU at 9 m → 2 lakh IU 6-monthly to 5 yrs (9 doses)**

Topic	Value
Vit K at birth	**1 mg IM** (0.5 mg preterm)
BCG dose	**0.1 ml ID (0.05 ml < 1 month)**
fIPV dose	**0.1 ml ID**
ILR / deep freezer	**+2 to +8 °C / −15 to −25 °C**
Open vial – BCG/ MR/ JE	**4 hrs** · others **28 days**
Neonatal hypothermia (moderate)	**32–35.9 °C**
Neonatal hypoglycaemia	**< 45 mg/dl** → 10 % D 2 ml/kg
Phototherapy wavelength	**450–460 nm**
Exchange transfusion volume	**160 ml/kg (double volume)**
Neonatal CPR ratio	**3 : 1**
Child CPR ratio	**30 : 2 (1 rescuer) / 15 : 2 (2 rescuers)**
Compression depth infant / child	**4 cm / 5 cm**
Adrenaline arrest	**0.01 mg/kg (0.1 ml/kg 1:10,000)**
Defibrillation	**2 → 4 J/kg**
ET tube	**(age/4) + 4**
Paracetamol	**15 mg/kg/dose**
Prednisolone (nephrotic)	**2 mg/kg/day × 6 wks**
Benzathine penicillin (RF)	**1.2 MU every 21 days**
Levothyroxine (CH)	**10–15 µg/kg/day**
DKA insulin	**0.05–0.1 U/kg/hr**
Parkland	**4 ml × kg × % TBSA**
Burns – infant head / leg	**18 % / 14 %**
Enuresis / encopresis after	**5 yrs / 4 yrs**
Febrile seizure age	**6 months – 5 yrs**
Status epilepticus	**≥ 5 min**
Mantoux positive	**≥ 10 mm**
IQ – mild / moderate / severe / profound	**50–69 / 35–49 / 20–34 / < 20**

19.2 Master Table – Signs & Eponyms

Sign / Name	Condition
Koplik's spots	Measles
Dew drop on rose petal	Chickenpox
Forchheimer spots	Rubella
Slapped cheek	Parvovirus B19 (fifth disease)

Sign / Name	Condition
Strawberry tongue	Scarlet fever, Kawasaki
Bull neck	Diphtheria
Risus sardonicus	Tetanus
Steeple sign	Croup
Thumb sign	Epiglottitis
Boot-shaped heart	TOF
Egg on string	TGA
Snowman heart	TAPVC
Machinery murmur	PDA
Rib notching	Coarctation of aorta
Olive-shaped mass	Pyloric stenosis
Red currant jelly stool / Dance's sign	Intussusception
Double bubble	Duodenal atresia
Ribbon stools	Hirschsprung
Setting-sun eyes / Macewen's sign	Hydrocephalus
Kernig's / Brudzinski's	Meningitis
Gowers' sign	Duchenne muscular dystrophy
Ortolani / Barlow	DDH
Galeazzi (Allis) sign	DDH (unequal knee height)
Adam's forward bend	Scoliosis
Leukocoria	Retinoblastoma
Raccoon eyes	Neuroblastoma
Hair-on-end skull	Thalassaemia
Craniotabes, rachitic rosary	Rickets
Bitot's spots	Vitamin A deficiency
Frog position	Scurvy
Flag sign	Kwashiorkor
Simian crease, sandal gap	Down syndrome
Rocker-bottom feet	Edwards syndrome
Webbed neck	Turner syndrome
Musty odour	PKU
Salty skin	Cystic fibrosis
Jones criteria	Rheumatic fever
New Ballard score	Gestational age of newborn
Silverman-Anderson	RDS score

Sign / Name	Condition
Kramer's rule	Neonatal jaundice spread
Ponseti	Clubfoot
Ramstedt	Pyloromyotomy
Kasai	Biliary atresia
Bryant's traction	Femur fracture < 2 yrs
Lund & Browder	Burns chart for children
Holliday-Segar	Maintenance fluids

19.3 150 One-Liners

One-liners 1–50

1. Father of Paediatrics – **Abraham Jacobi**.
2. Neonatal period – **first 28 days**.
3. Most sensitive health indicator – **IMR**.
4. Commonest cause of neonatal death – **prematurity/ LBW**.
5. Child helpline – **1098**.
6. ICDS started – **2 Oct 1975**.
7. RBSK – **4 D's, 0–18 yrs**.
8. RKSK – **10–19 yrs**.
9. HBNC visits (institutional) – **6**.
10. HBYC visits – **3, 6, 9, 12, 15 months**.
11. India polio-free – **27 March 2014**.
12. Mission Indradhanush – **25 Dec 2014**.
13. Children's Day – **14 November**.
14. Growth is – **quantitative**; development – **qualitative**.
15. Direction of development – **cephalocaudal, proximodistal**.
16. Fastest growth – **infancy**.
17. Birth weight doubles – **5 months**.
18. Head = chest – **1 year**.
19. AF closes – **18 months**.
20. First tooth – **lower central incisor, 6 months**.
21. First permanent tooth – **1st molar, 6 yrs**.
22. Social smile – **6–8 weeks**.
23. Neck holding – **3 months**.
24. Pincer grasp – **9–10 months**.
25. Walks alone – **12–13 months**.
26. Tricycle – **3 yrs**; hops – **4 yrs**.
27. Erikson – infant – **trust vs mistrust**.
28. Erikson – toddler – **autonomy vs shame & doubt**.
29. Erikson – school age – **industry vs inferiority**.
30. Piaget 0–2 yrs – **sensorimotor (object permanence)**.
31. Egocentrism – **preoperational**.
32. Toddler play – **parallel**.
33. School-age play – **cooperative**.
34. Stunting – **chronic malnutrition**.
35. Exclusive breastfeeding – **6 months**.
36. Complementary feeding from – **6 months**.
37. Whey : casein in breast milk – **60 : 40**.
38. SAM MUAC – **< 11.5 cm**.
39. Hallmark of kwashiorkor – **oedema**.
40. Old man face – **marasmus**.

41. F-75 – **stabilisation**.
42. Iron in SAM – **rehabilitation phase**.
43. Night blindness – **vitamin A**.
44. Earliest sign of rickets – **craniotabes**.
45. Scurvy – **vitamin C (frog position)**.
46. Pellagra – **3 D's (niacin)**.
47. Most common nutritional anaemia – **iron deficiency**.
48. IFA syrup (6–59 m) – **1 ml biweekly**.
49. Iodised salt (consumer) – **15 ppm**.
50. Pentavalent = **DPT + Hep B + Hib**.

One-liners 51–100

51. BCG route – **intradermal, left arm**.
52. MR route – **subcutaneous, right arm**.
53. Hep B birth dose – **within 24 hrs**.
54. Most heat-sensitive vaccine – **OPV**.
55. Most freeze-sensitive – **Hepatitis B**.
56. VVM detects – **heat exposure**.
57. Shake test detects – **freezing**.
58. Reconstituted MR discard – **4 hrs**.
59. Toxic shock syndrome – **reuse of reconstituted vial**.
60. AEFI observation – **30 min**.
61. Separation anxiety first stage – **protest**.
62. Preschool fear – **body mutilation**.
63. Pain scale 2 m–7 y – **FLACC**.
64. FACES scale – **≥ 3 yrs**.
65. IM site infant – **vastus lateralis**.
66. Ear drops < 3 yrs – **down & back**.
67. Young's rule – **age/(age + 12)**.
68. Microdrip – **60 drops/ml**.
69. Cleft lip restraint – **elbow**.
70. Preterm – **< 37 wks**.
71. LBW – **< 2500 g**.
72. Heat loss at birth – **evaporation**.
73. Non-shivering thermogenesis – **brown fat**.
74. KMC – **skin-to-skin + EBF**.
75. Jaundice < 24 hrs – **pathological**.
76. Kernicterus – **basal ganglia**.
77. Phototherapy – **cover eyes**.
78. Neonatal sepsis earliest sign – **refusal to feed**.
79. RDS – **surfactant deficiency**.
80. Pneumatosis intestinalis – **NEC**.
81. IMNCI in India – **2003**.
82. Pink – **urgent referral**.
83. Diarrhoea – **≥ 3 loose stools/24 hrs**.
84. Persistent diarrhoea – **≥ 14 days**.
85. Rotavirus – **commonest diarrhoea < 2 yrs**.
86. Dysentery – **Shigella**.
87. Skin pinch very slow – **> 2 sec**.
88. Pinworm test – **cellophane tape**.
89. Croup – **parainfluenza, steeple sign**.
90. Epiglottitis – **Hib, don't examine throat**.
91. Bronchiolitis – **RSV**.
92. Commonest bacterial pneumonia – **pneumococcus**.

93. Post-tonsillectomy bleeding sign – **frequent swallowing**.
94. Foreign body – **right bronchus**.
95. Cystic fibrosis test – **sweat chloride**.
96. Commonest CHD – **VSD**.
97. Commonest cyanotic CHD – **TOF**.
98. Tet spell position – **knee-chest**.
99. PDA closure – **indomethacin**.
100. Digoxin withhold (infant) – **HR < 90–110**.

One-liners 101–150

101. Rheumatic fever follows – **strep throat**.
102. RF prophylaxis – **penicillin every 21 days**.
103. Kawasaki – **IVIG + aspirin**.
104. Cleft lip repair – **3 months**.
105. TEF – **3 C's, drooling**.
106. Pyloric stenosis – **projectile non-bilious vomiting**.
107. Pyloric stenosis alkalosis – **hypochloraemic hypokalaemic**.
108. Intussusception – **red currant jelly stool**.
109. Hirschsprung – **no meconium in 48 hrs**.
110. Omphalocele – **covered sac**.
111. Biliary atresia – **Kasai < 60 days**.
112. Coeliac – **gluten-free diet**.
113. Nephrotic – **minimal change disease**.
114. AGN urine – **cola coloured**.
115. AGN C3 – **low**.
116. UTI organism – **E. coli**.
117. Wilms – **don't palpate**.
118. Hypospadias – **no circumcision**.
119. Orchidopexy – **6–12 months**.
120. Enuresis treatment – **alarm**.
121. Hydrocephalus earliest sign – **↑ OFC**.
122. VP shunt complication – **blockage**.
123. Meningomyelocele position – **prone**.
124. NTD prevention – **folic acid**.
125. Meningitis infant sign – **bulging fontanelle**.
126. Meningitis sequela – **deafness**.
127. Simple febrile seizure – **< 15 min**.
128. Status epilepticus first drug – **benzodiazepine**.
129. Phenytoin – **gum hypertrophy**.
130. Commonest CP – **spastic**.
131. Reye's – **aspirin + virus**.
132. Thalassaemia – **↑ HbF, hair-on-end**.
133. Iron chelator – **desferrioxamine**.
134. Sickle crisis – **hydration**.
135. Haemophilia A – **factor VIII**.
136. Commonest childhood cancer – **ALL**.
137. Leukocoria – **retinoblastoma**.
138. Congenital hypothyroidism – **TSH screening**.
139. DKA death – **cerebral oedema**.
140. PKU – **musty odour, Guthrie test**.
141. Down syndrome – **trisomy 21, AVSD**.
142. Turner – **45 XO, coarctation**.

143. Measles – **Koplik's spots, vitamin A**.
144. Chickenpox – **no aspirin**.
145. Diphtheria death – **myocarditis**.
146. Neonatal tetanus – **inability to suck**.
147. Clubfoot – **Ponseti**.
148. DDH – **Pavlik harness**.
149. Kerosene poisoning – **no emesis**.
150. Infant choking – **5 back blows + 5 chest thrusts**.



Exam strategy – Asha Didi ki last advice

- GSSSB paper me **Child Health se ~10–15 questions** aate hain – **growth & milestones, immunization schedule (dose/ route/ site), IMNCI & diarrhoea (ORS plans), newborn care, CHD aur common disorders** sabse important hain.
- **Numbers** (milestones ki age, vaccine doses, fluid formulas) ki tables roz 10 minute padho; **age-based questions** me pehle socho "ye kaunsa age group hai".
- **Priority questions**: Airway → Breathing → Circulation; bachche me **tachycardia shock ka earliest sign** hai aur **hypotension late sign**.
- MCQ Bank PDF ke saare questions **2 baar** solve karo, galtiyan notebook me likho.

— Child Health Nursing · GSSSB Staff Nurse Preparation · Padhte raho, revise karte raho! —